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From Electronic Couplings to Exciton Dynamics: Modeling Photophysical Processes in Rubrene Solids

Author:
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Universidad del País Vasco
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October 27, 2025

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Thesis presented in



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Aitor DIAZ-ANDRES

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A thesis submitted in fulfillment of the requirements
for the degree of Doctor of Theoretical Chemistry and Computational Modeling

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Declaration of Authorship

I, Aitor DIAZ-ANDRES, declare that this thesis titled, "From Electronic Couplings to Exciton Dynamics: Modeling Photophysical Processes in Rubrene Solids" and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

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"So, what do you say? Why not help one another on this lonely journey?"

Solaire of Astora - Dark Souls I

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Abstract

Doctor of Theoretical Chemistry and Computational Modeling

From Electronic Couplings to Exciton Dynamics: Modeling Photophysical Processes in Rubrene Solids

by Aitor DIAZ-ANDRES

This Thesis studies the triplet-triplet annihilation process in the crystalline and amorphous solids of rubrene. Triplet annihilation is the process that converts two triplet excitons into a high-energy singlet exciton (twice the energy of the triplets). However, other processes are important to understand the dynamics of these excitons in solids, such as exciton transport, absorption, and emission. In this Thesis, I have focused my efforts on understanding the transport processes of singlet and triplet excitons as well as the interaction of two triplets to form the TT state and the subsequent formation of the singlet state. All these processes have been studied first for the crystalline solid rubrene, and later, thermal vibrations have been included as an approximation of the effect of temperature on solids. With these results, a computational protocol has been designed to calculate all the mentioned couplings, with special mention to the RASCI method to calculate the couplings in which the TT state is involved. Following this protocol, the triplet fusion process in the amorphous solid rubrene was studied. The results have shown that the triplet fusion process is impeded because triplet transport is not allowed. However, the triplet fusion coupling indicates that the process would be allowed in systems where triplet transport is possible.

The second part of the thesis focuses on simulating the processes involved in triplet fusion. For this purpose, I have developed SEDyn, a software for simulating photophysical processes in molecular organic solids. Simulations have been performed on the crystalline and amorphous solids of rubrene. In this thesis, in addition to the exciton transport simulations, the triplet fusion and singlet fission processes in the crystalline solid are simulated, including the effect of temperature. The results show that in this solid, the predominant process is singlet fission, coinciding with the experimental studies. This tells us that SEDyn is capable of simulating complex processes efficiently to help the scientific community discover molecular organic solids (crystalline or amorphous) that can perform the triplet annihilation process efficiently.

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Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Contents

Declaration of Authorship iii

Abstract vii

Acknowledgements ix

1 Introduction 1
- 1.1 Organic Optoelectronics . . . 1
- 1.2 Rubrene: Structure, Photophysics, and Exciton Processes . . . 3
- 1.3 Motivation and Computational Challenges . . . 4
- 1.4 Objectives . . . 4
- 1.5 Structure of the thesis . . . 5

2 Theory and Methods 7
- 2.1 The Kinetic Monte Carlo Method . . . 8
- 2.2 Rate Constant and Fermi's Golden Rule . . . 8
- 2.3 Exciton Energy Transfer Couplings . . . 9
  - 2.3.1 Classical Dipole-Dipole Interaction . . . 9
  - 2.3.2 Half Energy Gap Rule . . . 10
  - 2.3.3 Fragment Difference Methods . . . 10
  - 2.3.4 Multistate generalization . . . 11
  - 2.3.5 Boys Exciton Diabatization of Excited States . . . 11
- 2.4 Triplet Fusion and Singlet Fission . . . 12
  - 2.4.1 Triplet-Pair State . . . 13
- 2.5 Restricted Active Space Configuration Interaction . . . 14
- 2.6 Solid State Morphology . . . 15
- 2.7 Electronic Structure Calculations . . . 16

3 SEDyn Software 17
- 3.1 Software Design Principles . . . 18
  - 3.1.1 Programming Language and Libraries . . . 18
  - 3.1.2 Beyond the APS Methodology . . . 19
  - 3.1.3 Data Representation and Storage . . . 19
  - 3.1.4 Implementation of Kinetic Monte Carlo . . . 22
  - 3.1.5 Classes and Relationships . . . 23
  - 3.1.6 Cutoff Radius & First-Neighbors . . . 34
  - 3.1.7 Trajectory Analysis . . . 34
- 3.2 Machine Learning Training and Input . . . 35
  - 3.2.1 Decision Tree . . . 35
  - 3.2.2 Kernel Ridge Regression . . . 37
  - 3.2.3 Feedforward Neural Network . . . 38
- 3.3 Installation and Execution . . . 40
  - 3.3.1 Installation Requirements . . . 40

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

3.3.2 Compilation and General Usage . . . 40
3.4 Example Simulation . . . 41

4 Electronic Couplings for Triplet-Triplet Annihilation Photon Upconversion in Crystal Rubrene 45
4.1 Singlet and Triplet Exciton Energy Transfer . . . 46
4.2 Triplet-Pair state . . . 48
4.3 Triplet Fusion and Singlet Fission . . . 49
4.4 RASCI Active Space Comparison . . . 51
4.5 Role of Thermal Fluctuations . . . 51
4.6 Structure - Coupling Analysis . . . 55
4.7 Conclusions . . . 56

5 Exciton Dynamics in Crystal and Amorphous Rubrene 57
5.1 Morphology of Amorphous Rubrene . . . 57
5.2 Electronic Couplings in Amorphous Rubrene . . . 58
5.3 Training of ML Models . . . 61
5.4 Simulations of Exciton Dynamics . . . 64
5.4.1 Simulation conditions . . . 65
5.4.2 Diffusion of singlet and triplet excitons . . . 65
5.4.3 Exciton dynamics beyond diffusion . . . 67
5.5 Conclusions . . . 68

6 Final Remarks 69

A Supplementary Material for Chapter 3 71
A.1 Advanced Simulation . . . 71

B Supplementary Material for Chapter 4 75
B.1 Symmetric RASCI Couplings . . . 75
B.2 Median Electronic Couplings . . . 75
B.3 Electronic Coupling Distributions . . . 76
B.4 Additional PCA of SEET Couplings . . . 79
B.5 PCA Analysis of TEET Couplings . . . 82

C Supplementary Material for Chapter 5 87
C.1 MSE and R^2 Values of the Discrete Exploration . . . 87
C.2 Direct and Indirect TF/SF Contributions in Amorphous Solid . . . 88
C.3 Additional Simulations . . . 88

Bibliography 89

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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30/10/2025 11:18
EHU2025E051403

List of Figures

2.1 Visual representation of the cumulative function of the rate constants. The arrow represents the random number, $r\Gamma_{\mathrm{Tot}}$. The chosen rate constant is $\Gamma_{m\to 2}$. 8
2.2 Schematic representation of TF (A) and SF (B) processes. Green arrows represent the TEET (photoabsorption) mechanism to obtain triplet (singlet) excitons in TF (SF). Blue arrows represent the formation (dissociation) of the ^{l}TT state through k_{TTF} (k_{TTD}) in Eq. (2.15). Small purple arrows refer to the TF (SF) process associated with the k_{TF} (k_{SF}) in Eq. (2.16). 12
2.3 Representation of RASCI orbital subspaces RAS1 (doubly occupied orbitals), RAS2 (fully correlated space) and RAS3 (virtual orbitals). . . 14
2.4 Crystal structure of rubrene along the *bc* (A) and *ac* (B) planes. 15

3.1 CM of (A) acetic acid and (B) glycolaldehyde. Hydrogen atoms are not used to compute the CMs. 21
3.2 Eigenvalues of the CMs of acetic acid and glycolaldehyde. Hydrogen atoms are disregarded. 22
3.3 Graphical representation of isolated and non-isolated states according to the first and second-neighbors approach. (A) is a non-isolated case of interacting states, (B) is a non-isolated case of non-interacting states and (C) is an isolated case. 23
3.4 Hierarchy of the main classes and their relationships. Solid arrows refer to strong dependence and dashed arrows refer to weak dependence. Attributes are marked with a light grey border. 24
3.5 Graphical representation of the spherical (A) and ellipsoidal (B) cutoff radius. The spherical cutoff includes X as a first-neighbor of itself, while the ellipsoidal cutoff does not. Note that in a real simulation, the spherical cutoff would include the diagonal molecules as well. . . . 34
3.6 DT graph with three evaluation steps and four return values. Each $T(x_n)$ refers to the evaluation of x_n to an optimized parameter. 35
3.7 FNN graph with three input nodes ($N_x = 3$) and two output nodes. The hidden layer contains four nodes ($N_h = 4$). 39
3.8 Square root variances of each eigenvalue indexes of the CMs of the four pairs in the unit cell of the orthorhombic rubrene crystal. Note that the Y axis is in logarithmic scale. 39

4.1 Crystal structure of rubrene along the *bc* (A) and *ac* (B) planes. Purple arrows indicate the crystal rubrene dimers studied. 46
4.2 Frontier natural orbitals and their electron occupancies for the ^{1}TT state of the stacked (A) and herringbone (B) first-neighbor rubrene crystal dimers computed at the RASCI/cc-pVDZ level. 49

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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30/10/2025 11:18
EHU2025E051403

4.3 Distribution of absolute electronic couplings (in meV) in π-stacked dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. Vertical red lines in (A)-(C) indicate values obtained at 0 K. . . . 53
4.4 Distribution of absolute TF/SF couplings (in meV) in π-stacked dimers computed from 200 MD frameworks. (A) Direct coupling, (B) 2nd order (mediated) correction. . . . 54
4.5 PCA analysis of SEET coupling (in meV) in stacked dimers computed from 200 MD frameworks. . . . 55

5.1 Radial distribution function ($g(r)$) as a function of intermolecular distance (r, in Å) obtained for the simulated amorphous rubrene solid. The red line represents the distance of the first local minimum. . . . 58
5.2 Caption . . . 59
5.3 Distribution of absolute electronic couplings (in meV) in amorphous dimers computed from 9400 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. Vertical red lines indicate mean values. . . . 60
5.4 DT input file (JSON format) for crystal SEET electronic couplings (in eV). $F(i)$ refers to the ith index of the input feature vector. . . . 61
5.5 Surfaces of MSE (A) and R^2 (B) of the discrete exploration of regularization strength (λ) and the width of the Gaussian (σ). Note that the MSE has been calculated in meV. . . . 63
5.6 Predicted vs Test SEET couplings for amorphous rubrene dimers with optimized KRR ($\lambda = 10^{-2.8}$, $\sigma = 100$) model. . . . 64
5.7 Simulation of 10 trajectories of TF/SF starting with 10 S_1 and 20 T_1 excitons. The exciton numbers represent the qualitative exciton density evolution. . . . 67

B.1 Distribution of absolute electronic couplings (in meV) in herringbone dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. . . . 76
B.2 Distribution of absolute electronic couplings (in meV) in a-dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. . . . 77
B.3 Distribution of absolute electronic couplings (in meV) in ac-dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. . . . 78
B.4 PCA analysis of SEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks. . . . 79
B.5 PCA analysis of SEET coupling (in meV) in a-dimers computed from 200 MD frameworks. . . . 80
B.6 PCA analysis of SEET coupling (in meV) in ac-dimers computed from 200 MD frameworks. . . . 81
B.7 PCA analysis of TEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks. . . . 82
B.8 PCA analysis of TEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks. . . . 83
B.9 PCA analysis of TEET coupling (in meV) in a-dimers computed from 200 MD frameworks. . . . 84
B.10 PCA analysis of TEET coupling (in meV) in ac-dimers computed from 200 MD frameworks. . . . 85

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	16/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

C.1 Distribution of absolute direct (A) and indirect (B) contributions of the TF/SF electronic couplings (in meV) extracted from 9400 MD frameworks of amorphous dimers. Vertical red lines indicate mean values. 88
C.2 Simulation of TF/SF starting with 20 S_1 excitons. 88

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	17/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	18/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

List of Tables

4.1 Intermolecular distances (in Å) between the molecular center of masses for the four unique first neighbor dimers in the orthorhombic crystal structure of rubrene. 46
4.2 Electronic couplings (in meV) for SEET and TEET processes in the four first-neighbor crystal dimers computed at the RASCI/cc-pVDZ level. 47
4.3 Electronic couplings (in meV) for SEET and TEET process in the four firt-neighbor crystal dimers computed at the rCAM-B3LYP/cc-pVDZ level. 48
4.4 Electronic couplings (in meV) computed at the RASCI and C-DFT level/cc-pVDZ level through Boys diabatization, and CT energies, E(CT) = (E(AC) + E(CA))/2 (in eV), computed at the C-DFT level/cc-pVDZ level for the four crystalline rubrene dimers. 50
4.5 Electronic couplings (in meV) computed for the crystal stacked structure for SEET, TEET, triplet-triplet binding (J) and ^{1}TT/S$_1$ (TF/SF) computed with RASCI with different RAS2 spaces, with and without hole and particle excitations (h, p), and with different basis sets. 51
4.6 Electronic couplings (in meV) computed for the crystal structure (cryst.) and averaged over 200 snapshots from MD (TA: thermal average) for SEET, TEET, triplet-triplet binding (J), and ^{1}TT/S$_1$ (TF/SF). 52
4.7 ^{1}TT/S$_1$ (TF/SF) electronic couplings (in meV) computed and averaged over 200 snapshots from MD with and without hole and particle excitations (h, p). 54

5.1 Characteristic parameters of crystalline and amorphous solids for the simulations. 65
5.2 Diffusion coefficients (D) in cm^2/s of the simulations of five singlet excitons of the crystal, thermal average and amorphous SEET couplings at 100 K and 300 K. D$_{a,b,c}$ represent the directional diffusion coefficient over a (x), b (y) and c (z) axes of the crystal (amorphous). . . 66
5.3 Diffusion coefficients (D) in cm^2/s of the simulations of ten triplet excitons of the crystal and thermal average TEET couplings at 100 K and 300 K. D$_{a,b,c}$ represent the directional diffusion coefficient over a, b and c axes of the crystal. 67

B.1 Electronic couplings (in meV) computed at the RASCI and C-DFT level/cc-pVDZ level through Boys diabatization, and CT energies, E(CT) = (E(AC) + E(CA))/2 (in eV), computed at the C-DFT level/cc-pVDZ level for the four crystalline rubrene dimers. 75
B.2 Electronic couplings (in meV) computed for the crystal structure (cryst.) and median of 200 snapshots from MD (TM: thermal median) for SEET, TEET, triplet-triplet binding (J), and ^{1}TT/S$_1$ (TF/SF). 75

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	19/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

C.1 MSE values of the discrete exploration for the optimization of the hyperparameters of the KRR model. Value in purple cell indicate the minimum with $\lambda = 10^{-3}$ and $\sigma = 100$. 87
C.2 R^2 values of the discrete exploration for the optimization of the hyperparameters of the KRR model. Value in purple cell indicate the minimum with $\lambda = 10^{-3}$ and $\sigma = 100$. 87

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	20/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

List of Abbreviations

COF	Covalent Organic Framework
CSD	Cambridge Structural Database
DET	Dexter Energy Transfer
DFT	Density Functional Theory
DT	Decision Tree
EDSS	Exciton Dynamics Simulation Software
ESP	Electrostatic Potential
ET	Electron Transfer
FCD	Fragment Charge Difference
FED	Fragment Excitation Difference
FNN	Feedforward Neural Network
FRET	Forster Resonance Energy Transfer
FSD	Fragment Spin Difference
GMH	Generalized Mulliken-Hush
HEG	Half Energy Gap
IC	Internal Conversion
IQR	Interquartile Range
ISC	Intersystem Crossing
KMC	Kinetic Monte Carlo
KRR	Kernel Ridge Regression
MD	Molecular Dynamics
ML	Machine Learning
MLP	Multilayer Preceptron
MP2	Moller-Plesset Perturbation Theory
MPI	Message Passing Interface
MSE	Mean Squared Error
OFET	Organic Field-Effect Transistor
OLED	Organic Light-Emitting Diodes
PCA	Principal Component Analysis
PDB	Protein Data Bank
PES	Potential Energy Surface
PV	Photovoltaic
RASCI	Restricted-Active-Space Configuration Interaction
RDF	Radial Distribution Function
ReLU	Rectified Linear Uniforrm
RISC	Reverse Intersystem Crossing
SEET	Singlet Exciton Energy Transfer
SF	Singlet Fission
SOC	Spin-Orbit Coupling
TADF	Thermally Activated Delayed Fluorescence
TDDFT	Time-Dependent Density Functional Theory
TEET	Triplet Exciton Energy Transfer

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	21/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

TF Triplet Fusion
TTA-UC Triplet Triplet Anihilation Photon Upconversion

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	22/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

List of Publications

Publications included in this thesis:

- "*Electronic Couplings for Triplet–Triplet Annihilation Upconversion in Crystal Rubrene*"
  Diaz-Andres, A.; Tonnelé, C.; Casanova, D. *J. Chem. Theory Comput.* **2024**, 20, 4288–4297

Publications not included in this thesis:

- "*A carnitine-based BODIPY photosensitizer*"
  Maierhofer, L.; Prieto-Montero, R.; Cubiella, T.; Diaz-Andres, A.; Morales-Benítez, N.; Casanova, D.; Martínez-Martínez, V.; Chiara, M.-D.; Mann, E.; Chiara, J. L. *J. Mater. Chem.* **2025**, 13, 4330–4340
- "*Halogen-free photosensitizers based on meso-enamine-BODIPYs for bioimaging and photodynamic therapy*"
  Prieto-Montero, R.; Diaz-Andres, A.; Prieto-Castañeda, A.; Tabero, A.; Longarte, A.; Agarrabeitia, A. R.; Villanueva, A.; Ortiz, M. J.; Montero, R.; Casanova, D.; Martínez-Martínez, V. *J. Mater. Chem. B* **2023**, 11, 169–179
- "*Rational design of anti-Kasha photoemission from a biazulene core embedded in an antiaromatic/aromatic hybrid*"
  Diaz-Andres, A.; Marín-Beloqui, J.; Wang, J.; Liu, J.; Casado, J.; Casanova, D. *Chem. Sci.* **2023**, 14, 6420–6429
- "*Deciphering the role of (anti)aromaticity in cofacial excimers of linear acenes*"
  Krishnan, A.; Diaz-Andres, A.; Sudhakaran, K. P.; John, A. T.; Hariharan, M.; Casanova, D. *J. Phys. Org. Chem.* **2023**, 36, e4438
- "*Synthesis of a Dicyclohepta[a,g]heptalene-Containing Polycyclic Conjugated Hydrocarbon and the Impact of Non-Alternant Topologies*"
  Wang, J.; Gordillo-Gámez, F.; Marín-Beloqui, J.; Diaz-Andres, A.; Miao, X.; Casanova, D.; Casado, J.; Liu, J. *Angew. Chem. Int. Ed.* **2023**, 62, e202217124
- "*Benzene Excimer and Excited Multimers: Electronic Character, Interaction Nature, and Aromaticity*"
  Diaz-Andres, Casanova, D. *J. Phys. Chem. Lett.* **2021**, 12, 7400–7408

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	23/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Dedico este trabajo a mi ama, que las adversidades de la vida no nos han alejado, y a Rubén, en quien he encontrado un compañero para la vida mortal y la vida eterna.

Este trabajo también es vuestro.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	25/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	26/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Chapter 1

Introduction

"The world is vast, but the heart knows its path. Follow it, and you shall find your way."

Grimm - Hollow Knight

Optoelectronic organic materials are those that, formed only by light atoms, form structures that have the capacity for light-matter interaction. In many cases, this interaction also includes electricity as an energy source or as a result of the interaction [1, 2]. Organic light-emitting diodes (OLEDs) are the quintessential example of optoelectronic organic materials, emitting light upon the application of an electric current [3]. Furthermore, the versatility of these materials allows tuning the wavelength, and thus the color, of the emitted light through molecular design [4]. Conversely, organic optoelectronic materials have redesigned photovoltaic (PV) technology, enabling flexible, lightweight, and durable solar cells that compete with traditional silicon-based panels [5]. In addition to OLEDs and PVs, significant progress has also been made in the development of bioimaging devices, photodetectors, and organic field-effect transistors in recent years [6–8].

1.1 Organic Optoelectronics

Organic optoelectronic materials can be classified mainly into three groups depending on their composition, size, and structure, as well as the properties that define them. Polymeric materials with optoelectronic properties are characterized by being high molecular weight structures that, by designing the molecular structure, allow modifying their energy levels and conductivity, among other physicochemical properties [1, 9]. Covalent organic frameworks (COFs) are described as crystalline and porous materials formed solely by organic elements

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
Sinatzaile / Firmante	AITOR DIAZ ANDRES (PUESTO DE TRABAJO PARA INCORPORACIÓN DE DOCUMENTOS FIRMADOS)		
Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	27/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

(H, B, C, N and O), held together by strong covalent bonds, reminiscent of the bonding network found in materials such as diamond and graphite [10, 11]. The inorganic counterparts of COFs are organic zeolites, which are also crystalline porous frameworks but composed of inorganic building units such as Si, Al, and O. The low density and porosity, together with the high surface area and functional groups of COFs, allow the synthesis of materials with applications in gas storage, optoelectricity, catalysis and even as functional devices [11]. Lastly, optoelectronic materials composed of small organic molecules are utilized for their ability to achieve high charge-carrier mobilities in organic field-effect transistor (OFET). These molecules can form structures with highly ordered crystallinity, typically adopting π-stacking or herringbone packing motifs. This ordering and interaction of π-orbitals results in very low reorganization energies and large transfer integrals. These features of the electronic structure provide the general properties of crystals and optoelectronic processes [12].

Organic molecular materials inherit the optoelectronic properties of their constituent molecules, such as their characteristic absorption and emission spectra. Upon absorption of photons, the system is promoted from the ground state to an electronically excited state ($\mathtt{S}_0 \rightarrow \mathtt{S}_n$), while emission typically involves the radiative decay from the lowest excited state, in accordance with Kasha's rule [13]. The ensemble of these electronic transitions gives rise to the absorption and emission spectra, which generally reflect the coupling between electronic excitations of interacting molecules and their intermolecular vibrational modes. In many cases, such interactions lead to shifts, broadening, or splitting of absorption and emission bands [12].

Organic chromophores, beyond their absorption and emission properties, can undergo a wide variety of photophysical processes. Upon population of electronically excited states, different radiative and nonradiative decay pathways become accessible, ultimately leading to the recovery of the ground-state population. The relaxation of photoexcited molecules may involve ultrafast (sub-picosecond) internal conversion (IC) between excited states, followed by nonradiative decay to the ground state. Alternatively, relaxation can proceed through radiative decay, i.e., fluorescence, which typically occurs on the nanosecond timescale. [14].

While IC and fluorescence in organic materials are spin-allowed processes within the singlet manifold, additional photophysical pathways arise through the population of spin-triplet states. Transitions between singlet and triplet states involve spin–orbit coupling (SOC) and are therefore relatively slow in purely organic materials, owing to the low atomic masses of their constituent elements. Specifically, the radiative and nonradiative singlet-to-triplet transitions, phosphorescence and intersystem crossing (ISC), respectively, occur on much longer timescales than their spin-allowed counterparts, fluorescence and IC [12].

Alternatively, the population of triplet states can proceed through singlet fission (SF) [15, 16], a spin-allowed photophysical process that converts the energy of one singlet exciton into two triplet excitons, $\mathtt{S}_0 + \mathtt{S}_1 \rightarrow \mathtt{T}_1 + \mathtt{T}_1$ (see Section 2.4 for further details). Since SF is spin-allowed, it can occur on ultrafast timescales and may be regarded as a particular case of IC involving coupled chromophores in suitable molecular assemblies [17].

An intriguing photophysical phenomenon involving reversible transitions between singlet and triplet excited states is thermally activated delayed fluorescence (TADF) [18]. TADF consists of a sequence of processes that ultimately lead to delayed fluorescent emission, originating from the thermal up-conversion of triplet excitons to singlets via reverse ISC (RISC), followed by radiative decay from the

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	28/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

singlet state. In organic molecular crystals used for light emission, such as OLEDs, the primary goal is to minimize triplet formation and maximize singlet emission to enhance light output. However, during charge recombination in OLEDs (and similar devices), the spin statistics of electron-hole recombination yield three triplet excitons for every singlet exciton, imposing a theoretical upper limit of 25% on internal quantum efficiency [19]. Molecules capable of undergoing TADF enable the reconversion of triplet excitons into singlets, thereby enhancing fluorescence efficiency beyond the spin-statistical limit.

In addition to the photophysical processes occurring at the single-molecule level, extended molecular systems enable the transfer of energy (and charge) between adjacent interacting molecular sites. Exciton transport in molecular organic solids is governed primarily by Förster resonance energy transfer (FRET) and Dexter energy transfer (DET) mechanisms. Both involve the transfer of energy from an excited donor molecule (D) to an acceptor molecule (A). FRET is the non-radiative process involving the transfer of energy between spin-singlet state (${}^1\mathrm{D}^* + {}^1\mathrm{A} \rightarrow {}^1\mathrm{D} + {}^1\mathrm{A}^*$) [20]. This process is mainly determined by the transition dipole moments of the molecules, their relative orientation, and the intermolecular distance [21]. DET, in contrast, is a triplet transport mechanism (${}^3\mathrm{D}^* + {}^1\mathrm{A} \rightarrow {}^1\mathrm{D} + {}^3\mathrm{A}^*$). It decays exponentially with distance and is highly dependent on the electronic structure and relative orientation of the molecules involved [22].

1.2 Rubrene: Structure, Photophysics, and Exciton Processes

Rubrene (5,6,11,12-tetraphenylnaphthacene) has attracted significant attention in recent years due to its distinctive physicochemical properties. It displays an intense absorption spectrum spanning the blue-violet to green range, and it crystallizes in three polymorphs: monoclinic, triclinic, and orthorhombic [23]. Among these, only the orthorhombic phase displays notable optical activity and efficient exciton propagation, especially along the short axis of the crystal [24]. This behavior arises from the coplanar arrangement of tetracene backbones in the orthorhombic structure, which maxizes orbital overlap and facilitates energy transport [25].

Recent studies have revealed the ability of crystalline rubrene to efficiently perform SF, converting high-energy photons into pairs of lower-energy triplet excitons [26]. Indeed, in rubrene crystals, SF is often the dominant decay pathway of singlet excitons, outcompeting fluorescence [27, 28]. The coplanar crystal structure further provides long exciton diffusion lengths and high hole mobility [29], making rubrene crystals an effective medium for generating triplets.

Beyond SF, the reverse process, i.e., triplet fusion (TF), where two triplet excitons combine to produce a singlet exciton ($\mathrm{T}_1 + \mathrm{T}_1 \rightarrow \mathrm{S}_1 + \mathrm{S}_0$), has also been investigated in rubrene solids. However, in the orthorhombic crystals, SF is strongly favored over TF, limiting the material's use in applications aiming to upconvert low-energy photons via TF. Strategies to promote TF include using acceptor molecules, such as perylene derivatives, to capture singlets via FRET and outcompete SF [30]. Another approach is to modify intermolecular distances and reduce the orbital overlap, as SF efficiently depends critically on the intermolecular coupling, which is determined by both intermolecular distance and orientation [31]. These findings have motivated the study of amorphous rubrene as a promising medium for photon upconversion via TF [32–35], offering a complementary pathway to harness the unique exciton dynamics of this material.

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1.3 Motivation and Computational Challenges

Understanding the mechanistic details of photophysical processes in organic materials remains a major scientific challenge. Processes such as exciton transport, SF, and TF are strongly influenced by the molecular environment, yet experimental techniques often struggle to disentangle the microscopic factors at play. In this context, computational tools have become essential for simulating, rationalizing, and predicting photophysical phenomena in the solid state [36, 37].

At the computational level, however, several difficulties arise. First, the inherently multiscale nature of exciton dynamics demands the combination of different theoretical approaches, from quantum-chemical methods to mesoscale simulations. Second, achieving reliable predictions requires accurate evaluation of excited-state energies and intermolecular couplings, quantities that are highly sensitive to the choice of method and level of theory. Third, the solid-state morphology, defined by intermolecular arrangements, plays a decisive role in exciton dynamics, yet is often complex and challenging to model. Accounting for thermal fluctuations adds an additional layer of difficulty, as structural distortions strongly modulate electronic properties. Finally, efficient simulations require a delicate balance: electronic structure calculations must remain computationally feasible while still providing the input necessary to describe exciton dynamics over relevant time and length scales.

The present thesis aims to address some of these challenges, as explicitly defined in the research objectives outlined in the following section.

1.4 Objectives

- Study the processes involved in TF and SF in rubrene solid.
  - Analyze the singlet and triplet transfer couplings for the different dimers that form the unit cell of the rubrene crystalline solid.
  - Relate the values of the singlet and triplet transfer couplings with the physical and electronic characteristics of the dimers.
  - Explore the effect of temperature on the couplings involved in singlet and triplet transport and TF/SF.
- Design a protocol to calculate TF/SF couplings in rubrene dimers.
- Analyze the electronic couplings in the dimers of amorphous rubrene solid.
  - Rationalize the results obtained with experimental and theoretical results from the literature.
- Design software to simulate exciton dynamics in crystalline and amorphous molecular organic solids.
- Perform simulations of singlet and triplet transport and TF/SF in crystalline and amorphous solids.
  - Study the effect of temperature on exciton diffusion constants.
  - Compare the exciton dynamics performed in crystalline and amorphous solids.

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1.5 Structure of the thesis

To address the objectives of this work in an structured and coherent manner, the present thesis is organized into six chapters. Following the introductory chapter (Chapter 1), the general theoretical concepts of exciton dynamics will be reviewed in Chapter 2, along with the theoretical methods used to calculate them computationally, as well as the various theoretical chemistry methods that will be used throughout the thesis to compute the couplings involved in exciton dynamics processes. Special mention will be made of the RASCI method and its importance in this thesis.

The next chapter (Chapter 3) introduces the software designed to simulate exciton dynamics in rubrene solids. It is important to mention that the software is not restricted to the simulation of rubrene. The chapter discusses the motivations for the software design and compares it with other existing programs. Therefore, the main classes that comprise the program, as well as their main interactions, will be discussed. The software has been designed to support machine learning models, and therefore, the theory of these models and their usefulness for simulations of exciton dynamics are reviewed. This chapter concludes with instructions for compiling and running this program and a simple example.

Chapter 4 presents the results of the study of the electronic couplings of the processes surrounding triplet fusion in the rubrene crystal. This study compares different methods for calculating these couplings with the goal of defining a protocol. For this reason, RASCI is explored as a method for describing and calculating triplet fusion couplings. In addition, a section will be devoted to studying the effect of temperature on the couplings calculated in the crystalline rubrene solid. These results will be compared with the couplings calculated at 0 K to study the impact of molecular vibrations. To conclude the chapter, a brief statistical analysis of the relationship between temperature-distorted structures and couplings will be performed, thus exploring any correlations.

The next chapter (Chapter 5) studies the morphology of amorphous rubrene, its structures, and intermolecular distances, thus exploring the morphological differences with the crystal. From this analysis of the amorphous rubrene solid, the electronic couplings involved in triplet fusion will be analyzed using the previously designed protocol. In this efficient manner, approximately 10k rubrene molecular pairs extracted from the amorphous solid will be calculated, and the results obtained will be statistically analyzed. In the second half of the chapter, simulations of exciton dynamics in the crystalline and amorphous solids will be performed. In the simulations of the amorphous solid, machine learning models will be used to predict couplings at runtime.

Finally, Chapter 6 is dedicated to gathering all the results and conclusions that have been obtained in the previous chapters to make a summary of the thesis, in addition to proposing possible ways to continue with the work carried out in the thesis.

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Chapter 2

Theory and Methods

"I may be but small, but I will die a colossus."

Ludleth of Courtland - Dark Souls III

This thesis investigates exciton transport in organic molecular solids by accounting for the key dynamical mechanisms, including singlet and triplet diffusion, exciton fission, and exciton fusion. The theoretical framework adopted is the incoherent exciton transport (or hopping approximation), where energy transfer is modeled as stochastic hopping events between molecular sites, without memory of the exciton's previous states. This stands in contrast to the coherent transport regime, where excitons propagate in a wave-like manner and quantum coherence between sites must be explicitly considered. Therefore, the evolution of exciton transfer dynamics can be naturally simulated using the kinetic Monte Carlo (KMC) algorithm, which provides a stochastic framework for modeling the time evolution of processes governed by known transition rates. This approach is particularly well suited for exciton transport, as it captures the probabilistic nature of incoherent hopping events and enables simulations over time and length scales that are inaccessible to fully quantum-mechanical treatments.

In this chapter, I present the KMC method from both theoretical and computational point of views. Building on this framework, I present the rate constant concept and discuss the theoretical approaches used to compute them in the systems under study. Then, I dive into the electronic couplings between states. More specifically into the exciton energy transfer couplings and the singlet fission/triplet fusion couplings that describe all the processes of relevance for this work. Afterwards, I shortly present the restricted-active-space configuration interaction (RASCI) method for electronic structure calculations and its importance in the calculation of the triplet fusion couplings. Other electronic structure

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calculation methods will be mentioned. By the end of the chapter, I briefly describe the morphology of the organic molecular solids used in this work and their relevance to exciton transport and triplet fusion.

2.1 The Kinetic Monte Carlo Method

The KMC algorithm is well explained in literature from the most fundamental perspective [38] to more computationally affordable approaches [39]. Besides, there are plenty of studies that implement KMC for different application such as the simulation of quartz dissolution [40] and exciton diffusion [41, 42].

The central idea of the KMC algorithm is to consider an initial state m and determine the rates of all possible processes to other states $(\Gamma_{m\to1}, \Gamma_{m\to2}, ..., \Gamma_{m\to N})$. Next , the total rate $\Gamma_{\mathtt{Tot}}$ is calculated, and a random number $r \in (0, 1]$ is drawn from a uniform distribution. The transition whose cumulative rate satisfies $\Gamma_{m\to n-1} < r\Gamma_{\mathtt{Tot}} \leq \Gamma_{m\to n}$ is then selected. Figure 2.1 illustrates the cumulative distribution and the chosen transition. Afterwards, the system is updated according to the process associated with the selected rate and the simulation time is updated by

$$t = t + \Gamma_{\mathtt{Tot}}^{-1} \ln\left(\frac{1}{r'}\right) \tag{2.1}$$

where r' is a second random uniform number ($r' \in (0, 1]$).

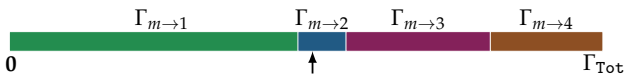


FIGURE 2.1: Visual representation of the cumulative function of the rate constants. The arrow represents the random number, $r\Gamma_{\mathtt{Tot}}$. The chosen rate constant is $\Gamma_{m\to2}$.

2.2 Rate Constant and Fermi's Golden Rule

The general quantum mechanical framework to describe transition probabilities between states was first derived by Paul Dirac [43], and it was later named Fermi's Golden Rule after Enrico Fermi, who referred to it "golden rule No. 2" [44]. This equation describes the probability of a transition per unit time between eigenstates due to a weak perturbation and is defined by

$$\Gamma_{m\to n} = \frac{2\pi}{\hbar}|V_{mn}|^2\rho(E_n) \tag{2.2}$$

where V_{mn} is the matrix element of the coupling between initial (m) and final (n) states and $\rho(E_n)$ refers to the density of states of the final state n at the E_n energy. The Fermi Golden Rule applies in the weak-coupling regime, where transitions between quantum states are induced by a perturbation that is small compared to the unperturbed Hamiltonian. In this regime, the perturbation drives transitions incoherently and with low probability per unit time, such that higher-order effects and memory of past states can be neglected.

A paradigmatic example of the application of the Fermi Golden Rule is Marcus theory [45], which offers a classical approximation to the Franck–Condon weighted density of states that enters the expression for electron and exciton transfer rates.

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The Marcus model reformulates Eq. (2.2) in a closed analytical form, allowing the transfer rate to be written as

$$\Gamma_{m\to n} = \frac{2\pi}{\hbar}|V_{mn}|^2 \frac{1}{\sqrt{4\pi\lambda_n k_B T}} \exp\left(-\frac{(\lambda_n + \Delta G^\circ)^2}{4\lambda_n k_B T}\right) \tag{2.3}$$

where ΔG° is the total Gibbs free energy change and λ_n is the reorganization energy of the final state, which is the energy difference between initial and final structure coordinates.

In the following sections, I briefly outline the theoretical methods used to calculate the couplings that enter the Fermi's Golden Rule expressions for the different photophysical processes investigated in this thesis.

2.3 Exciton Energy Transfer Couplings

In this section, we focus on the couplings that govern exciton energy transfer, considering both the singlet and triplet manifolds. Exciton transfer processes are central to the photophysics of molecular materials, as they dictate how electronic excitations migrate between chromophores and thus influence the efficiency of light-harvesting, photonic, and optoelectronic applications.

To describe these couplings, we examine a range of established approaches that vary in complexity and physical rigor. At the classical level, the dipole–dipole interaction provides a simple and widely used approximation for long-range exciton transfer of bright states. Moving beyond this, the half energy gap (HEG) rule offers a heuristic estimate of transfer couplings based on energy differences between interacting states. More advanced quantum-chemical approaches include the fragment difference methods, which evaluate electron-transfer and exciton couplings within the two-state approximation, and the Boys exciton diabatization procedure, which provides a rigorous framework for constructing localized excited states and computing couplings between them.

Together, these methods span from approximate, physically intuitive models to systematically derived quantum-chemical approaches, offering a hierarchy of tools for analyzing exciton energy transfer in molecular systems.

2.3.1 Classical Dipole-Dipole Interaction

In the long-range limit, the overlap between the electronic wave functions of the excited donor and the ground state acceptor molecules sites vanishes. Hence, their interaction can be described as a purely electrostatic dipole-dipole interaction [46] and under the point dipole approximation, the equation is described by

$$V^{\mathrm{d\text{-}d}} = \frac{1}{4\pi\varepsilon_0\eta^2 r^3}\left[(\vec{\mu}_{\mathrm{D}}\cdot\vec{\mu}_{\mathrm{A}}) - \frac{3}{r^2}(\vec{\mu}_{\mathrm{D}}\cdot\vec{r})(\vec{\mu}_{\mathrm{A}}\cdot\vec{r})\right] \tag{2.4}$$

where $\vec{\mu}_{\mathrm{D/A}}$ are the transition dipole moments of the donor (D) and acceptor (A) chromophores, $\vec{r}$ is the intermolecular distance vector, ε_0 is the vacuum permittivity, and η is the medium refractive index. In the present study, we compute $V^{\mathrm{d\text{-}d}}$ under vacuum ($\eta = 1$). However, the point dipole approximation is no longer valid for pairs of molecules that are close and the overlap between orbitals cannot be neglected. Moreover, the simultaneous excitation and deexcitation processes are

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restricted to bright states, hence, Eq. (2.4) cannot describe dark states, e.g., triplet states.

2.3.2 Half Energy Gap Rule

In the cases of equivalent pair of molecules, the intermolecular electronic couplings can be heuristically estimated by the HEG rule

$$V_{mn}^{\mathtt{HEG}} = \frac{\Delta E_{mn}}{2} \tag{2.5}$$

where ΔE_{mn} is the energy difference between the states involved as computed in the dimer. This rule provides a rough estimate of the strength of the interaction at stake in excitation energy transfer processes.

2.3.3 Fragment Difference Methods

Under the two-state appoximation, the generalized Mulliken-Hush (GMH) [47] method defines diabatic states as the states that diagonalize the dipole moment operator. The electron-transfer (ET) coupling is defined by

$$V_{mn}^{\mathtt{GMH}} = \frac{\Delta E_{mn}\mu_{mn}}{\sqrt{\Delta\mu_{mn}^2 + 4\mu_{mn}^2}} \tag{2.6}$$

where ΔE_{mn} is the energy difference between eigenstates $|\phi_m\rangle$ and $|\phi_n\rangle$, μ_{mn} is the transition dipole moment and $\Delta\mu_{mn}$ is the difference of the eigenstate dipole moment. Later, Voityuk and Rösch developed the fragment charge difference (FCD) [48] method that replaces the difference in dipole moment by the charge difference in adiabatic states, $\boldsymbol{\Delta q}$,

$$V_{mn}^{\mathtt{FCD}} = \frac{\Delta E_{mn}\left|\overline{\Delta q_{mn}}\right|}{\sqrt{\left(\Delta q_m - \Delta q_n\right)^2 + 4\overline{\Delta q_{mn}}^2}} \tag{2.7}$$

where $\overline{\Delta q_{mn}} = (\Delta q_{mn} + \Delta q_{nm})/2$. With the same strategy, Hsu et al. formalized the fragment excitation difference (FED) approach [49], where the charge differrence, $\boldsymbol{\Delta q}$, of the FCD method is replaced by the difference in excitation population matrix, $\boldsymbol{\Delta x}$, and is defined with its element

$$\Delta x_{mn} = \int_{r\in\mathtt{D}} \sigma_{mn}^{(\mathtt{ex})}(r)\,\mathrm{d}r - \int_{r\in\mathtt{A}} \sigma_{mn}^{(\mathtt{ex})}(r)\,\mathrm{d}r \tag{2.8}$$

where $\sigma_{mn}^{(\mathtt{ex})}$ is the sum of the electron ($\rho_{mn}^{(\mathtt{elec})}(r)$) and hole ($\rho_{mn}^{(\mathtt{hole})}(r)$) densities for the $m \to n$ transition. These quantities are calculated using the attachment and detachment densities exctracted from the difference of the ground state and excited state densities [50]. The FED coupling is then defined by

$$V_{mn}^{\mathtt{FED}} = \frac{\Delta E_{mn}\left|\overline{\Delta x_{mn}}\right|}{\sqrt{\left(\Delta x_m - \Delta x_n\right)^2 + 4\overline{\Delta x_{mn}}^2}} \tag{2.9}$$

which in symmetrical situations, the equation becomes half of the energy gap (Section 2.3.2). Similarly, the Fragment Spin Difference (FSD) approach replaces the excitation population matrix, $\boldsymbol{\Delta x}$, with the spin population matrix $\boldsymbol{\Delta s}$ [51]. The

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elements of this matrix are calculated by replacing from Eq. (2.8) the measure of excitation, $\sigma_{mn}^{(\mathtt{ex})}$, with the difference between one-electron α and β electron densities $\sigma_{mn}^{(\mathtt{sp})}(r) = \rho_{mn}^{\alpha}(r) - \rho_{mn}^{\beta}(r)$. The equation to calculate the FSD coupling is defined by

$$V_{mn}^{\mathrm{FSD}} = \frac{\Delta E_{mn} \left| \overline{\Delta s_{mn}} \right|}{\sqrt{\left(\Delta s_m - \Delta s_n\right)^2 + 4\overline{\Delta s_{mn}}^2}} \tag{2.10}$$

In Eqs. (2.7), (2.9) and (2.10), the overline refers to quantities from symmetrized matrices.

2.3.4 Multistate generalization

The methods described above for evaluating electronic couplings in electron and energy transfer processes are based on the two-state approximation. In this framework, it is assumed that the diabatic states of interest, namely, the initial and final states involved in the process, can be adequately represented by two adiabatic states. However, this approximation breaks down in multistate scenarios, where more than two (and often several) adiabatic states contribute significantly to the process. To address such cases, I have extended, implemented, and applied fragment-based difference methods to treat general multistate systems. In particular, this section focuses on the development and application of the multistate FED approach, although the same theoretical framework can be readily adapted to other fragment difference methods.

The expressions for the different couplings are obtained by diagonalizing the target fragment difference matrix. In the FED framework, this involves the transformation of the adiabatic excitation population matrix, $\boldsymbol{\Delta x}_{adiab}$, of dimension 2×2:

$$\boldsymbol{\Delta x}_{diab} = \mathbf{C} \boldsymbol{\Delta x}_{adiab} \mathbf{C}^{-1} \tag{2.11}$$

where $\boldsymbol{\Delta x}_{diab}$ is diagonal, and $\mathbf{C}$ is the rotation matrix containing the eigenvectors of $\boldsymbol{\Delta x}_{adiab}$. The FED coupling between diabatic states is then obtained by applying the same unitary transformation to the diagonal adiabatic Hamiltonian:

$$\mathbf{H}_{diab} = \mathbf{C} \mathbf{H}_{adiab} \mathbf{C}^{-1} \tag{2.12}$$

in which the analytical expression of the coupling in Eq. (2.9) corresponds to the off-diagonal elements of $\mathbf{H}_{diab}$.

This procedure can be straightforwardly generalized to an n-state system by diagonalizing an $n \times n$ adiabatic fragment difference matrix, $\boldsymbol{\Delta x}_{adiab}$, and applying the same transformation to the corresponding n-state diagonal adiabatic Hamiltonian. The off-diagonal elements of the resulting $\mathbf{H}_{diab}$ matrix yield the interstate FED couplings, amounting to $n(n-1)/2$ distinct coupling terms.

2.3.5 Boys Exciton Diabatization of Excited States

The fragment based methods (Section 2.3.3) design general strategies to calculate diabatic states by defining donor and acceptor fragments for the construction of population matrices, e.g., the excitation population matrix in Eq. (2.8). However, it is difficult to define the donor and acceptor fragments for many systems, specially for intramolecular electron transfer processes, where there is no clear physical criterion for fragmenting the molecule [52]. Moreover, in many cases,

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
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the two-state approximation used in fragment-difference methods may not be adequate, particularly in systems with multiple charge or excitation centers. The Boys localization method offers an alternative strategy [53], in which diabatic states are constructed by maximizing the spatial separation of their charge distributions

$$f_{\text{Boys}} = \sum_{kl} |\langle \phi_k|\vec{\mu}|\phi_k\rangle - \langle \phi_l|\vec{\mu}|\phi_l\rangle|^2 \tag{2.13}$$

where $\{\phi_i\}$ are the Boys diabatic states and $\vec{\mu}$ is the dipole moment operator. Here, the interstate couplings can be directly evaluated as the off-diagonal elements of the electronic Hamiltonian in the diabatic basis. However, indirect contributions to other states can become important if the electronic couplings are large enough [15, 54]. In terms of perturbation theory, electronic couplings can be expressed as the sum of direct (first-order) or indirect (second-order) contributions

$$V_{\text{Boys}} = \langle \phi_n|H|\phi_m\rangle - \sum_{k\neq m,n} \frac{\langle \phi_n|H|\phi_k\rangle \langle \phi_k|H|\phi_m\rangle}{E_k - (E_m + E_n)/2} \tag{2.14}$$

where $|\phi_m\rangle$ and $|\phi_n\rangle$ correspond the initial and final diabatic states, respectively, and the $|\phi_k\rangle$ state refers to an intermediate diabatic state. In exciton energy transfer processes, the potentially relevant mediator states $|\phi_k\rangle$ are, in many cases, of charge transfer (CT) nature, although they are not directly populated during the process [15].

2.4 Triplet Fusion and Singlet Fission

TF and SF are two bimolecular processes that are two faces of the same coin [55]. While in the SF a spin singlet excited state is converted into two-triplet excitons, in the TF two-triplet excitons are converted into one singlet exciton [15, 56]. The former can be used as a downconversion process of high energy frequency photons into lower energy excitons [15], while the later can mediate the upconversion of low energy frequency photons into high energy excitons. In Figure 2.2 SF and TF processes are schematically represented.

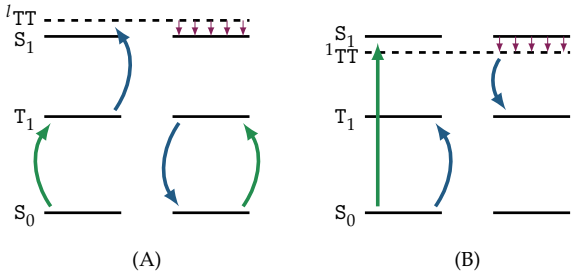


FIGURE 2.2: Schematic representation of TF (A) and SF (B) processes. Green arrows represent the TEET (photoabsorption) mechanism to obtain triplet (singlet) excitons in TF (SF). Blue arrows represent the formation (dissociation) of the ^{l}TT state through k_{TTF} (k_{TTD}) in Eq. (2.15). Small purple arrows refer to the TF (SF) process associated with the k_{TF} (k_{SF}) in Eq. (2.16).

SF and TF are described as two consecutive step processes where the intermediate triplet-triplet (TT) coupled state is formed [56]. In SF the excited

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singlet state is usually generated through photoabsorption, either directly or from a higher S_n state, which rapidly decays to the lowest energy singlet state. Then, the excited singlet state transitions to the intermediate ^1TT state delocalized over two neighboring molecules (k_{SF} in Eq. (2.16)). Note that, in SF, the TT state is coupled as an overall spin singlet state ($l = 1$), so that the process is spin allowed. Last, the coupled ^1TT state can dissociate into two triplet states via triplet exciton energy transfer (TEET), k_{TTD} in Eq. (2.15).

$$T_1 + T_1 \underset{k_{TTD}}{\overset{k_{TTF}}{\rightleftharpoons}} {}^lTT \tag{2.15}$$

$${}^lTT \underset{k_{SF}}{\overset{k_{TF}}{\rightleftharpoons}} S_1 + S_0 \tag{2.16}$$

Contrary, TF follows the reverse steps to form high energy singlet excited state. The triplet excitons are usually injected by another molecule species (sensitizer) that generates singlet excited states through photoabsorption and undergoes efficient ISC forming the triplet exciton [55]. This triplet exciton is then, transferred to the annihilatior through TEET. The interaction between two triplet (S = 1) spins leads to a coupled ^lTT with three possible overall spins: singlet ($l = 1$), triplet ($l = 3$) or quintet ($l = 5$), through the triplet-triplet formation process (k_{TTF} in Eq. (2.15)). The ^1TT (singlet spin) coupled state can transition to the first excited state via triplet fusion, k_{TF} in Eq. (2.16).

SF and TF processes require special conditions to perform efficiently. The interaction of triplet states located in different molecules depends on the orbital overlap between them, which decreases exponentially with distance. Besides, the energy of the singlet excited state must be lower (higher) than the energy of the TT state for TF (SF) to perform (Figure 2.2). Nevertheless, the decay of higher energy singlet states to the lowest singlet state is, in general, faster than TF/SF processes and the vibrational relaxation is even faster. Therefore, the relative energies should be evaluated at the minimum of the potential energy surfaces (PES) in the lowest excited states [15].

2.4.1 Triplet-Pair State

The TT manifold, comprising the nine spin projections arising from the coupling of two $S = 1$ states, can be effectively modeled using a spin Hamiltonian formalism. In the absence of an external magnetic field (i.e., no Zeeman interaction) and neglecting relativistic effects, the spin Hamiltonian simplifies to

$$H_{spin}^{(TT)} = J\vec{S}_A \cdot \vec{S}_B \tag{2.17}$$

where J is the inter-triplet exchange interaction, and subindices A and B refer to interacting centers (molecules). This approach holds in the regime of strong exchange coupling, which is well-justified in tetracene-based molecular materials as exchange interactions are significantly larger than both intra- and inter-triplet-triplet zero-field splittings [57–60].

The eigenstates of the spin Hamiltonian in Eq. (2.17) correspond to nine spin-adapted states (eigenstates of $\hat{S}^2$), resulting from the coupling of the two triplets [61]. Therefore, their eigenenergies can be correlated with those obtained from ab initio calculations and J can be quantified by the energy difference between

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the high (quintet) and low (singlet) solution.

$$E(^5\mathrm{TT}) - E(^1\mathrm{TT}) = 3J \tag{2.18}$$

Evaluating J via Eq. (2.18) necessitates a balanced computation of ^{1}TT and ^{5}TT states. The high-spin state ($M_s = \pm 2$) is predominantly governed by a single configuration, making it well-suited for accurate treatment with single-reference post-HF methods, such as MP2 or CCSD, or even Kohn-Sham DFT approximations. In contrast, the calculation of ^{1}TT is considerably more challenging due to its pronounced spin correlation [62]. Additionally, the pristine ^{1}TT state can exhibit mixing with other configurations, which may account for approximately 10% of the wave function in organic crystals [63]. This intricate electronic structure prevents the applicability of standard single-configurational approaches. In the following section, I present the multiconfigurational method that is used for electronic structure calculations and the description of the ^{1}TT and ^{5}TT states, and their energetic gap.

2.5 Restricted Active Space Configuration Interaction

RASCI is a single reference wave-function approach for electronic structure calculations that emerges as an alternative to other multi-reference wave-fuction methods [64, 65]. The single reference approach greatly decreases computational costs as the number of electrons and orbitals increase. RASCI splits the orbital space into three subspaces (Figure 2.3): doubly occupied orbitals (RAS1), fully correlated orbital space (RAS2) and virtual orbitals (RAS3). In RASCI, excitations are classified upon the number of electron vacancies generated in the doubly occupied subspace (RAS1), called holes (h), and the number of electrons promoted to the virtual space (RAS3), called particles (p).

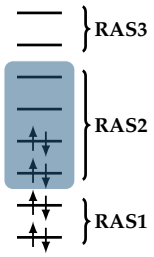


FIGURE 2.3: Representation of RASCI orbital subspaces RAS1 (doubly occupied orbitals), RAS2 (fully correlated space) and RAS3 (virtual orbitals).

The performance of RASCI, as a single reference method, highly depends on the reference wave function, $|\Psi_0\rangle$, which, in general, is the Hartree-Fock determinant. The target states, $|\psi_i\rangle$, are expressed as linear combinations of Slater determinants (Eq. (2.19)) and the associated amplitudes are obtained by the diagonalization of the CI Hamiltonian (Eq. (2.20)).

$$|\psi_i\rangle = \hat{R}\,|\Psi_0\rangle = \sum_j c_j\,|\Psi_j\rangle \tag{2.19}$$

$$\sum_k c_k \langle \Psi_k|\hat{H}|\Psi_j\rangle = E c_j \tag{2.20}$$

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$\hat{R}$ is the RASCI excitation operator, which has a huge impact on the target states along with the reference state. In this project, I use the single hole and particle excitation operator, $\hat{R} = \hat{R}_0 + \hat{R}_h + \hat{R}_p$ (Eqs. (2.21), (2.22), and (2.23)), which truncates the possible excitations to configurations holding up to a single electron vacancy in the RAS1 space or a single electron in the RAS3 space.

$$\hat{R}_0 = \sum_{\{s\}} r^{\{s\}} \left(\prod_{s \in \mathtt{RAS2}}^{n_0} a_s^\dagger \right) \left(\prod_{p \in |0\rangle}^{n_0} a_p \right) \tag{2.21}$$

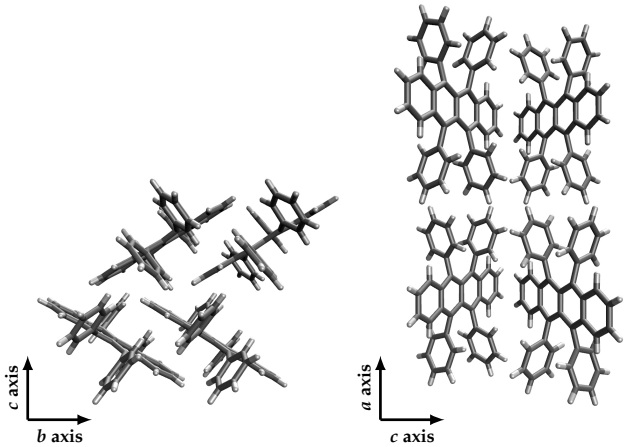
$$\hat{R}_h = \sum_{\{s\},i} r_i^{\{s\}} \left(\prod_{s \in \mathtt{RAS2}}^{n_0+1} a_s^\dagger \right) \left(\prod_{p \in |0\rangle}^{n_0} a_p \right) \tag{2.22}$$

$$\hat{R}_p = \sum_{\{s\},a} r^{a\{s\}} \left(\prod_{s \in \mathtt{RAS2}}^{n_0-1} a_s^\dagger \right) \left(\prod_{p \in |0\rangle}^{n_0} a_p \right) \tag{2.23}$$

Within the RASCI(h, p) framework, the particle- and spin-converving excitation operators and the splin-flip operators are used. The later generate target states with different M_s with respect to the reference state, $|\Psi_0\rangle$ [66].

2.6 Solid State Morphology

Rubrene crystal structure (Figure 2.4) forms an orthorhombic crystal [67] and its cell parameters are: $a = 26.789(4)$ Å, $b = 7.1700(10)$ Å, $c = 14.211(2)$ Å, $\alpha = 90°$, $\beta = 90°$, and $\gamma = 90°$. The employed crystal structure is extracted from the Cambridge Structural Database (CSD) [68]. All the calculations in the crystal structure are performed on the frozen molecular structure, i.e., with no further geometry optimizations.



(A) (B)

FIGURE 2.4: Crystal structure of rubrene along the *bc* (A) and *ac* (B) planes.

The frozen molecular structure, although it facilitates the interpretation of intermolecular electronic couplings, it does not cover the effect of temperature and molecular vibrations. The symmetry of the rubrene's crystal structure can

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be disrupted by thermally activated molecular vibrations. In fact, it has been shown that slight symmetry breaking distortions can produce large fluctuations in intermolecular interactions, and in particular for the one-electron terms [69, 70]. Therefore, to account for thermally induced distortions, molecular dynamics (MD) simulations of the rubrene crystal is carried out using the NAMD program [71].

Beyond the frozen crystal and thermally distorted structures, intermolecular interactions and electronic couplings are further examined in amorphous rubrene solids, with representative topologies obtained from MD simulations. In order to generate amorphous rubrene structures, rubrene was modeled using a partially modified version of the general AMBER force field [72–74]. Namely, electrostatic potential (ESP) fitted atomic charges were obtained from IEF-PCM/B3LYP/6-311G(d,p) DFT calculation with benzene as implicit solvent. New torsional potential parameters were derived from relaxed potential energy scan of the dihedral angle between the tetracene core and phenyl moieties calculated at the B3LYP/6-311G(d,p) level of theory on a reduced fragment (biphenyl-tetracene) following the methodology reported in [75, 76]. To generate the amorphous bulk morphology of rubrene, 500 molecules were randomly arranged in a 140 Å side simulation box using Packmol [77]. Four independent starting conformations were prepared. These low-density samples were first annealed at 1000 K before being quenched to 300 K in the NpT ensemble and further equilibrated for 10 ns. Pressure and temperature controls were achieved using the Berendsen's barostat and the velocity scaling thermostat, respectively.

2.7 Electronic Structure Calculations

All electronic structure calculations presented in this thesis, including excited-state energy computations, and evaluations of electronic couplings, were carried out using a development version of the Q-Chem software package [78].

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Chapter 3

SEDyn Software

"The true enemy of humanity is disorder."

Symmetra - Overwatch

Electronic structure calculations based on the time-independent Schrödinger equation are essential for characterizing the eigenstates of molecular systems, ranging from individual molecules to molecular oligomers, aggregates, and even extended structures like molecular crystals. However, while invaluable for understanding static properties, these methods are insufficient for describing time-dependent phenomena such as exciton or charge dynamics, which are crucial for applications ranging from photovoltaics to organic light-emitting diodes in molecular solids.

The study of exciton and charge dynamics in molecular materials has a rich history, rooted in multiple fields, including condensed matter physics, quantum chemistry, and materials science. Foundational works and theories have significantly shaped our understanding of these phenomena. Key examples include the Frenkel exciton model [79], FRET [80], and Marcus theory [45]. Despite this strong theoretical foundation and sustained interest from the scientific community in simulating these processes, it is only in recent years that open-source, standardized, and reliable software has become available to perform such simulations. For decades, researchers around the world relied on custom-built software, which posed challenges for new scientists entering the field. Recently, tools like Kimonet [81] and Excimontec [82] provide accessible solutions. Kimonet, written in Python, enables straightforward simulations of exciton and charge dynamics in crystalline structures. In contrast, Excimontec, developed in C++, offers a versatile platform for more complex simulations and dynamic processes. Both tools utilize the Approximation of Points in Space (APS) methodology, which simplifies molecular

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or atomic information by representing them as discrete spatial points, referred to as *sites*. This approach reduces the complexity of crystalline structures to a manageable number of sites, enabling efficient and rapid computation of long simulations. However, despite its efficiency, the APS methodology has certain limitations, which motivated the development of SEDyn.

3.1 Software Design Principles

The SEDyn program has been developed with the objective of providing an efficient and flexible framework for simulating exciton and charge dynamics in molecular materials, including both crystalline and amorphous systems, using parameters derived from ab initio calculations. The code is designed to capture a broad range of processes, from linear exciton or charge diffusion in singlet and triplet manifolds to nonlinear phenomena such as recombination, exciton fusion, or singlet fission.

A key feature of SEDyn is its modular architecture, which ensures flexibility and ease of extension. This design makes it straightforward to incorporate new theoretical models (e.g., rate-based approaches), introduce additional electronic states, or implement new dynamical processes without disrupting the core structure of the program.

Finally, SEDyn has been released as an open-source project, with the aim of fostering collaboration, transparency, and adoption within the scientific community. This open development model encourages contributions that will broaden the scope and capabilities of the software while ensuring its long-term relevance and usability.

3.1.1 Programming Language and Libraries

SEDyn is written in C++, an extension of C developed by Bjarne Stroustrup [83, 84]. In Stroustrup's words, "C++ is based on the idea of providing both: direct mappings of built-in operations and types to hardware to provide efficient memory use and efficient low-level operations, and; affordable and flexible abstraction mechanisms to provide user-defined types with the same notational support, range of uses, and peformance of built-in types" [84]. In other words, C++ was developed for fast and efficient built-in operations at the hardware level, and it was designed with abstraction mechanisms for the user that are as fast as built-in types, enabling abstract and complex type designs without compromising efficiency. These features of C++ encouraged the scientific community to develop new numerical computation software in this programming language. In fact, according to Stroustrup's book written in 2013, "much numerical, scientific, and engineering computation is done in C++". Therefore, C++ was the language of choice for the development of SEDyn.

Besides, and in addition to C++ standard libraries, SEDyn uses external libraries to boost performance and simplify code development and readability. We can group external libraries by their purposes. SEDyn implements OpenMP [85] for parallel execution and to achieve scalability in large simulations. For numerical calculations, Armadillo is included, which provides a good balance between speed and simplicity [86, 87]. Besides, Armadillo has dependencies on OpenBLAS [88, 89] and LAPACK [90], which are libraries with highly optimised functions for numerical linear algebra. These libraries play a key role in eigenvalue problems, as we will explore later. Last, SEDyn requires HDF5 [91], a high-performance data management and storage suite, in order to read/write information to disk efficiently.

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This is highly useful to store all the information from the trajectories that will be used in the statistical analysis at the end of the simulations.

3.1.2 Beyond the APS Methodology

SEDyn does not rely on the APS and stores all the atomic coordinates in memory. This is one of the key differential aspects of the software and the core strategy proposed for exciton dynamics simulations. In crystalline systems, exciton dynamics simulation software (EDSS) typically exploits translational symmetry. By leveraging the periodicity of the structure, EDSS can significantly reduce computational costs by restricting calculations to the unit cell of the crystal, thereby limiting the number of interactions to a manageable subset. With only a few possible interactions per unit cell and an infinitely repeating crystal structure, it becomes computationally feasible to parametrize all processes using electronic structure calculations performed with quantum chemical software packages. By explicitly obtaining these interactions from quantum chemical calculations, EDSS eliminates the need to store molecular information. Instead, molecules are replaced by discrete points in space, referred to as sites. These interaction parameters remain constant regardless of the system size, enabling efficient simulations over extended time scales or for arbitrarily large crystal structures.

However, the APS breaks down for non-periodic systems or systems whose interactions cannot be reduced to their unit cell. This is the case of amorphous solids, where translational symmetry is completely lost. In such systems, each site may possess unique properties arising from its specific molecular structure and local environment, leading to distinct state energies and inter-site interactions. Applying APS to an amorphous solid would require labeling every molecule as a distinct site. Consequently, the number of interactions that would need to be precomputed would become prohibitively large. There would be no computational efficiency advantage in applying the APS for non-crystalline systems. My solution to this challenge was to eliminate the use of sites in the software and instead store the molecular geometry and spatial positions. With this new approach, the interactions between molecules in a crystalline solid preserve the periodicity inherent to the crystal structure; that is, interactions within the unit cell are repeated due to the periodic boundary conditions of crystals. By retaining atomic position information, it becomes possible to establish structure–property relationships through the pre-computation of a representative set of cases, which can then be used to rapidly evaluate molecule- and interaction-specific properties across the system. Instead of relying on site-to-coupling relationships, this method utilizes molecule-to-coupling relationships.

3.1.3 Data Representation and Storage

SEDyn's new strategy of replacing the sides with the molecules' geometries required a reinterpretation of the objects that store this information and the functions to calculate the energies. The data structure to store the geometry information was designed as an object that mimics a molecule, that is, a class that stores the atomic coordinates, the elements and computes different properties such as the centre of mass. In addition, this structure also includes a vector with the references of the neighbours and a flag to store the electronic (or charge) state of the molecule. These layers of complexity to store the information of the molecules have no advantage in an algorithm that uses APS, since all the functions that calculate energies only

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require knowing the position of the sides, that is, the centres of mass. However, in the algorithm that I present, in addition to the functions we find in the EDSS, I implemented new functions that require knowledge of all the coordinates and atoms of each molecule.

In particular, these can be integrated with Machine Learning (ML) algorithms, which are capable of using a wide variety of molecular representations as input, ranging from the centre of mass to two-dimensional representations of atomic positions. In 2012, Rupp et al. [92] designed the Coulomb Matrix (CM) to represent molecules for ML models. In their study, they demonstrate that the CM can be constructed with the same information used to construct a Hamiltonian for an electronic structure calculation, the set of coordinates, $\{\mathbf{R}_i\}$, and the nuclear charges, $\{Z_i\}$. Later, the CM was extended to build a 2D representation of dimers and aggregates by expanding the CM to the combination of molecular geometries. The CM takes the expression of

$$\mathbf{M}_{ij} = \begin{cases} 0.5 Z_i^{2.4}, & i = j \\ \dfrac{Z_i Z_j}{\|\mathbf{R}_i - \mathbf{R}_j\|}, & i \neq j \end{cases} \tag{3.1}$$

where the diagonal elements represent a polynomial fit of atomic energies to nuclear charges, and off-diagonal elements are defined by the Coulomb repulsion between nuclei i and j. Implementation of Eq. (3.1) is shown in Algorithm 1. The CM is a two-dimensional representation of a three-dimensional structure, which is invariant with respect to translation and rotations, but is atom index ordering dependent. The diagonalization of the CM is a straightforward procedure to remove this dependence, though it results in some loss of information due to the reduction in dimensionality. However, despite the reduced dimensionality, optimization of ML models have shown to give better results when using the one-dimensional representation of pairs of molecules, despite the loss of information [93]. Therefore, input data for ML algorithms is defined as the eigenvalues of the CMs, $\gamma = \left[\gamma^{(1)}, \gamma^{(2)}, ..., \gamma^{(m)}\right]^\top$.

Algorithm 1 The Coulomb Matrix algorithm

```
function CoulombMatrix(C, m)                    ▷ Coordinates and Masses
    d ← dim(m)
    R(d, d)                                     ▷ Coulomb Matrix
    for i ← 0, d do
        for j ← i + 1, d do                     ▷ If i = d this block is not executed
            R(i, j) ← m(i) · m(j) / ‖C(i) − C(j)‖
            R(j, i) ← R(i, j)
        end for
        R(i, i) ← 0.5 · pow(m(i), 2.4)
    end for
end function
```

For a visual showcase, in Figure 3.1 I present two examples of the CM of acetic acid and glycolaldehyde. For the sake of simplicity, the hydrogen atoms are removed from the calculation of the CMs, because the relevant information resides in the carbon and oxygen atoms. These molecules are structural isomers, which means that they contain the same atoms, but they are arranged differently. Following the atomic

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	46/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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labeling in Figure 3.1, the only changes in the CM of glycolaldehyde compared to acetic acid correspond to interactions involving the O_4 atom, while all other terms in the CM remain unchanged.

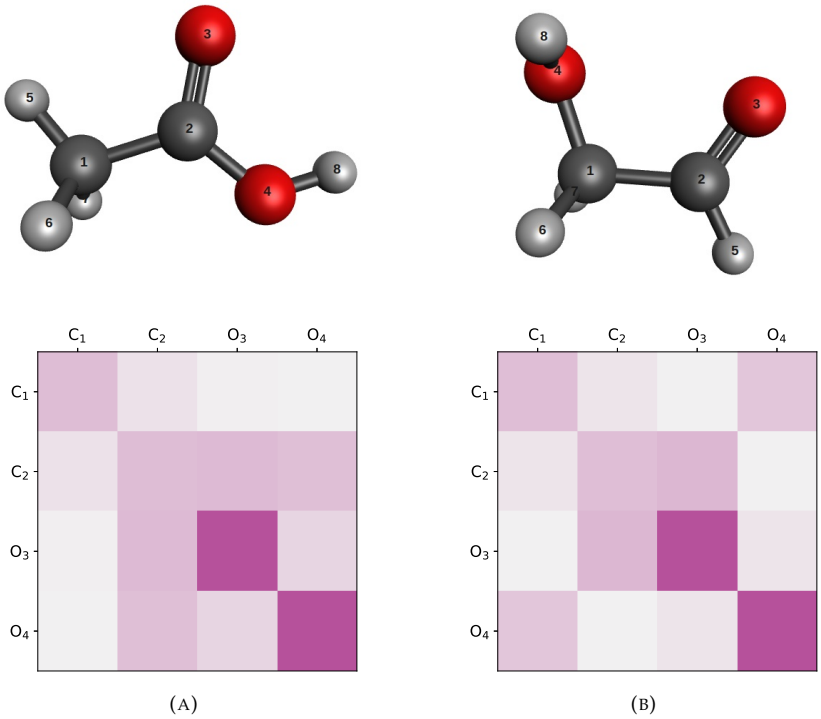


FIGURE 3.1: CM of (A) acetic acid and (B) glycolaldehyde. Hydrogen atoms are not used to compute the CMs.

We can further analyze the difference between these CMs by computing their eigenvalues (Figure 3.2). The plotted graph shows that the differential data is stored mostly in the second and third eigenvalues, while the first and last are nearly the same for the two molecules. As it has been explained before, CMs provide a two-dimensional representation of three-dimensional structures that is invariant to rotation and translation. The diagonalisation of the CMs reduces the information to a one-dimensional space to obtain order invariance at the cost of losing information. However, I hypothesize that, for the purpose of this study, the eigenvalues of the CM retain sufficient molecular-specific physical information to be meaningfully correlated with relevant properties.

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	47/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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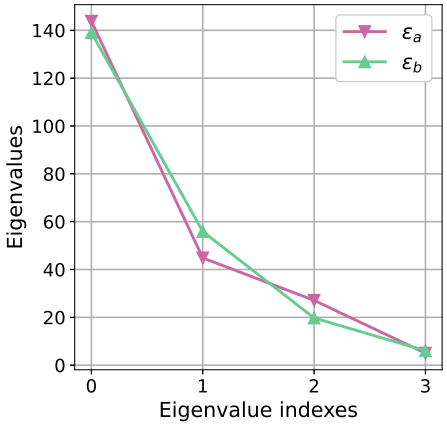


FIGURE 3.2: Eigenvalues of the CMs of acetic acid and glycolaldehyde. Hydrogen atoms are disregarded.

3.1.4 Implementation of Kinetic Monte Carlo

SEDyn is designed to simulate the time evolution of multiple states coexisting in a solid. However, the explanation of the KMC algorithm presented in Section 2.1 assumes that a simulation step is equivalent to one of the processes selected for a state. This approach to implementing KMC is effective for simulations comprising a single state, with trajectories consisting of only a few steps. However, it becomes less efficient when multiple states coexist. Therefore, SEDyn implements an algorithm that overcomes this limitation by allowing more than one state to evolve at each step.

SEDyn classifies states into isolated and non-isolated groups. Isolated states are those that, upon undergoing any of the evolutions dictated by their processes, do not change the environment of other states or the evolution of those states. To clarify this, in Figure 3.3 I have graphically represented three situations that can occur in a simulation with multiple states. Isolated states are considered to be those in which there is no state other than the ground state in their neighboring molecules and second neighbors. Thus, if an isolated state moves to a neighboring center, it does not alter the environment of other states. Therefore, for each isolated state, KMC selects a process, and the system updates accordingly. In the case of non-isolated states, that is, states with one or more other states as their first or second neighbors, KMC selects a process from among all possible processes in all non-isolated states. This is because in non-isolated processes, there is a possibility that two states may "want" to move to a molecular site that is a first neighbor for both of them, which would lead to an inconsistency and, therefore, a crash of the simulation. Once a state and a process are selected, the process time is calculated using only the possible processes in that state to ensure that the KMC algorithm is not conditioned by the number of non-isolated states.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

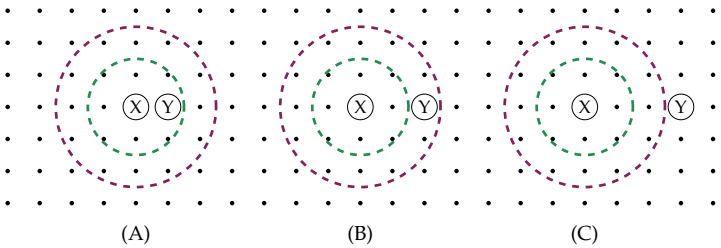


FIGURE 3.3: Graphical representation of isolated and non-isolated states according to the first and second-neighbors approach. (A) is a non-isolated case of interacting states, (B) is a non-isolated case of non-interacting states and (C) is an isolated case.

This approximation, while enabling the simultaneous and computationally efficient propagation of multiple excitons, introduces an intrinsic limitation: the total simulation time becomes ill-defined. Since each exciton evolves independently and the process time is computed locally for each selected event, the effective time experienced by each exciton may differ at any given simulation step. Consequently, the reported total evolution time should be interpreted as an approximate measure, representative of the ensemble dynamics rather than a strictly synchronized temporal progression across all excitons.

3.1.5 Classes and Relationships

SEDyn software is formed by multiple classes arranged in five layers according to their interdependence. Figure 3.4 shows the hierarchy of the main classes and their relationships, where solid and dashed arrows represent different degrees of dependency. For example, the `System` class is linked to `Kinetic Monte Carlo`, `Molecule`, `Simulation State` and `Process` classes by solid arrows; hence, the `System` class depends on these other classes to compile and execute. On the other hand, the `Process` class does not have any solid arrows, but is linked to the label attribute through a dashed arrow. This implies that processes "know" about the label attribute of the `State` class, but do not depend on the `State` class to compile and execute. In the following, I will introduce and discuss the key features of the main classes in SEDyn, starting from the top-level `System` class and proceeding down to the `Electronic Function` class. In cases where polymorphism is present, these will be discussed in more detail.

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	49/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

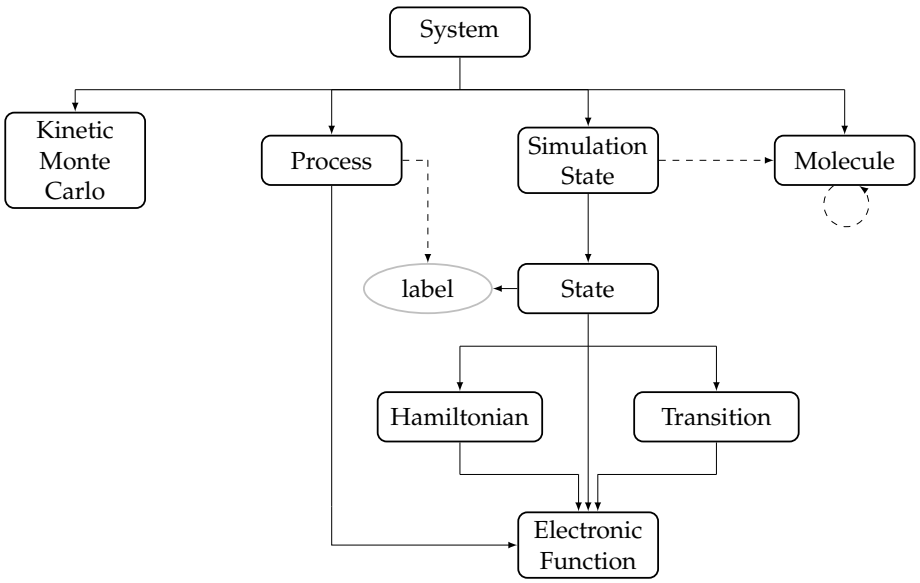


FIGURE 3.4: Hierarchy of the main classes and their relationships. Solid arrows refer to strong dependence and dashed arrows refer to weak dependence. Attributes are marked with a light grey border.

System

The `System` class is designed to be the top-level manager of the simulation, from holding general information, such as simulation steps, to storing the trajectories for each state in the system. The class is constructed at the very beginning of the execution of the software as the root of everything, even before all the input files are processed. In fact, most of the information collected from the input files is transferred to the `System` class, but for the statistical analysis instructions. After interpreting the input data and checking the consistency of the information, molecules are processed to read or identify interacting neighbors. More concretely, the `System` will tag each molecule's neighbors by reading the information from a save file or by computing the distances between molecules and checking if their distance is within a cutoff declared in the input file. Finally, the requested trajectories are run in sequence. Each trajectory is initialised by constructing the requested initial states, which are evolved through the steps until the trajectory is finalised, and the information is stored to be analysed later. Currently, in each simulation step, the calculation of all the possible processes, e.g., exciton energy transfer, exciton fission, exciton fusion, ISC, etc., is parallelised with OpenMP. However, it will be possible to parallelise all trajectories by making use of the Message Passing Interface (MPI) parallelisation algorithm. This allows the software to either increase the number of trajectories computed in the same amount of time or greatly reduce the time needed for existing simulations. The only downside of this new parallelisation approach is the increase in computational resources that are needed.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

CODE 3.1: Source code for the `System` class.

```
class System     // system.hpp
{
...
public:
    ~System()
    {
        for (auto &it : m_processes)
        {
            for (ProcessBase const *proc : it.second)
            {
                delete proc; proc = nullptr;
            }
        }
        m_processes.clear();
    }
...
private:
    arma::dvec m_cell;
    arma::dvec m_cutoff;
    arma::ivec m_pbc;
    KineticMonteCarlo m_kmc;
    SimulationState m_ground_state;
    double m_temperature;
    int m_step;
    int m_trajectory;
    int m_print_level;
    vector<Molecule> m_molecules;
    vector<State> m_states_ref;
    unordered_map<string, vector<ProcessBase const *>> m_processes;
    unordered_map<string, StateTrajectory> m_state_trajectories;
    map<string, size_t> m_state_count_index;
    arma::umat m_state_count;
    unordered_map<string, size_t> m_id_track;
    vector<SimulationState> m_states;
...
};
```


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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Processes

The `System` class above stores a vector attribute of type `ProcessBase`. The `ProcessBase` class serves as the base class for all other process types, such as the `DistanceRate` and `FermiGoldenRule` classes. In C++ and many other programming languages, this is called polymorphism and allows the `System` to store multiple different classes as part of the same base class. In addition, the `ProcessBase` class contains pure virtual functions that must be overwritten by child classes. There are many advantages to this type of design, as it allows the `System` to execute functions declared in the base class, while in practice the function is executed by the child class. For example, the system calls the *get_rate_constant* function of the `ProcessBase` class many times through the simulation. However, the "virtual ... const = 0;" syntax declares that this method must be overwritten by all derived classes. This means that every time the system calls the *get_rate_constant* function, the overwritten function above will be executed if and only if the derived class is of type `DistanceRate`. In order to compile the software, all derived classes must have overwritten this function (and all pure virtual functions); therefore, the `FermiGoldenRule` class will have its own overwritten function to execute when the system calls it.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

CODE 3.2: Source code for the `ProcessBase`, `DistanceRate`, and `FermiGoldenRule` classes.

```
class ProcessBase   // process_base.hpp
{
...
public:
    virtual double get_rate_constant(State const &sa, State const &sb,
        vector<Molecule *> const &molecules) const = 0;
...
protected:
    string m_int_state;
    vector<string> m_tar_states;
    double m_proc_time;
    int m_ham_coeff;
...
};

class DistanceRate : public ProcessBase     // distance_rate.hpp
{
...
public:
    virtual double get_rate_constant(State const &sa, State const &sb,
        vector<Molecule *> const &molecules) const
    {
        Molecule const &ma = *molecules[0];
        Molecule const &mb = *molecules[1];

        double dist = arma::norm(ma.center_of_mass()-mb.center_of_mass());

        if (dist <= m_distance) { return m_rate_constant; }
        else { return 0.0; }
    }
...
private:
    double m_distance;
    double m_rate_constant;
...
};

class FermiGoldenRule : public ProcessBase      // fermi_golden_rule.hpp
{
...
private:
    ElecBase *m_elec_func;
...
};
```


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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	53/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Molecule

The `Molecule` class, as its name suggests, stores all the molecular information relevant for the simulations. From the protein data bank (PDB) files, the coordinates and the atomic elements are read and stored, and the centre of mass of the molecule is computed. This is crucial for many calculations that must be performed before the simulation begins, such as the identification of interacting neighbors. The system loads all the molecules in the structure file and stores them in the *m_molecules* attribute. Afterwards, for each molecule, the neighbors are evaluated and a pointer reference to each neighbor is stored in a vector (*m_neighbours*).

The molecules read from the PDB file are immutable from the beginning of the simulation until the end. I refer to immutable, when an object in SEDyn cannot be modified from their creation to their destruction. Object creation in C++ corresponds to allocating memory to store its data, while destruction refers to deallocating that memory, thereby erasing the stored information. In general, and always according to the source code, this information may be modified directly for public attributes or through functions for private attributes. Therefore, immutable object are protected against any form of external modifications. Although the `Molecule` class is not immutable by design, those objects created by the system that hold the information of the solid, are considered as constant, hence, cannot be modified. The only molecules that can be modified are virtual ones, created on the fly either from existing molecules or by combining different molecules. These virtual molecules do not exist in the solid, they are not protected from structural modifications and their lifetime is restricted to the function in which they were created. This is the case of any process of interacting molecules. In order to get the coupling for the process, a virtual molecule is created from two interacting molecules and the CM between them is calculated which, then, is passed to the target electronic function. This allows the software to create and destroy any number of molecules, build CMs and obtain energies from Electronic Functions without modifying the "real" molecules of the solid.

CODE 3.3: Source code for the `Molecule` class.

```
class Molecule  // molecule.hpp
{
...
private:
    arma::dmat m_coordinates;
    vector<string> m_elements;
    bool m_occupied;
    arma::dvec m_center_of_mass;
    vector<Molecule const *> m_neighbours;
...
};
```


Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	54/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

SimulationState

The `SimulationState` structure serves to associate electronic states with a group of molecules. Each state in the system must be located in at least one molecule; however, some states, such as the triplet-triplet (`TT`) state, involve two (or more) molecules. Therefore, this class links one state to a group of molecules. Besides, the design of the `SimulationState` structure allows the simulation to associate electronic states with molecules across different unit cells. This would be the case, for example, in a `TT` state where the two neighboring molecules are located in adjacent unit cells of the solid.

CODE 3.4: Source code for the `SimulationState` class.

```
struct SimulationState  // simulation_state.hpp
{
    State m_state;
    vector<Molecule *> m_molecules;
    string m_id;
...
};
```

State

For each `SimulationState` in the system, there is one `State` object that holds the second most relevant information to run the simulation. In addition to the label, size and other attributes, the state owns an electronic function, a Hamiltonian and multiple transitions declared in the input file. These attributes will be used to compute the energy of the state at any time with only the information of the molecules stored in its `SimulationState`. In fact, at the beginning of each trajectory and every time the state suffers any modifications (unless it is destroyed), its *update* function is called and the vector of molecules stored in the `SimulationState` is passed to compute the energy of the state. The energy is computed through the electronic function and the Hamiltonian (see below), if this is requested by the user.

CODE 3.5: Source code for the `State` class.

```
class State     // state.hpp
{
...
private:
    string m_label;
    int m_size;
    int m_init_num;
    ElecBase *m_elec_func;
    double m_energy;
    Hamiltonian m_hamiltonian;
    unordered_map<string, Transition> m_transitions;
...
};
```


Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	55/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Hamiltonian

Each `State` has an associated `Hamiltonian` object, which is set to invalid by default. The `Hamiltonian` is constructed and used for energy calculations only for states where it is explicitly requested. Defining a state-specific Hamiltonian allows differentiation between multiple microstates, such as those arising from the combination of two spin-triplet states during the formation of a `TT` state. Currently, the `Hamiltonian` only supports the contribution of the exchange Hamiltonian of the interaction of two triplet states. In the future, the user will be able to specify different Hamiltonian contributions and a wide range of microstate interactions.

CODE 3.6: Source code for the `Hamiltonian` class.

```
class Hamiltonian    // hamiltonian.hpp
{
...
private:
    ProbabilityType m_type;
    ElecBase *m_elec_func;
    arma::dvec m_character;
    arma::dmat m_Hex;
    arma::dmat m_singlet;
    arma::dmat m_triplet;
    arma::dmat m_quintet;
    vector<HamiltonianBasis> m_basis;
...
};
```

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	56/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Transition

The `Transition` class is designed to store a single electronic function that defines the transition energy between the holder state and the target state. This transition is declared in the input file, and it must contain the information of the origin state label, target state label and the information regarding the electronic function which will compute the state energy, interaction energy or coupling.

CODE 3.7: Source code for the `Transition` class.

```
class Transition     // transition.hpp
{
...
private:
    string m_int_state;
    ElecBase *m_elec_func;
...
};
```

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	57/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Electronic Functions

Many classes we have visited contain the attribute `ElectBase`. This attribute makes reference to any electronic function class that derives from the `ElecBase` class and will compute some state energy, interaction energy or coupling. The `ElecBase` class is defined as an abstract class with only pure virtual functions, which means that no object of this type can be constructed, because the pure virtual functions are not declared, hence, cannot be executed. Therefore, all derived classes, such as `ConstantValue` and `KernelRidgeRegression`, must contain the definition of the pure virtual functions in order to compile.

The most crucial function to be overwritten is the *get_energy* method, which, in fact, computes the requested state energy, interaction energy or coupling according to the input molecules. Above, I show two examples of this function overwritten by derived classes. The `ConstantValue` class returns the state energy, interaction energy or coupling stored. If a deviation is allowed, a random value within a user-specified range centered on the constant value is applied. Currently, the random value is computed from a uniform distribution, but it is planned to allow the user to decide the distribution type, such as the normal distribution. The `KernelRidgeRegression` class will compute the CM between all the molecules passed to the function as an argument. Afterwards, it will build the Kernel matrix and perform the dot product between the matrix and the preoptimized parameters. More information regarding this machine learning method will be explained in the next section.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	58/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

CODE 3.8: Source code for the `ElecBase`, `ConstantValue`, `DecisionTree`, and `KernelRidgeRegression` classes.

```
class ElecBase  // elec_base.hpp
{
...
public:
    virtual double get_energy(vector<Molecule *> const &mols) const = 0;
...
};  // Abstract class with only pure virtual functions

class ConstantValue : public ElecBase   // constant_value.hpp
{
...
public:
    virtual double get_energy(vector<Molecule *> const &mols) const
    {
        if (m_dev < 1e-8) { return std::abs(m_energy); }
        else
        {
            auto distr = arma::distr_param(-m_dev, m_dev);
            double res = m_energy + arma::randu(1, distr)(0);
            return std::abs(res);
        }
    }
...
private:
    double m_energy;
    double m_dev;
...
};

class DecisionTree : public ElecBase    // decision_tree.hpp
{
...
private:
    DecisionTreeLayer m_root;
...
};

template <class T>
class KernelRidgeRegression : public ElecBase   // kernel_ridge_regression.hpp
{
...
public:
    virtual double get_energy(vector<Molecule *> const &mols) const
    {
        Molecule target(mols);
        arma::dmat CM = target.get_coulomb_matrix();
        arma::dvec eval = arma::sort(arma::eig_sym(CM), "dencend");

        arma::dmat kernel = m_kernel.get_kernel_matrix(eval, m_feat_space);
        return arma::dot(kernel.t(), m_amplitudes);
    }
...
private:
    T m_kernel;
    arma::dmat m_feat_space;
    arma::dvec m_amplitudes;
...
};
```


Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	59/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

3.1.6 Cutoff Radius & First-Neighbors

The cutoff radius is the parameter that allows SEDyn to map molecules into first-neighbor spheres, thus reducing the possible interactions to a computationally feasible number. Typically, this value determines the radius of a sphere, and those molecules that lie within this sphere are considered first neighbors (Figure 3.5). However, the spherical cutoff may include unwanted molecules in the first-neighbor list for solids whose unit cells are neither spherical nor cubic. SEDyn employs an ellipsoidal criterion to define first neighbors, adjusting its shape along all three axes to account for anisotropy in non-cubic unit cells, such as in the orthorhombic crystalline phase of rubrene.

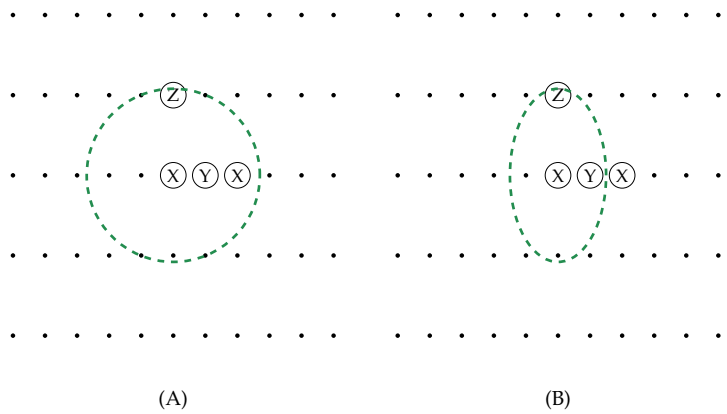


FIGURE 3.5: Graphical representation of the spherical (A) and ellipsoidal (B) cutoff radius. The spherical cutoff includes *X* as a first-neighbor of itself, while the ellipsoidal cutoff does not. Note that in a real simulation, the spherical cutoff would include the diagonal molecules as well.

3.1.7 Trajectory Analysis

The trajectories of the states requested in the input file are analyzed in the `Analysis` class, and although the code of this class is not entirely important, SEDyn implements a trick that allows you to artificially increase the number of trajectories to obtain more reliable statistical results. SEDyn is designed to perform long simulations with many states and few trajectories because, currently, code parallelization is focused on individual trajectories. However, the number of states in the simulations cannot be huge due to the computational cost that this would entail. Therefore, I have implemented the *diff_coeff_subtraj_length* parameter to divide long simulations into shorter fragments, effectively increasing the number of trajectories available for analysis. For example, in a single-state transport simulation with 100k steps, setting *diff_coeff_subtraj_length* to 1k steps results in 100 separate trajectories of 1k steps each. At the end of the simulation, the diffusion constant is then computed from these 100 shorter trajectories rather than from a singlet long trajectory, improving statistical sampling. Naturally, the optimal value of this parameter depends on the specific system under study, the number of states included in the simulation, and the total number of steps in each trajectory.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	60/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

3.2 Machine Learning Training and Input

ML models implemented in SEDyn require external information to calculate energies and/or couplings from the molecular representation discussed above. In general, ML models are parameterised mathematical models that return a calculated or expected value from numerical inputs. For example, the Decision Tree (DT) model takes input values and compares specific index values with optimised parameters. When one of these evaluations is true, the model returns an expected value. Another example is the Kernel Ridge Regression (KRR) model, which calculates a value by projecting the input values into a high-dimensional feature space and multiplying optimised parameters. This feature space is built from the training values. KRR is a mathematical tool that enables linear regression in a transformed feature space, allowing it to model non-linear relationships in the original data. In the following, I will briefly discuss the details of the ML models implemented in SEDyn and how to create external files to use ML in the exciton and charge dynamics simulations.

3.2.1 Decision Tree

According to Blockeel et al. [94], in their article on the role of decision trees in ML, a DT is defined as a representation of a procedure for computing the outcome of a function $f(x)$. The procedure consists of repeatedly performing tests on the input x, where the outcome of each test determines the next test, until $f(x)$ is known with certainty. In Figure 3.6, a very simple graph is shown to illustrate how the DT evaluates any input data until it returns a certain value.

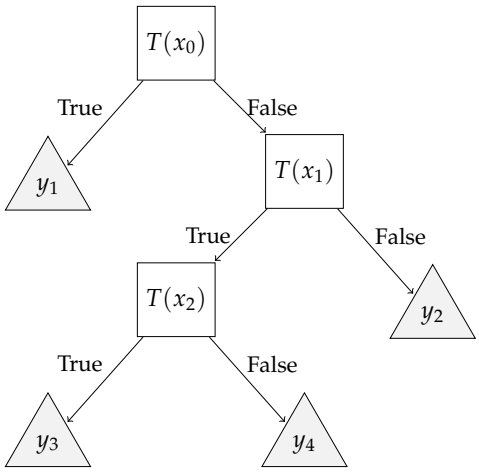
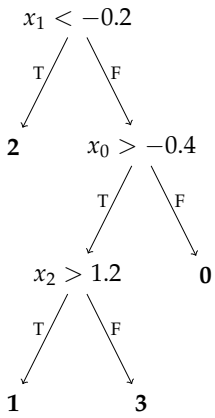


FIGURE 3.6: DT graph with three evaluation steps and four return values. Each $T(x_n)$ refers to the evaluation of x_n to an optimized parameter.

For a more visual example, I built three input vectors and a simple DT. For each input vector, the DT evaluates the input indexes iteratively until there are no more tests and a final result is returned. In the first case, the first test is true and returns the number 2, while in the second and third cases, the first test returns false; therefore, more tests are required. In the next test, the second input vector results in false, while the third vector's test is true. The latter, then, needs the last test to determine the return value, which happens to be 3, after the final test. This dummy example illustrates how classification DTs work.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	61/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403



$v_{in} = \{ x_0 , x_1 , x_2 \}$

Case 1: $v_{in} = \{ 1.4 , -0.3 , 1.6 \}$
1. $T(-0.3 < -0.2) =$ `True` $\rightarrow$ **2**

Case 2: $v_{in} = \{ -0.6 , 3.0 , 2.4 \}$
1. $T(3.0 < -0.2) =$ `False`
2. $T(-0.6 > -0.4) =$ `False` $\rightarrow$ **0**

Case 3: $v_{in} = \{ 1.3 , 0.6 , -0.7 \}$
1. $T(0.6 < -0.2) =$ `False`
2. $T(1.3 > -0.4) =$ `True`
3. $T(-0.7 > 1.2) =$ `False` $\rightarrow$ **3**

In the next section, I will show the applications of this ML model in this work and how this classification algorithm can be useful for exciton dynamics in molecular solids. In fact, this approach can be used to mimic the APS methodology used in Kimonet and Excimontec programs, which makes SEDyn able to reproduce the simulations run in these softwares with ease. However, before presenting the real applications of the DTs in this work, I will show how we can provide SEDyn with the necessary information. SEDyn will require a JSON file with all the data from the DT. I will use the dummy example above as a guide.

CODE 3.9: DT json file example.

```
{
    "lh": "1",
    "condition": "lt",
    "rh": "-0.2",
    "true": 2,
    "false": {
        "lh": "0",
        "condition": "gt",
        "rh": "-0.4",
        "true": {
            "lh": "2",
            "condition": "gt",
            "rh": "1.2",
            "true": 1,
            "false": 3
        },
        "false": 0
    }
}
```

In Code 3.9, "lh" (left hand) means the index of the input vector, "condition" is the mathematical evaluation written as a string (e.g., "lt" indicates "<") and "rh" (right hand) means the optimised parameter. "true" and "false" will either return a value (coupling energy) or initialise another evaluation. In order to tell SEDyn that it needs to read a DT from a JSON file, we need to declare the "elec_func" keyword as follows:

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	62/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

CODE 3.10: Code to indicate the use of external file with DT data.

```
$section
...
elec_func dectree path/to/dt_model          ! Do not include the .json
...
$$
```

3.2.2 Kernel Ridge Regression

KRR, as described by Allerbo [95], "is a generalisation of linear ridge regression that is non-linear in the data, but linear in the model parameters". This linear regression model takes advantage of the kernel trick to optimise a linear function in a high-dimensional feature space. Given the excellent elaboration of the KRR mathematical details in the paper by Allerbo, I do not provide a detailed definition of the model here and instead refer the reader to that paper [95].

For a new data, $\gamma_* = \left[\gamma_*^{(1)}, \gamma_*^{(2)}, ..., \gamma_*^{(m)}\right]^\top$, the prediction of the KRR model is described by the prediction function, $\hat{\mathbf{f}}$, defined by

$$\hat{\mathbf{f}}(\gamma_*, \mathbf{\Gamma}) = \boldsymbol{\kappa}_*^\top \boldsymbol{\alpha} \quad \text{and} \quad \boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y} \tag{3.2}$$

where $\mathbf{\Gamma}$ is the design matrix formed by stacking column vectors of training data, $\mathbf{\Gamma} = [\gamma_1, \gamma_2, ..., \gamma_n]^\top$. $\boldsymbol{\kappa}_*$ and $\mathbf{K}$ denote a kernel vector and matrix defined as $(\boldsymbol{\kappa}_*)_i = k(\gamma_*, \gamma_i)$ and $\mathbf{K}_{i,j} = k(\gamma_i, \gamma_j)$ according to the kernel function, $k(\gamma, \gamma')$. λ is the regularization strength hyperparameter and $\mathbf{y}$ is the vector of training electronic coupling values. Here, I am using superindexes to denote the elements for each feature to clearly distinguish them from the subindexes, which denote different data instances. In a preliminary benchmark, the Gaussian kernel function has proved to give the best r^2 values in the linear regression. This kernel projects the data by computing the Euclidean distances between them and is defined by

$$k(\gamma, \gamma') = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\|\gamma - \gamma'\|^2}{2\sigma^2}} \tag{3.3}$$

where σ is a hyperparameter, which needs to be optimised in the learning process. In our case, the instances of the training data, $\mathbf{\Gamma}$, and new data, γ_*, are the eigenvalues of the CMs, Eq. (3.1).

KRR models can be optimised in any programming language, although, in general, Python is used because there are plenty of mathematical tools to perform such ML optimisations. However, the user is free to use any external package or build their own module to implement the algorithm. Once the KRR model is optimized, SEDyn will require the optimized $\boldsymbol{\alpha}$ parameters, the design matrix $\mathbf{\Gamma} = [\gamma_1, \gamma_2, ..., \gamma_n]^\top$ and the Gaussian parameter σ. In order to make SEDyn compatible with any KRR model, the software will read the data from a single HDF5 file containing three datasets, according to the following naming:

- *amplitudes*: $\boldsymbol{\alpha}$ vector.
- *feature_space*: $\mathbf{\Gamma}$ matrix.
- *gaussian_parameters*: Vector containing σ parameter.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	63/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Users who will use Python as their main tool for KRR model optimisations can find in Code 3.11 the simple script that will save the data structures inside an HDF5 file,

CODE 3.11: Python script to save data structures to a HDF5 file.

```
import numpy as np
import h5py

# Create new file
h5f = h5py.File(f'/path/to/krr_model.h5', 'w')

# Save data into file in the higher level
h5f.create_dataset('amplitudes', data = alpha)
h5f.create_dataset('feature_space', data = F)
h5f.create_dataset('gaussian_parameters', data = np.array([s]))

# Close the file
h5f.close()
```

and to inform SEDyn that we want to use a KRR model from an external file, we declare the following lines in the target electronic function in the input file.

CODE 3.12: Code to indicate the use of external file with KRR data.

```
$section
...
elec_func krr path/to/krr_model              ! Do not include the .h5
...
$$
```

3.2.3 Feedforward Neural Network

According to the "Feedforward Neural Network Methodology" book by Terrence L. Fine [96], "the architecture of a feedforward net is defined by a directed, acyclic graph and a choice of node functions". This graph is usually composed of multiple nodes grouped in layers and linked to other layers, that is, connected by directed links. Besides, this type of graph is called acyclic because "for no node is there a directed path that departs from that node and returns to it", in other words, there are no closed "cycles". The first layer with no incoming links contains the input nodes, while the last layer with only incoming links contains the output nodes. In this work, we focus on the special case of the Multilayer Perceptron (MLP), in which each layer is only connected to the immediately preceding layer, called Feedforward Neural Network (FNN). For the sake of simplicity, in what follows, I will refer to MLP simply as FNN. The predicted values of the schema shown in Figure 3.7 are obtained with the equation

$$y_m = f\left(\sum_{j=1}^{N_h} b_{2:j} + w_{2:j,m} \cdot f\left(\sum_{i=1}^{N_x} b_{1:i} + w_{1:i,j} \cdot x_i\right)\right) \tag{3.4}$$

where y_m is any output node and x_i is the ith input node, N_h and N_x represent the number of nodes in the hidden layer and input layer, respectively. w_1 and w_2 are matrices with the weights that link nodes, while b_1 and b_2 are the vectors with the corresponding biases (shifts). The f denotes the activation function, applied to

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
Araua / Normativa	Elektronikoki sinatutako dokumentua (203/2021 ED). Adierazitako URLan lokalizatzaile bidez egiazta daitekeen egiazkotasuna/ Documento firmado electrónicamente (R.D.203/2021). Autenticidad verificable mediante localizador en la url indicada.	Dokumentuaren baliozkotasuna / Validez del documento	Otros
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	64/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

the weighted sum computed at each node. While f is typically nonlinear, linear activations may also be used in specific layers, such as the output layer.

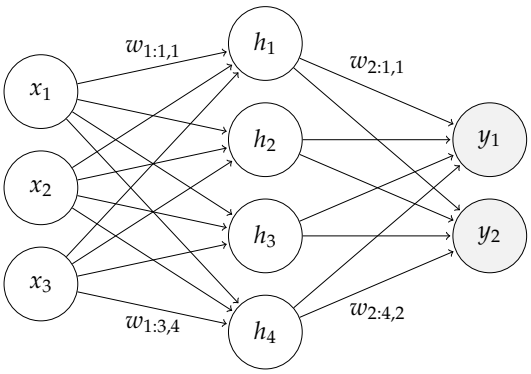


FIGURE 3.7: FNN graph with three input nodes ($N_x = 3$) and two output nodes. The hidden layer contains four nodes ($N_h = 4$).

In this work, the Rectified Linear Uniform (ReLU) activation function has been chosen due to its simplicity and improved performance [97, 98]. As defined by Dubey et al., ReLU "is a simple function which is the identity function for positive input and zero for negative input", described by,

$$\mathrm{ReLU}(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{3.5}$$

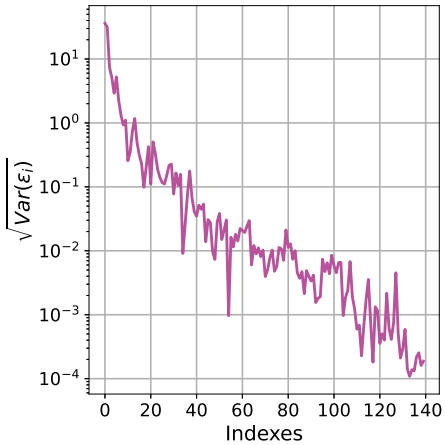


FIGURE 3.8: Square root variances of each eigenvalue indexes of the CMs of the four pairs in the unit cell of the orthorhombic rubrene crystal. Note that the Y axis is in logarithmic scale.

In other words, the ReLU function returns the maximum value between input and zero, that is, $max(0, x)$, where x is the input value. This makes the range of ReLU $[0, \infty)$, whose downside is "with the vanishing gradient problem for the negative inputs". Preliminary explorations of the eigenvalues of the CMs of crystal structures

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

qualitatively reveal that the highest eigenvalues contain most of the differential information.

The calculation of the variances for each of the eigenvalue indices roughly quantifies the weight of each parameter in the training of the FNN, as shown in Figure 3.8. While the values are not significant, I can argue that the negative eigenvalues should not contribute largely to the training of the net, as their square root variance decreases exponentially. Ideally, one should prove this hypothesis by training the same FNN with and without the n negative eigenvalues, instead of relying on this rough approximation. Nevertheless, it is out of the scope of this work to test this, as the KRR discussed above shows better performance and only needs to tune a few hyperparameters.

3.3 Installation and Execution

In this section, I will explain how to install and execute SEDyn Software on a Linux system. Readers who have previously installed a GitHub package or repository may skip to the next section and proceed directly to the examples provided. SEDyn can be downloaded from my Gitlab repository (*gitlab.com/sanerdor/sedyn*). Then the user must uncompress it and follow the installation steps below.

3.3.1 Installation Requirements

The compilation of SEDyn requires some packages listed below, most of which are available on Linux systems. Apart from these packages, Armadillo is also required to compile and run SEDyn, as this library is the one responsible for all vector and matrix data structures, as well as fast and easy-to-manage mathematical operations between these data structures. The user can find in *arma.sourceforge.net/download.html* the package and all the information on how to compile Armadillo.

- CMake `sudo apt install cmake`
- OpenMP `sudo apt install libomp-dev`
- OpenBLAS `sudo apt install libopenblas-dev`
- LAPACK `sudo apt install liblapack-dev`
- HDF5 `sudo apt install libhdf5-dev`

3.3.2 Compilation and General Usage

After every required package has been installed in the machine, the compilation of SEDyn can be done in two steps: configuration and compilation. For this, we execute the following commands inside the extracted directory.

```
./configure release gcc
cd build
make install
```

This is the simplest way to compile SEDyn. However, the software is thought to benefit from OpenMP parallelisation to speed up some of the processes. Therefore, I strongly recommend including the OpenMP flag in the configuration and parallelising the compilation if possible.

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	66/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403


```
./configure release gcc openmp
cd build
make -jX install              # where X is the number of OpenMP threads
```

Once the installation is completed, the environment variables must be declared, and SEDyn's execution file path must be included in the PATH. Both of these steps can be done in the command line, or the user can create a custom environment in their preferred shell's startup file, e.g., .bashrc, that sets all these variables automatically.

```
export SDN=/path/to/SEDyn
export PATH=$PATH:$SDN/bin
export OMP_NUM_THREADS=X            # where X is the number of threads
```

On the other hand, the execution of SEDyn requires at least two input files: input and molecular system files. In summary, the input file (see Section 2.3) contains the parameters for the simulation, ranging from the number of steps per trajectory to the declaration of all processes available to each state. While the input file may initially appear complex, the accompanying examples are intended to clarify its structure and usage. The molecular system files provided in standard PDB format, containing the atomic coordinates and atom types for all the molecules. The use of a PDB file is compulsory, because SEDyn requires a format that explicitly distinguishes between different molecules. While coordinate formats such as *.xyz* contain most of the necessary information, they lack explicit molecular separation. Although there exist multiple tools to infer molecular boundaries from geometric criteria such as bond lengths, I designed the software to avoid misinterpretations by requiring an unambiguous input format.

In addition to the input and molecular system files, the output and save files can be declared in the command line. If any of them are not declared, the default output and save files are created in the currenty directory with the file names *input*.out and *input*.h5, respectively. Besides, the name of the scratch directory can be declared to store all the files created by SEDyn.

```
sedyn infile.in molfile.pdb
---
sedyn -nt X infile.in molfile.pdb outfile.out h5file.h5 scratchdir
        # where X is the number of OpenMP threads
```

3.4 Example Simulation

In this section, a dumb example is presented for the user to be able to compare and check if SEDyn has compiled correctly. All of the energies and results in this simulation have been made up for the sake of the example. Due to the stochastic nature of the KMC algorithm, the user should expect similar results, although exact coincidences occur only by chance. The system is formed by 1000 atoms of hydrogen with an interatomic distance of 3 Å in all directions. Four excitons are spawned at the beginning of the simulation and are evolved for 10k simulation steps. Here is the input example:

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

CODE 3.13: Input file example.

```
! ******
! SYSTEM
! ******

$run
    max_step 10000          ! maximum number of simulation steps
    trajectories 1          ! number of trajectories
    cutoff 3.5, 3.5, 3.5    ! Elipsoid cutoff radius
    boundaries 1, 1, 1      ! 3D vector of PBC
    temperature 298         ! Temperature in K
$$

! ******
! SCHEMA
! ******

$schema                     ! Compulsory
    system standard         ! Save system data information
    trajectory standard     ! Save trajectory data information
$$

! ********
! ANALYSIS
! ********

$analysis
    diff_coeff st           ! State label to analyse diffusion coeff.
    diff_coeff_subtraj_length 100   ! Linear regression vector length
    lifetime st             ! State label to analyse lifetime
$$

! **********
! EXCITON ST
! **********

$state
    label st                ! State label (any string, unique)
    initial_num 4           ! Number of states at the beginning
    size 1                  ! Size of the state
    elec_func const 2.23    ! Electronic function with the energy
$$

$process
    label st                ! Label of the origin state
    type fermigoldenrule    ! Type of process
    interaction_state gs    ! Label of the interacting state
    target_states gs, st    ! Label of the target states ordered by
                            ! origin molecules + interacting molecules
    elec_func const 0.1     ! Electronic function with the coupling
$$

$transition
    label st                ! Label of the origin state
    interaction_state gs    ! Label of the interacting state
    elec_func const 0.1     ! Electronic function with the energy
$$
```

The output of the simulations contains the information requested in the input file. After a small section with general information, SEDyn prints the consistency of the requested simulation, some information regarding the solid structure and the

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	68/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

process of linking neighbours, and the simulations results. For each trajectory, the number of states in each step is monitored. At the end of the simulation, the statistical analysis of all the trajectories is done, and the results are printed in the output file. Another example can be found in Appendix A.1.

```
1
2
3                ========================================
4
5                            SEDyn (3.0.0)
6
7                ========================================
8
9                   ---Solid State Exciton Dynamics---
10
11
12 Authors : Diaz-Andres, A.
13
14 SEDyn begins on Mon Aug 04, 10:40:44, 2025
15
16 Number of OpenMP threads: 7
17
18
19 Checking system consistency
20 ---------------------------
21 Number of molecules in the system: 1000
22 Number of states in the system: 1
23 Number of processes for state ST: 1
24
25
26 Simulation general information
27 ------------------------------
28 Cell size: [27, 27, 27]
29 Cutoff radius: [3.5, 3.5, 3.5]
30 Periodic Boundary Conditions: [1, 1, 1]
31 Temperature: 298
32 Number of steps: 10000
33 Number of trajectories: 1
34
35
36 Neighbour linking
37 -----------------
38 Max number of pairs to evaluate: 499500
39 Number of neighbours to link: 5116
40 ... Neighbours writen to archive file
41
42 Linking elapsed time: 0h 0m 0s
43
44
45 Trajectory : 0
46 ------------
47 Number of states initialized: 4
48 Number of molecules occupied: 4
49
50     Step |  ST |
51  --------------------
52        0 |   4 |
53     1000 |   4 |
54     2000 |   4 |
55     3000 |   4 |
56     4000 |   4 |
```


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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	69/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403


```
    5000 |   4 |
    6000 |   4 |
    7000 |   4 |
    8000 |   4 |
    9000 |   4 |
 --------------------

Trajectory elapsed time: 0h 1m 12s


Analysis
--------

** State : ST

Diffusion Coefficient ...  2.6390e+01 cm^2/s
  x:  2.1802e+01  y:  3.7078e+01  z:  2.0290e+01

Lifetime ...  4.5816e-03 ns



  SEDyn simulation time: 0h 1m 12s


               ----------------------------------------

                  Tranquility lies in human's heart

               ----------------------------------------
```

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	70/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Chapter 4

Electronic Couplings for Triplet-Triplet Annihilation Photon Upconversion in Crystal Rubrene

"If you find yourself in a hole, the first thing to do is stop digging."

John Marston - Red Dead Redemption II

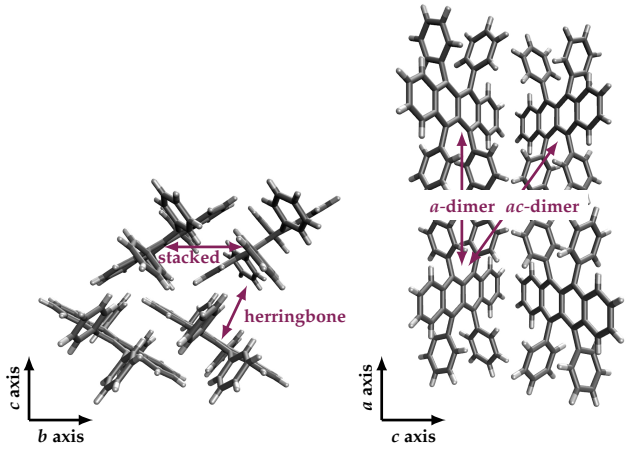
Triplet-triplet annihilation photon upconversion (TTA-UC) is a process able to repackage two low-frequency photons into light of higher energy. This transformation is typically orchestrated by the electronic degrees of freedom within organic compounds possessing suitable singlet and triplet energies, and electronic couplings. In this work, I propose a computational protocol for the assessment of electrornic couplings crucial to TTA-UC in molecular materials. This protocol applied the RASCI multiconfigurational method to effectively characterize the triplet-triplet coupled stated with overall singlet spin, which has become challenging for other quantum chemical methods. This methodology integrates sophisticated yet computationally affordable approaches to quantify couplings in singlet and triplet energy transfer, the binding of triplet pairs, and the fusion to the singlet exciton. I investigate intermolecular interactions in the four unique first neighbor rubrene dimers within the orthorhombic crystal (Figure 4.1). In Table 4.1 the intermolecular distances between molecular center of masses are presented. More information of the crystal can be found in Chapter 2.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	71/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

TABLE 4.1: Intermolecular distances (in Å) between the molecular center of masses for the four unique first neighbor dimers in the orthorhombic crystal structure of rubrene.

dimer	distance
stacked	7.17
herringbone	7.96
a-dimer	13.86
ac-dimer	15.16



(A) (B)

FIGURE 4.1: Crystal structure of rubrene along the *bc* (A) and *ac* (B) planes. Purple arrows indicate the crystal rubrene dimers studied.

By the end of the chapter, the effect of temperature on the electronic couplings is studied to assess how molecular distortions and symmetry breakings influence their magnitude and behavior. These results are already published in an indexed journal [99]. Note that numerical results may vary from the published paper because, in the present results, additional molecular calculations have been included in the study of thermal effects.

4.1 Singlet and Triplet Exciton Energy Transfer

The study of electronic couplings governing singlet and triplet exciton energy transfer is performed using the methods presented in Chapter 2. The couplings for singlet exciton energy transfer (SEET) values obtained with the different approaches (Table 4.2), typically on the order of tens of meV, are in very good agreement with each other, except for those predicted by the point dipole approach for the dimers with short intermolecular separation (stacked and herringbone dimers), which are considerably larger. This discrepancy is somehow expected, as the point dipole approach breaks down at short distances due to the omission of molecular size effects and intermolecular orbital overlaps. On the other hand,

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Egiaztapenerako URLa / Url Verificación	https://app.ehu.eus/verifirma/code/IVVH2KPLYEB6YSVKU5MS6UST6E	Orrialdea / Página	72/119

REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

dipole-dipole couplings for *a*-dimer and *ac*-dimer are much closer to those from electronic structure calculations.

Interestingly, for all four dimers, the methods that take electronic structure considerations into account, namely HEG, FED and Boys(n) diabatization (with n the number of states considered), produce nearly identical results, with the herringbone dimer exhibiting the largest SEET interaction. The couplings computed with HEG, FED and Boys(2) account for exciton coupling between localized singlets, that is, interaction between S_1S_0 and S_0S_1 configurations, but they also implicitly contain potential anion-cation (AC) and cation-anion (CA) contributions partially mixed in the two lowest singlet eigenstates. In order to explore the role of CT states in SEET couplings, I expand the dimension of the diabatization scheme to five states (Boys(5) in Table 4.2), adding the two lowest singlets with strong CT character and the multiexcitonic state mostly corresponding to the triplet-pair state (^{1}TT). Within this approach, the couplings can be described in terms of direct (first order) and mediated (second order) contributions (Eq. (2.14)).

The results clearly indicate that the direct exciton coupling dominates in all cases except for the stacked dimer, for which the CT-mediated terms partially cancel the direct interaction. Therefore, despite the favorable intermolecular disposition for the exciton interaction in the stacked dimer (as indicated by the dipole-dipole interaction and the Boys(5) direct coupling), the mixing of local excitons with charge separated terms notably diminishes the total coupling. I thus conclude that, in this case, CT excitations are detrimental for the singlet exciton diffusion in crystal rubrene, specifically along the *b*-axis.

TABLE 4.2: Electronic couplings (in meV) for SEET and TEET processes in the four first-neighbor crystal dimers computed at the RASCI/cc-pVDZ level.

dimer	dip-dip	HEG	FED	Boys(2)	Boys(5)
			SEET		
stacked	50.6	13.1	21.3	17.7	26.8 (48.0)
herringbone	37.0	33.3	34.9	31.8	31.8 (31.7)
a-dimer	13.9	12.7	13.0	12.5	12.5 (12.5)
ac-dimer	7.2	7.1	7.3	7.0	7.0 (7.0)
			TEET		
stacked	-	11.4	11.2	12.9	-
herringbone	-	4.1	0.2	0.4	-
a-dimer	-	0.0	0.0	0.0	-
ac-dimer	-	0.0	0.0	0.0	-

The dark character of triplet excitons (zero transition dipole moment) prevents the use of dipole-dipole approach for TEET. Moreover, here I do not consider the diabatization scheme with five states, since ^{3}CT (spin-triplet CT states) and ^{3}TT are energetically much higher than the monomeric triplet (T_1S_0 and S_0T_1), and state mixing can be safely disregarded. TEET couplings are notably smaller compared to those obtained for singlet excitons. Additionally, they manifest a substantial decline with increasing intermolecular distance, in accordance with the exponential decay predicted by the Dexter energy transfer rate formula [100], which arises from the decay of the exchange integral governed by the overlap between donor and acceptor

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

wavefunctions,

$$\Gamma = KJ \exp\left[\frac{-2R_{DA}}{L}\right] \tag{4.1}$$

where R_{DA} is the distance between donor and acceptor molecules, L is the sum of Van der Waals radius of the donor and acceptor, K is an experimental constant and J is the normalized spectral overlap integral. The strongest TEET interactions are along the b-axis (11-13 meV), since the π-π overlap is favored in the coplanar stacked dimer. On the other hand, triplet exciton diffusion is predicted to be very inefficient on the ac crystal plane, and specially along the a-direction.

It is worth noticing that the SEET results in Table 4.2 seem to be in contradiction with experimental measurements, which indicate that exciton diffusion in the crystal is preferred along the molecular stacking direction (b-axis) [101, 102]. But, it has been shown that the anisotropic exciton diffusion in rubrene single crystals is driven by triplet exciton transport, available by singlet-triplet interconversion via singlet fission and triplet fusion annihilation processes [103, 104]. In fact, this is in agreement with our TEET results, indicating the tendency of triplet excitons to move along axis b.

SEET and TEET electronic couplings are then calculated with time-dependent density functional theory (TDDFT). The results (Table 4.3) come in agreement with the results obtained with RASCI level of theory. Classical dipole-dipole interaction fails to describe the SEET electronic couplings of stacked and herringbone dimers due to the orbital overlap between molecules. HEG, FED and Boys(2) diabatization methods for the calculation of exciton energy transfer couplings result in very similar results.

TABLE 4.3: Electronic couplings (in meV) for SEET and TEET process in the four firt-neighbor crystal dimers computed at the rCAM-B3LYP/cc-pVDZ level.

dimer	dip-dip	HEG	FED	Boys(2)
		SEET		
stacked	60.8	13.3	18.2	18.2
herringbone	44.4	31.0	30.6	30.5
a-dimer	15.1	17.8	17.8	17.8
ac-dimer	8.6	9.8	9.8	9.8
		TEET		
stacked	-	7.2	7.2	7.2
herringbone	-	3.0	0.3	0.4
a-dimer	-	0.0	0.0	0.0
ac-dimer	-	0.0	0.0	0.0

The results obtained with RASCI and TDDFT bring the possibility to effectively calculate these couplings with either of these methods. However, the innacuracy of RASCI to describe excited state energies favors the choice of TDDFT for the calculation of SEET and TEET electronic couplings.

4.2 Triplet-Pair state

The TTA process involves the fusion of two triplet excitons, which can be regarded as the reverse photophysical reaction of SF. It proceeds through the formation of the

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triplet-pair state (TT), as described in Section 2.4. In the following, I characterize the properties of TT states in rubrene crystal dimers in line with the framework presented in Section 2.4.1.

The results of the triplet-pair coupling (J) at the RASCI level indicate a substantial exchange interaction favoring the spin-singlet state for the π-stacked rubrene dimer, with $J = 7.5$ meV (antiferromagnetic coupling), while the singlet, triplet, and quintet TT states are nearly degenerated ($J \approx 0$ meV) in the three other dimers. These results can be rationalized by the degree of intermolecular overlap in the natural orbitals of the triplet-pair state (see Figure 4.2), which is significantly larger in the stacked dimer compared to the others. In other words, the electronic triplet-triplet binding energy is only significant in the π-stacked dimer, for which orbital interaction stabilizes ^{1}TT with respect to ^{5}TT. This behavior is also manifested by the distribution of natural orbital occupancies of the ^{1}TT state, with values slightly departing from four orbitals with a single electron each, which is a sign of (small) electron pairing.

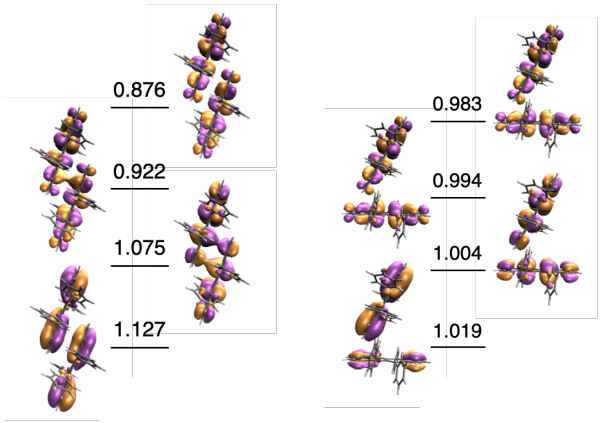


FIGURE 4.2: Frontier natural orbitals and their electron occupancies for the ^{1}TT state of the stacked (A) and herringbone (B) first-neighbor rubrene crystal dimers computed at the RASCI/cc-pVDZ level.

4.3 Triplet Fusion and Singlet Fission

Next, I explore the interactions controlling the spin-allowed transition from ^{1}TT to S_1 in crystal rubrene. It is well-known that the direct coupling between ^{1}TT and S_1 states is, in many cases, very small [105, 106], since it involves a change of two electrons. Within the two-electrons-in-two-orbitals model, i.e., only considering HOMO and LUMO of each chromophore, direct couplings correspond to the difference between two-electron integrals [54, 107]. On the other hand, charge separated states can strongly interact with both ^{1}TT and S_1, mediating their interconversion through the second order term in Eq. (2.14). Therefore, here we employ the second order perturbative approach with ^{1}TT and S_1S_0 as initial and final states, respectively. Electronic states and couplings have been obtained with the Boys diabatization of five excited singlet states computed at the RASCI/cc-pVDZ level, resulting in five diabatic states with pristine S_1S_0, S_0S_1, AC, CA, and ^{1}TT nature, respectively.

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Evaluation of the second order contribution to the total coupling requires diabatic energies. Although RASCI with one-hole and one-particle contributions has shown excellent performance in the characterization of electronic states, it is well understood that the lack of higher order terms can have an important impact on the computed relative energies [65]. Hence, to mitigate inaccuracies arising from RASCI energies, I take experimental energies for the lowest singlet state ($E(\mathrm{S}_1) = 2.23$ eV) and approximate the triplet-pair energy to twice the energy of the lowest triplet ($E(^1\mathrm{TT}) \approx 2E(\mathrm{T}_1) = 2.28$ eV) states [108]. The relative CT energies computed for isolated dimers have been corrected in order to take into account the surrounding polarizable medium as prescribed by previous polarization energy calculations in the bulk [109]. The computed interstate couplings and CT energies are shown in Table 4.4 (results considering $\mathrm{S}_0\mathrm{S}_1$ the target state can be found in the Appendix B.1).

TABLE 4.4: Electronic couplings (in meV) computed at the RASCI and C-DFT level/cc-pVDZ level through Boys diabatization, and CT energies, E(CT) = (E(AC) + E(CA))/2 (in eV), computed at the C-DFT level/cc-pVDZ level for the four crystalline rubrene dimers.

	stacked	herringbone	*a*-dimer	*ac*-dimer
$\langle ^1\mathrm{TT}\vert H\vert \mathrm{S}_1\mathrm{S}_0\rangle$	0.0	0.0	0.2	0.0
$\langle ^1\mathrm{TT}\vert H\vert \mathrm{AC}\rangle$	0.0	0.0	2.4	0.6
$\langle ^1\mathrm{TT}\vert H\vert \mathrm{CA}\rangle$	0.0	0.0	2.4	0.6
$\langle \mathrm{AC}\vert H\vert \mathrm{S}_1\mathrm{S}_0\rangle$	-173.1	-25.7	2.2	0.4
$\langle \mathrm{CA}\vert H\vert \mathrm{S}_1\mathrm{S}_0\rangle$	90.7	-8.1	1.6	0.5
E(CT)	2.65	3.03	3.87	2.71

Computed direct couplings between initial and final states are very small in all four first neighbor crystal dimers (0.2 meV in *a*-dimer and below 0.1 meV for the rest). On the other hand, some of the interactions between initial/final diabats and CT states appear to be rather significant. In the two short-distance dimers (stacked and herringbone), the coupling of the localized S_1 with CT configurations is rather strong and in rather good agreement with previous calculations on the stacked dimer at the multireference second order perturbation theory level (-175 and 86 meV, respectively) [106], with the ZINDO approach (67 meV) [27], and with evaluations of the HOMO-HOMO transfer integral (100-200 meV) [101, 106, 110, 111]. In contrast, the $^1\mathrm{TT}/\mathrm{AC}$ and $^1\mathrm{TT}/\mathrm{CA}$ couplings vanish for the stacked and herringbone dimers. The effectively zero interaction between the spin-singlet triplet-pair configuration and CT states in the π-stacked dimer has been rationalized by orbital symmetry arguments [105, 106, 108, 112].

Electronic couplings in *a*-dimer and *ac*-dimer involving charge separated terms are $\sim$2 meV in the former and $\sim$0.5 meV in the latter, but their contribution to the total coupling is hindered by the larger energy gap between initial/final states and CT terms, especially in the *a*-dimer. Moreover, the (small) second order term in these dimers partially cancels out the direct coupling contribution (opposite signs in Eq. (2.14)).

All in all, total electronic couplings for the interconversion between the triplet-pair state and S_1 in the crystal dimers are very weak (smaller than 0.1 eV in all cases). These results seem to prevent the fusion/fission process (Eq. (2.16)) in the rubrene crystal, which is in strong contradiction with experimental evidence of TF and SF in rubrene crystals.

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4.4 RASCI Active Space Comparison

The RASCI target states are susceptible to the quality of the reference wave function and the RASCI operator, $\hat{R}$ (Section 2.5). Moreover, the size of the active space directly impacts the target states as larger active spaces describe the states with more configurations, hence, stabilizing their energies and properties. Besides, hole and particle excitation operators can significantly contribute to the description and stabilization of the target states with new single excitation configurations. Therefore, I study the variations in the computed electronic couplings by comparing different active spaces, analyzing the role of hole and particle contributions, and by testing various basis sets to describe the orbitals.

The use of (h, p) has almost no impact on the TEET couplings and systematically diminish SEET interactions, although the results remain semi-quantitatively similar to those obtained without (h, p) terms. On the other hand, the presence of (h, p) contributions is rather important in the relative stabilization of the ^{1}TT state with respect to the ^{5}TT, as hole and particle terms in the minimal (4,4) RAS2 space preferentially mix with the singlet triplet-pair state and are decisive in the electronic binding of the two triplets.

TABLE 4.5: Electronic couplings (in meV) computed for the crystal stacked structure for SEET, TEET, triplet-triplet binding (J) and ^{1}TT/S_1 (TF/SF) computed with RASCI with different RAS2 spaces, with and without hole and particle excitations (h, p), and with different basis sets.

			SEET				
RAS2	(h, p)	basis	FED	Boys(5)	TEET	J	TF/SF
(4,4)	no	6-31G	22.7	15.6 (52.6)	11.9	1.2	0.0
(4,4)	yes	6-31G	18.5	0.5 (69.4)	10.3	6.8	0.0
(4,4)	no	cc-pVDZ	21.3	26.8 (48.0)	12.2	1.6	0.0
(4,4)	yes	cc-pVDZ	13.0	15.2 (65.7)	11.4	7.5	0.0
(6,6)	no	cc-pVDZ	43.7	0.6 (74.3)	9.3	1.7	0.0

The SEET Boys(5) coupling at the RAS(4,4)/6-31G (with hole and particle) and RAS(6,6)/cc-pVDZ (without hole and particle) are very small (< 1 meV). This result can be related to the inability of the diabatization scheme to produce diabatic states with the pristine local or CT character in these cases.

Therefore, considering the reliability of the methods and their computational cost, I decide to perform RASCI calculations on SEET, TEET and TF/SF with a (4,4) RAS2 space without hole and particle excitations. Only for triplet-triplet binding energies I include de (h, p) terms. Although results with the 6-31G basis do not present major discrepancies, I prefer to use the larger cc-pVDZ basis set, as I anticipate that polarization functions might be important to describe some molecular arrangements.

4.5 Role of Thermal Fluctuations

In the following, I examine the role of thermal fluctuations in the electronic interactions responsible for the singlet and triplet exciton diffusion, triplet-triplet binding and in the interconversion between the triplet-pair and the singlet exciton (fusion/fission). For that, I employ classical MD simulations to generate a

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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representative sample of thermally distorted structures, for which I compute interstate couplings and CT energies.

Finite temperature MD simulations of bulk rubrene crystal were carried out, using a force field specifically parameterized for rubrene [113]. The torsional parameters associated with the dihedral angle between the tetracene and phenyl moieties were derived from DFT calculations at the PBE0/def2-TZVP level, while atomic charges were obtained at the PBE0/TZVP level using benzene as the implicit solvent. Additionally, the carbon and hydrogen atoms' Lennard-Jonnes parameters were tuned in a trial-and-error procedure running simulations at both $T = 100$ K and $T = 300$ K at ambient pressure, systematically comparing simulated and experimental crystal lattice parameters a, b and c, and cell volume. A quasi cubic sample containing 256 rubrene molecules was first equilibrated in the NpT ensemble ($p = 1$ atm) at 298.15 K for 9 ns before a 1 ns production run, from which 20 frames were extracted at regular time intervals. Pressure and temperature controls were achieved using the Berendsen's barostat and the velocity scaling thermostat, respectively.

In the calculation of electronic couplings, $\mathtt{S}_1$ and $\mathtt{T}_1$ diabatic energies have been fixed to the experimental values used above, while CT states have been recomputed for each dimer. This approach does not take into consideration the role of crystal defects or energetic disorder (variations in site energies), which can be of relevance in actual devices [114–117]. Moreover, despite that the relative signs of the computed electronic couplings for a given structure are fully consistent, the obtained phase of diabatic wavefunctions is not fixed to a specific criterion, and might change between separate calculations. Therefore, here we analyze unsigned couplings.

TABLE 4.6: Electronic couplings (in meV) computed for the crystal structure (cryst.) and averaged over 200 snapshots from MD (TA: thermal average) for SEET, TEET, triplet-triplet binding (J), and $^1\mathtt{TT}/\mathtt{S}_1$ (TF/SF).

	SEET		TEET		J		TF/SF	
dimer	cryst.	TA	cryst.	TA	cryst.	TA	cryst.	TA
stacked	18.2	26.8	7.2	6.9	7.5	6.9	0.0	126.7
heringbone	30.6	34.3	0.3	0.2	0.0	0.1	0.0	1.1
a-dimer	17.8	19.2	0.0	0.0	0.0	0.0	0.0	0.0
ac-dimer	9.8	10.6	0.0	0.0	0.0	0.1	0.0	0.0

The average couplings over 200 structural MD frames at 298 K are shown in Table 4.6. Median values can be found in Table B.2. The SEET average values are slightly larger than those in the frozen crystal structure, especially for the π-stacked dimer, for which small displacements might involve sizeable changes in the orbital interactions. Thermal average TEET couplings and triplet-triplet binding energies are very close to the 0 K values. The most impressive changes upon consideration of thermal fluctuations are for the $^1\mathtt{TT}/\mathtt{S}_1$ interaction. While the fusion/fission coupling remains below 0.1 meV for the long-distance dimers, it increases to the order of 1 meV in the herringbone dimer and explodes for the stacked dimer (127 meV). This huge enhancement in the b-axis mostly emerges from breaking the symmetry of the molecular dimers responsible for the vanishing coupling between $^1\mathtt{TT}$ and $\mathtt{CT}$ configurations in the frozen crystal. These results unequivocally identify thermal fluctuations as being critical for TF and SF photophysical processes (Eq. (2.16)) in crystal rubrene.

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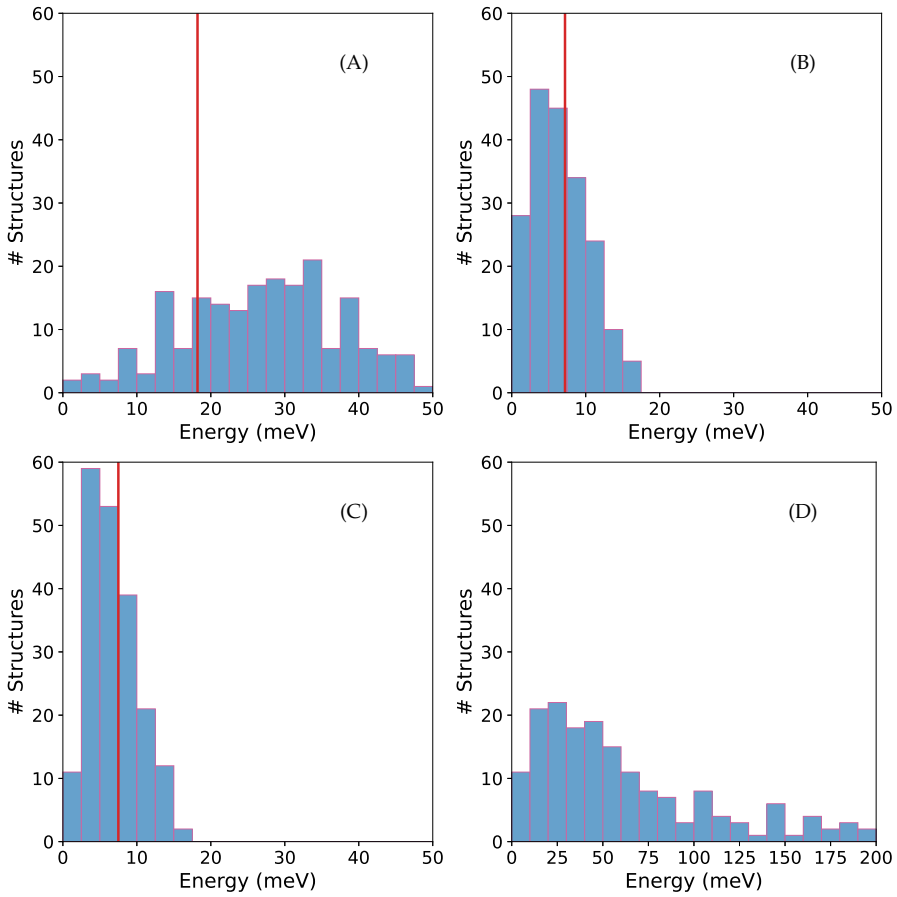


FIGURE 4.3: Distribution of absolute electronic couplings (in meV) in π-stacked dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. Vertical red lines in (A)-(C) indicate values obtained at 0 K.

Besides the average and median values, it is important to realize the distribution of couplings explored by thermal fluctuations in order to recognize the range of achievable values. Figure 4.3 shows the set of couplings related to the different studied processes in π-stacked dimers. Distributions for other first-neighbor dimers can be found in the Appendix B.3.

SEET couplings are quite symmetrically distributed around the average value, with a small number of arrangements with nearly zero interaction and a few structures reaching values as large as 50 meV. The couplings controlling triplet exciton diffusion present a different distribution, with a rather large density of structures below the average value and a decaying distribution tail of couplings larger than 8 meV. Most of the triplet-pair binding energies of stacked dimers appear within the 0-12 meV range, with few conformations exhibiting stronger triplet-triplet interactions. Finally, although the vast majority of structures exhibit vanishing $^1\mathrm{TT}/\mathrm{S}_1$ couplings, the distribution extends to very large interactions ($>$ 200 meV). This result indicates that thermally induced motions are very efficient to promote TF or SF in the b-direction of the crystal.

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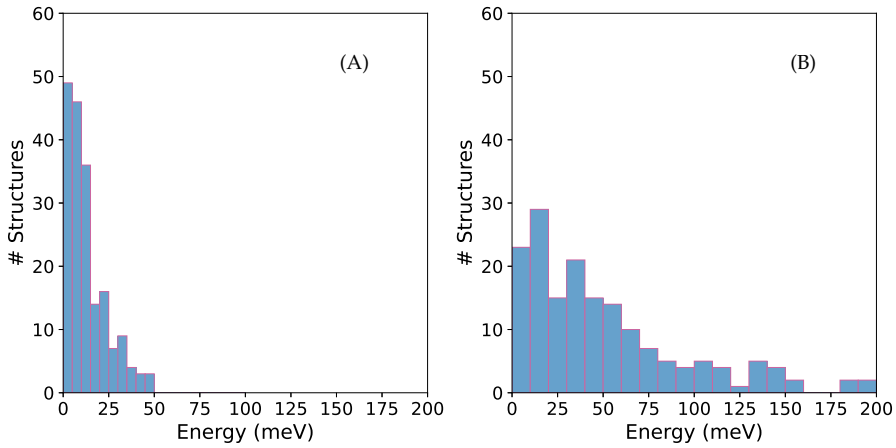


FIGURE 4.4: Distribution of absolute TF/SF couplings (in meV) in π-stacked dimers computed from 200 MD frameworks. (A) Direct coupling, (B) 2nd order (mediated) correction.

To decipher the underlying factors contributing to the notable enhancement in the interaction between S_1 and ^1TT states in stacked dimers, we conduct a detailed analysis of both first and second-order contributions (Figure 4.4). Examination of the distributions for direct and CT-mediated couplings distinctly reveals that the CT-mediated mechanism predominantly accounts for the large TF/SF couplings observed in numerous MD π-stacked dimers. While the direct contributions may not reach the same magnitude as the second-order terms, a substantial number of sampled structures exhibit considerable first-order interactions, with many exceeding 10 meV. This observation suggests that there are configurations in which the direct pathway plays a significant role.

A more comprehensive study has been conducted to evaluate the values of the TF/SF couplings in distorted structures. For this purpose, these energies have been calculated using the multistate-FED method at the RASCI level with hole and particle contributions. Previously, I argued that the hole and particle excitations had an important weight in the energy stabilization of the ^1TT state and calculated the triplet-triplet pair binding. However, no variations of the energies for the TF/SF couplings were found and therefore, these contributions were not included (Table 4.6) in addition to the high cost of the Boys(5) dibatization method with h and p configurations. The later development of the multistate-FED method allowed the calculation of the TF/SF coupling with hole and particle contributions without increasing the computational cost. The results are shown in the Table 4.7.

TABLE 4.7: $^1TT/S_1$ (TF/SF) electronic couplings (in meV) computed and averaged over 200 snapshots from MD with and without hole and particle excitations (h, p).

method	(h, p)	stacked	herringbone	a-dimer	ac-dimer
Boys(5)	no	126.7	1.1	0.0	0.0
Multi-FED(5)	yes	9.4	2.5	0.0	0.0

A significant decrease in the magnitude of the coupling can be observed when including hole and particle contributions in the stacked dimers, while the values of

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
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the herringbone dimers increase. The new configurations include excitations that stabilize the energy of the ^{1}TT state and influence the coupling value. Note that in the Eq. (2.9), the energies of the adiabatic states play an important role in the coupling value between them, so a good description of these states and their energies should, in principle, result in more reliable values. Furthermore, this difference is much more noticeable in the stacked dimers due to the orbital overlap and the strong interaction of the triplets, which require more configurations for a good description of the ^{1}TT state.

4.6 Structure - Coupling Analysis

To rationalize the results in terms of structure-coupling relationships, a principal component analysis (PCA) of selected structural parameters was performed to uncover correlations between molecular geometries and electronic couplings. The structural distortions analysed with the thermal fluctuations were not strong enough to be relevant for this analysis. Therefore, a PCA of molecular orbital energies (HOMO-1, HOMO, LUMO and LUMO+1) was performed. This analysis did reveal relevant principal components that have an impact on the electronic couplings.

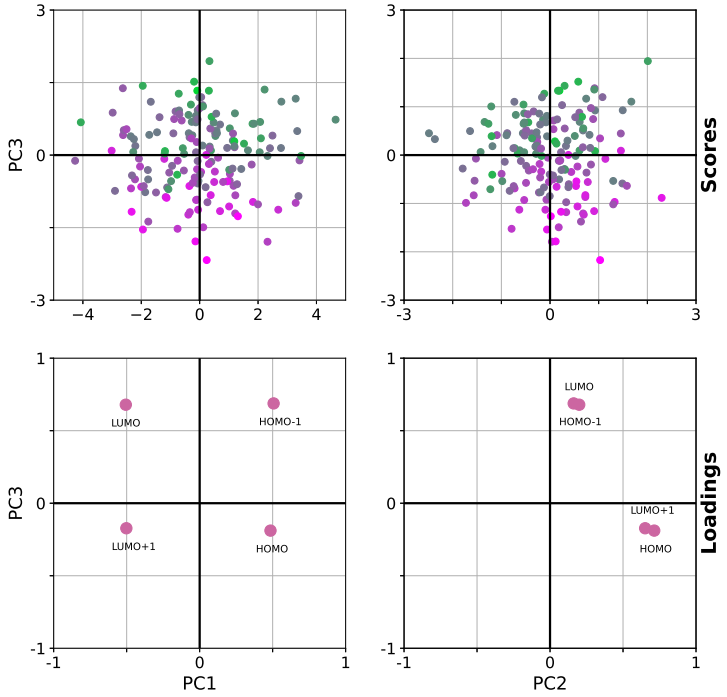


FIGURE 4.5: PCA analysis of SEET coupling (in meV) in stacked dimers computed from 200 MD frameworks.

$$\mathrm{PC1} = 0.51 \cdot E_{\mathtt{HOMO\text{-}1}} + 0.48 \cdot E_{\mathtt{HOMO}} - 0.50 \cdot E_{\mathtt{LUMO}} - 0.50 \cdot E_{\mathtt{LUMO+1}} \tag{4.2}$$

$$\mathrm{PC2} = 0.16 \cdot E_{\mathtt{HOMO\text{-}1}} + 0.71 \cdot E_{\mathtt{HOMO}} + 0.20 \cdot E_{\mathtt{LUMO}} + 0.65 \cdot E_{\mathtt{LUMO+1}} \tag{4.3}$$

$$\mathrm{PC3} = 0.68 \cdot E_{\mathtt{HOMO\text{-}1}} - 0.19 \cdot E_{\mathtt{HOMO}} + 0.68 \cdot E_{\mathtt{LUMO}} - 0.17 \cdot E_{\mathtt{LUMO+1}} \tag{4.4}$$

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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The PCA shows that there is a molecular orbital energies to couplings relationship. In Figure 4.5 it can be observed that the energies of HOMO-1 and LUMO from one side and HOMO and LUMO+1 from the other side describe the principal component number 3 (PC3). This principal component explains the relationship between SEET values and molecular orbital energies. The PCA of SEET couplings for herringbone, *a*-dimer and *ac*-dimer can be found in Section B.4, while those related to TEET are described in Appendix B.5.

4.7 Conclusions

The study of electronic couplings reveals that TEET couplings in crystal rubrene is notably more efficient along the *b*-axis, in agreement with experimental exciton diffusion measurements. Similarly, triplet-triplet binding energy is significant for π-stacked molecules due to the interaction of π-orbitals of neighboring molecules, while vanishing in the other directions. At 0 K TF/SF couplings diminish as the direct coupling between ^{1}TT and S_1 is extremely weak and the interaction of the triplet-pair with CT states is symmetry forbidden. Temperature induced symmetry-breaking distortions activate TF/SF processes predominantly through the CT-mediated mechanism. These results clearly evidence the crucial role of thermal fluctuations in comprehending the photophysics and exciton transport in crystal rubrene.

In summary, I have described and applied a set of quite sophisticated but computationally affordable electronic structure methods for the evaluation of electronic couplings in TTA-UC in molecular materials. The strategy provides a unified multielectron approach to account for the complexity of the studied states beyond the one-electron picture. The methodology can be used to identify potential mechanisms by quantifying the role of different diabatic states, such as CT configurations and multiexcitonic states. The low computational cost of the designed protocol allows for the fast evaluation of different couplings for a large number of molecular arrangements, as those required in the study of thermal fluctuations. Despite that here I direct our investigation to TTA-UC in rubrene single crystal, I believe that the validity of these approaches can be extrapolated to a large variety of organic molecular crystals, and that it represents a feasible and yet accurate approach to quantify electronic couplings.

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Chapter 5

Exciton Dynamics in Crystal and Amorphous Rubrene

"Hope is what makes us strong. It is why we are here. It's what we fight with when all else is lost."

Pandora - God of War III

The first part of this chapter addresses the structural characterization of amorphous rubrene and investigates the electronic couplings relevant to exciton dynamics in this disordered phase. To this end, I combine the methodologies previously applied to evaluate couplings in crystalline rubrene (Chapter 4) with the optimization of ML models aimed at reducing computational cost. In the second part, I present exciton dynamics simulations performed for crystalline rubrene, using couplings computed both at 0 K and including thermal vibrations, as well as for amorphous rubrene, where couplings were obtained from amorphous molecular pairs.

5.1 Morphology of Amorphous Rubrene

The lack of translational symmetry in the amorphous phase gives rise to an infinite variety of non-equivalent molecular sites and intermolecular interactions. Consequently, there is no clear distinction of first-neighbor molecules, which means that alternative methods are required to determine the first neighbors' distance. Therefore, to stablish an average distance for identifying first-neighbor molecules, I calculated the radial distribution function (RDF) in the simulated amorphous

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morphology (see Section 2.6 for details), which characterizes how the molecular density varies with distance.

The RDF calculated for the MD-generated disordered rubrene assembly (Figure 5.1) exhibits a distinct peak at around 10 Å, which can be interpreted as the average separation between a molecular site and its nearest neighbors. On the other hand, the the local minimum at shortest distance of the RDF appears at 15 Å (red vertical line in Figure 5.1) can be taken as the limit of the first-neighbor shell. Accordingly, all intermolecular interactions within this radius (15 Å) are considered in the simulations, under the assumption that molecules beyond this sphere vanish or are effectively screened by closer neighbors.

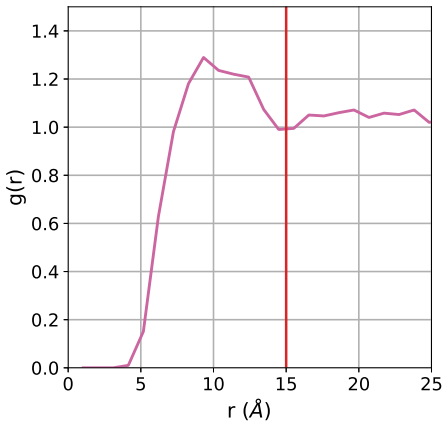


FIGURE 5.1: Radial distribution function ($g(r)$) as a function of intermolecular distance (r, in Å) obtained for the simulated amorphous rubrene solid. The red line represents the distance of the first local minimum.

5.2 Electronic Couplings in Amorphous Rubrene

From the RDF analysis, it is clear that electronic couplings in amorphous rubrene should generally be smaller than in the thermally distorted crystal, due to the larger average intermolecular distances. Considering the magnitudes of couplings obtained for the crystalline solid, it can be anticipated that SEET couplings will remain on the order of ~ 20 meV, while TEET, J, and TF/SF couplings will approach zero because the increased separation strongly suppresses orbital overlap. Nevertheless, at distances around 10 Å, TF/SF couplings may not vanish entirely, particularly as molecular distortions could enhance indirect contributions to these couplings, as observed in thermally distorted crystal configurations.

In Chapter 4, a protocol was proposed to calculate the intermolecular couplings governing singlet and triplet transport, triplet-triplet pair binding, and TF/SF conversion. This protocol involves the calculation of SEET and TEET couplings with FED and FSD with rCAM-B3LYP, and the interaction between triplet pairs with RASCI with hole and particle contributions. In thermally activated structures, TF/SF coupling was calculated with RASCI and Boys diabatization (without h, p). However, it is difficult to calculate TF/SF couplings for amorphous structures. Due to molecular distortions, C-DFT fails to separate charges in most of the selected pairs, so in this case it is not a reliable approach to evaluate CT state energies. To solve this

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problem, the CT energies are approximated with the function in Eq (5.1),

$$E_{\mathrm{CT}}(r) = \frac{s}{r} + E_{\mathrm{EA}} + E_{\mathrm{IP}} + i \tag{5.1}$$

where E_{EA} is the electron affinity of rubrene, E_{IP} is the ionization potential of rubrene, and r is the distance between the centers of mass of two rubrene molecules. s and i are parameters that have been fitted to the energies calculated with C-DFT of stacked dimers as a function of distance (Figure 5.2).

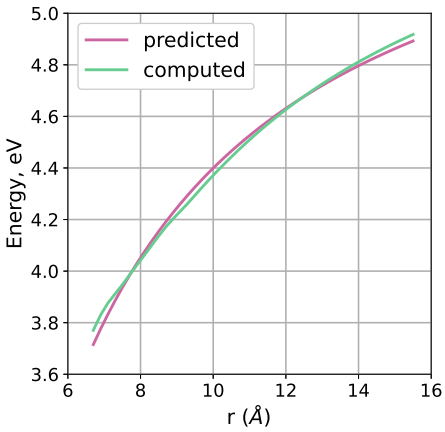


FIGURE 5.2: Caption

In the characterization of crystal dimers of Chapter 4, Boys(5) diabatization yields five localized diabatic states per dimer: two with LE character, two with CT character, and one state delocalized over both molecules. However, for distorted dimers in the amorphous structure, similar to the challenges encountered with C-DFT, well-localized states are not consistently obtained. Therefore, for amorphous dimers, diabatic states are constructed using the multistate FED method.

From the bulk rubrene amorphous, around 9400 unique pairs of rubrene molecules are extracted. For each rubrene dimer, I computed all the electronic couplings potentially involved in exciton dynamics in this system. The results are then analysed and the outliers are removed statistically according the the interquartile range (IQR). This statistical method removes samples from the data that are outside the range determined by the quartiles. The distributions of the electronic couplings can be found in Figure 5.3.

The results of the statistically relevant data show that the SEET coupling value averages to 13.4 meV, indicating that the singlet states are capable of moving efficiently in any direction of the amorphous solid. The decrease in this value with respect to the crystal can be associated with the distortion of the molecules and their orientation. That is, the orientation between the transition dipole moments of the dimers of the amorphous solid can influence the magnitude of this coupling.

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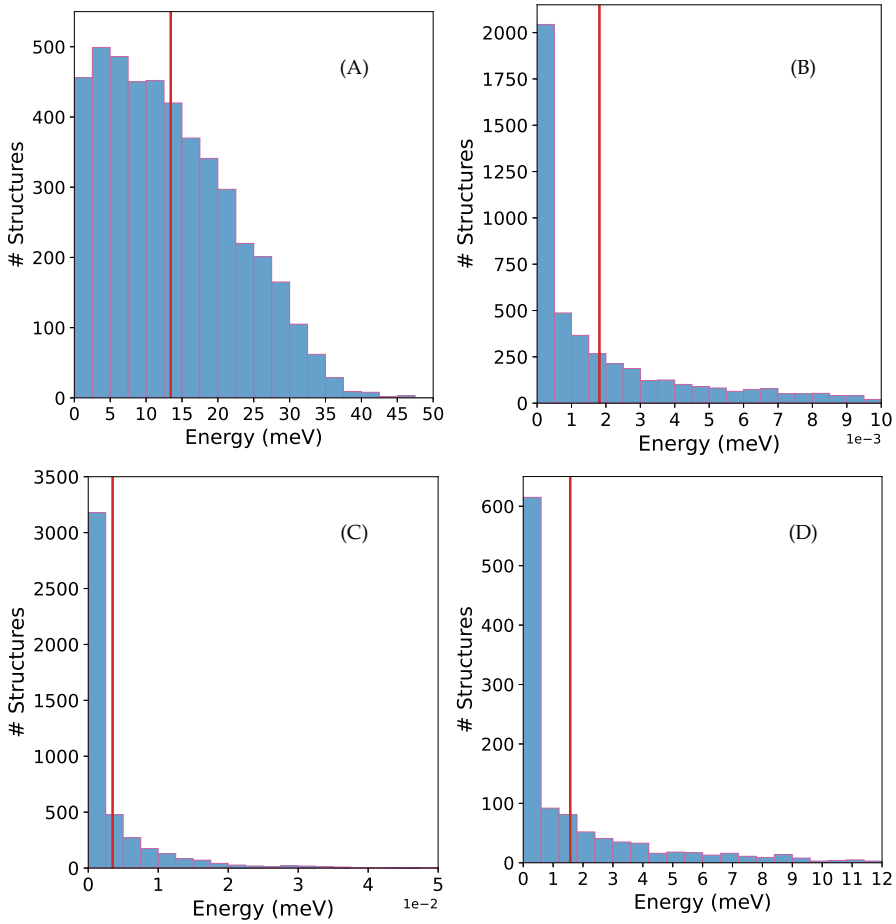


FIGURE 5.3: Distribution of absolute electronic couplings (in meV) in amorphous dimers computed from 9400 MD frameworks. (A) SEET, (B) TEET, (C) J, and (D) TF/SF. Vertical red lines indicate mean values.

On the other hand, the TEET coupling and the triplet-triplet pair binding energy result in values below 0.1 meV. Therefore, unlike the crystal and the thermally activated structures, the triplet states transfer process is strongly hindered in this amorphous rubrene solid. Likewise, there is no stabilization of the spin singlet TT state, ($J \approx 0$). This is because the intermolecular separation between the rubrene molecules in the amorphous solid and the symmetry breaking in the tetracene backbones decreases the orbital overlap. As previously stated, processes involving triplet states require a certain orbital overlap, and their interaction decreases exponentially with distance (Eq (4.1)). The same phenomenon had been previously observed in thermally activated structures.

On the other hand, TF/SF couplings exhibit an average value greater than 1 meV. This phenomenon has also been observed in thermally activated structures, where TF/SF coupling reaches values greater than TEET and triplet-triplet binding. Furthermore, the distribution of this coupling presents structures with values even greater than 10 meV, allowing TF/SF coupling in the amorphous solid.

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The distributions of the direct and indirect contributions of the TF/SF electronic couplings in the amorphous dimers can be found in Appendix C.2.

5.3 Training of ML Models

Based on the theoretical background of the ML models presented in Chapter 3, in this section, I focus on presenting the results of their training. In this context, they enable the efficient prediction of specific inter-site couplings relevan to the photophysical processes explored during exciton time-evolution simulations. These predictions are based on the coupling computed and analyzed in the previous section for a dataset of 9400 rubrene molecule pairs.

In the simulation of exciton transport and conversion processes in the crystal, both the calculated couplings at 0 K and the couplings from the thermally distorted structures are reduced to four pairs. The ML model that can easily distinguish four situations (also called classes) is the DT. The hierarchy of DT models optimized for estimating couplings consists of at most three operations that evaluate specific indices of the input vectors (Figure 5.4). All optimized DT models converge to the same scheme, where only the values returned by the model vary, that is, the couplings of each pair. This is because the DT model looks for three operations to distinguish between four pairs of rubrene molecules that can be easily distinguished by the distance between their centers of mass.

```
{
    "lh": "2",                                  } F(2) ≤ 156.8901
    "condition": "le",                          }
    "rh": "156.89012909",                       }
    "true": 0.0098,    ⟶ ac-dimer
    "false": {
        "lh": "13",                             } F(13) ≤ 63.9815
        "condition": "le",                      }
        "rh": "63.98154068",                    }
        "true": {
            "lh": "89",                         }
            "condition": "le",                  } F(89) ≤ −0.3173
            "rh": "-0.31733672",                }
            "true": 0.0306,    ⟶ herringbone
            "false": 0.0182    ⟶ stacked
        },
        "false": 0.0178    ⟶ a-dimer
    }
}
```

FIGURE 5.4: DT input file (JSON format) for crystal SEET electronic couplings (in eV). $F(i)$ refers to the ith index of the input feature vector.

In this specific case, the DT model proposed for simulating excitons in crystalline solids could be replaced by a SEDyn function that allows defining couplings based on the distance between the centers of mass (proposal in Algorithm 2). This algorithm could speed up the simulation, since it avoids having to build a CM that has to be diagonalized to obtain the input vector. However, in the following example, this approach fails to return the correct coupling. For a crystal formed with glucolaldehyde (gly) and acetic acid (acet), where the intermolecular separation is constant, this algorithm would not be able to differentiate a gly-acet interaction from

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a gly-gly interaction, because the distance between the centers of mass would not be a distinctive data. This makes the DT model a generalization of the intermolecular distance, since the CM not only contains information about the distances between atoms, but also the identity of the atoms that compose the molecules (Section 3.1.3). Therefore, distance and atomic composition complement each other, enabling the model to distinguish structural arrangements with greater accuracy.

Algorithm 2 The distance to coupling algorithm

```
function DistanceCoupling(M_A, M_B, r, c)        ▷ M_n: Molecules
    n ← dim(r)                                   ▷ r: vector of distances
    d ← Distance(M_A, M_B)                       ▷ c: vector of couplings
    for i ← 0, n do
        if d ≤ r(i) then
            return c(i)
        end if
    end for
    Throw Error
end function
```

The DT model is a good tool for systems in which interactions can be grouped into clusters or classes. However, in systems where there is no pattern for grouping interactions, this model fails, and we must resort to ML models that can estimate new values from those learned. These can be regression models, such as KRR or prediction models (FNN) that are trained with known data to estimate or predict values using mathematical operations.

In the amorphous rubrene solid, it is not possible to group the dimers into clusters since there is no ordering. Furthermore, any attempt to relate the distance between the centers of mass with the electronic couplings would fail since in the amorphous solid, most of the dimers that form the first neighbors are separated by approximately 10 Å, that is, there are no significant differences in the intermolecular distances that group numbers into clusters so that the DT could be used to predict these couplings. This has already been explained previously with the RDF. Therefore, the simulation of exciton dynamics in the amorphous system requires predictive models. In preliminary attempts to optimize FNN models, I have not been able to get a network capable of predicting the SEET coupling, because the amount of data available to optimize this type of model is scarce. The depth and complexity of the FNN model grow with the number of nodes needed to read the input vector. For couplings between dimers of rubrene, the input vector would contain 140 values, and although some are almost zero, they can influence the values that the model predicts. The number of input values can be reduced by operations that preprocess the input vector to decrease the dimension of the network. However, I have not found any operation that, by reducing the input vector, maintains the relevant information that links the structure to the coupling. An alternative strategy to address this problem is to increase the number of samples used for training the network. However, this approach is impractical considering the computational cost, as each calculation can take up to 7 hours. Considering that in the previous section I used 9400 pairs for the statistical analysis of couplings, further increasing the number of pairs to train a more complex FNN model may be less efficient than directly calculating the couplings for each molecular pair at runtime. After all, if data generation and model training require more time than the on-the-fly calculations for

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the pairs involved in the exciton processes, the use of ML models no longer serves its purpose.

KRR offers an alternative to the FNN model, relying on kernel-based linear regression. This technique is more versatile than the FNN model since it only requires selecting a kernel that transports the non-linear data to a space in which they are linear, and in this space, performs a linear regression. Likewise, as explained in Section 3.2.2 and considering the Gaussian kernel, KRR only requires optimizing two hyperparameters. This is a big difference with respect to the FNN model, since in this one, it is necessary to optimize from the total number of layers in the net and how many nodes each layer contains, to what activation functions are applied in each layer, as well as the possibility of adding a function that preprocesses the data. The large number of hyperparameters affects the effectiveness of the model so much that, in many cases, it is usually a matter of trial and error until the appropriate combination is found, as long as there are enough samples for the number of features they have.

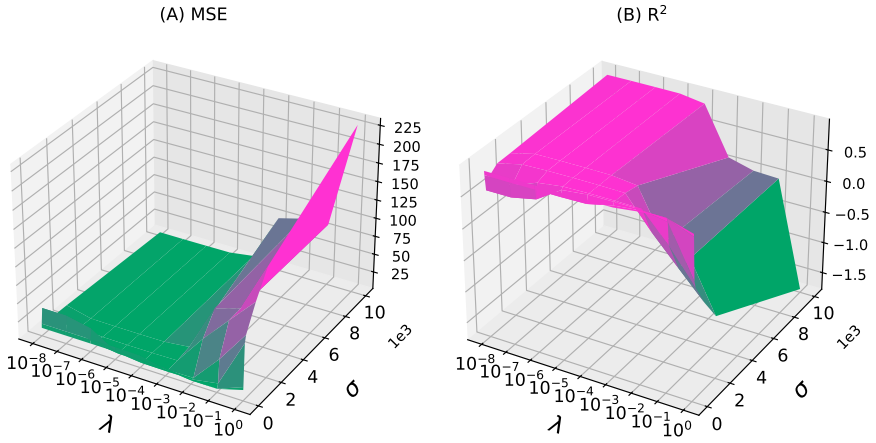


FIGURE 5.5: Surfaces of MSE (A) and R^2 (B) of the discrete exploration of regularization strength (λ) and the width of the Gaussian (σ). Note that the MSE has been calculated in meV.

The hyperparameters of the KRR model for SEET coupling have been optimized through a discrete exploration. The Mean Squared Error (MSE) and the R^2 of the regression between the predicted and test values are used to evaluate the model's accuracy. The surface of the results obtained in the exploration can be seen in Figure 5.5, and the table with the surface values is found in Appendix C.1. The combination of hyperparameters with the highest R^2 and lowest MSE has resulted in $\lambda = 10^{-2.8}$ and $\sigma = 100$ for the regularization strength and the width of the Gaussian, respectively. Even so, in the figure, we can observe that there are other combinations of parameters that result in a high R^2 and low MSE that could be candidate models for the simulation. The prediction of the KRR model with the optimized values versus the test values is shown in Figure 5.6. These results indicate that the model is able to predict with high accuracy (R^2 = 0.94 and MSE = 6.18 calculated in meV) the SEET couplings for structures outside the training set.

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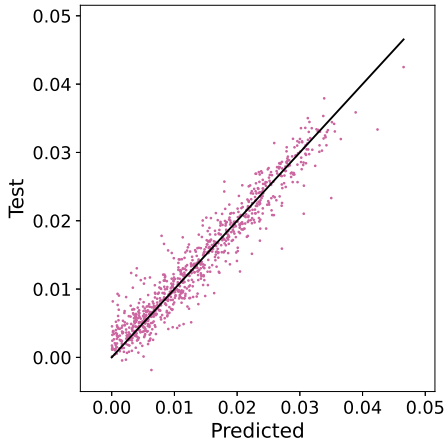


FIGURE 5.6: Predicted vs Test SEET couplings for amorphous rubrene dimers with optimized KRR ($\lambda = 10^{-2.8}$, $\sigma = 100$) model.

The same process has been attempted with TEET, triplet-triplet pair binding, and TF/SF couplings to obtain different KRR models. However, the evaluation of these models always results in R^2 values below 0.4 and MSE values above 50. A more exhaustive exploration of the hyperparameters reveals no signs of improvement. From these results, I deduce that the Gaussian kernel is not able to transport the data to a space in which they are linear or that the CMs do not contain information relating the structures to these couplings. In fact, as previously mentioned, couplings involving triplet states cannot be directly related by CMs, because the interaction of these states is strongly related to the orbital overlap and decreases exponentially with distance. Therefore, in the triplet exciton dynamics simulations, the couplings are treated as constant values for any molecular pair, independent of distance and CM shape. This approximation is justified in the amorphous system, since the first-neighbor intermolecular distances are typically within 10 Å, making it reasonable to assume nearly uniform couplings. Moreover, both TEET and the triplet-triplet pair binding couplings are essentially negligible, while only the TF/SF coupling reaches values about 0.1 meV. Therefore, it is appropriate to approximate the couplings as constant for any rubrene dimer.

5.4 Simulations of Exciton Dynamics

A central objective of this project is the simulation of exciton dynamics in rubrene molecular solids, with particular attention to the processes involved in TTA-UC. In this section, I present the most relevant results of these simulations performed in both the crystalline and amorphous solids. First, I define the simulation conditions and technical details, such as temperature or the set of couplings considered for the study of exciton dynamics in crystalline and amorphous rubrene. Next, I analyze the independent diffusion of singlet and triplet excitons under different conditions. Finally, I explore exciton dynamics in the thermally activated rubrene crystal, accounting for the possibility of SF and TF processes.

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5.4.1 Simulation conditions

All simulations were performed by running 10k steps per trajectory over a total of 10 independent trajectories. PBC were applied in both the crystal and the amorphous systems to prevent excitons from being artificially trapped at the edges of the simulation box and to better approximate the behavior of real molecular solids. The main structural characteristics of the crystalline and amorphous solids are summarized in Table 5.1. The initial number of excitons in the transport simulations (SEET and TEET) and in the TTA-UC process simulations is set to 96, randomly distributed throughout the simulation box. This corresponds to an initial density of approximately one exciton per 2592 molecules, a proportion consistent with absorbed photon densities reported for rubrene crystals in experimental investigations [118, 119].

Rate constants for the processes included in the KMC scheme implemented in SEDyn were obtained from Ferrmi's Golden Rule within the Marcus framework. Singlet and triplet excitation energies were fixed at their experimental values, 2.23 and 1.14 eV, respectively [61, 120]. Reorganization energies of singlet at triplet states were set to 0.1 eV, a standard value for organic chromophores [121, 122]. Electronic couplings used for the time evolution of excitons in the crystalline solid are those discussed in Chapter 4 (Table 4.3). Couplings for the amorphous system were obtained following the procedures described in Sections 5.2 and 5.3. Simulations were carried out at two temperatures, 100 K and 300 K, in order to assess how temperature-induced changes in the processes' rates influence exciton transport and related processes.

TABLE 5.1: Characteristic parameters of crystalline and amorphous solids for the simulations.

	crystal	amorphous
simulation box, nm^3	$1.6192 \cdot 10^5$	$2.0994 \cdot 10^5$
# molecules	237276	256000
cutoff distance, Å	20.09, 8.96, 10.66	15.00 (spherical)

5.4.2 Diffusion of singlet and triplet excitons

Table 5.2 summarizes results of the singlet exciton transport simulations, reporting the diffusion coefficients obtained along the three crystal axes and spatial directions in the amorphous system, as well as their average value. In all cases, the simulation temperature has a large impact on the diffusion coefficients, with a systematic increase of approximately three orders of magnitude between simulations at 100 K and 300 K, as a direct consequence of the temperature dependence in the Marcus model (Eq. (2.3)).

In the crystal, the results show that singlet excitons diffuse most efficiently along the *a* axis, with diffusion coefficients up to one order of magnitude larger than along the other two axes. This anisotropy arises not only from the electronic couplings associated with the *a*- and *ac*-dimers, but also from the influence of intermolecular distance on transport along this direction. Notably, the electronic coupling of the *a*-dimer (17.8 meV) is comparable to that of the stacked dimer (18.2 meV). Therefore, the significantly higher diffusion constant observed along the *a* axis can be attributed to the larger intermolecular separation in the *a*-dimer, which is approximately twice that of the stacked dimer.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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Comparison of the diffusion constants (D) and the directional components ($D_{a,b,c}$) between simulations using couplings calculated at 0 K and those including thermal vibrations reveals a direct correlation between coupling variations and the resulting transport properties. In the case of SEET, structural distortions lead to an overall enhancement of the couplings, which in turn translates into a 58% increase in the diffusion constant.

The results of SEET in the amorphous rubrene assembly show that, in general, the diffusion constant, D, and the directional diffusion constants decrease with respect to those in the crystal. This can be justified by the average reduction of electronic couplings between molecular sites due to the increase of the intermolecular distances in the amorphous solid with respect to the crystalline solid. In the crystalline solid, the intermolecular distance of the a dimer is 13.86 Å, while in the amorphous solid, in any direction, the most frequent first neighbor distance is 10 Å. Furthermore, the average SEET couplings obtained in the amorphous solid is $\sim$4 meV lower than in the crystalline solid ($\sim$3 meV for the couplings in the crystal with thermal vibrations), which results in lower rate constants. All this justifies the decrease of the diffusion constant. Even so, the diffusion constants along the b and c axes are greater than in the crystal, mainly due to a considerable increase in the intermolecular distances in those directions.

TABLE 5.2: Diffusion coefficients (D) in cm^2/s of the simulations of five singlet excitons of the crystal, thermal average and amorphous SEET couplings at 100 K and 300 K. $D_{a,b,c}$ represent the directional diffusion coefficient over a (x), b (y) and c (z) axes of the crystal (amorphous).

	Temperature	D	D_a	D_b	D_c
Crystal	100 K	$1.27 \cdot 10^{-4}$	$2.64 \cdot 10^{-4}$	$2.57 \cdot 10^{-5}$	$8.96 \cdot 10^{-4}$
	300 K	$1.49 \cdot 10^{-1}$	$3.67 \cdot 10^{-1}$	$1.89 \cdot 10^{-2}$	$6.10 \cdot 10^{-2}$
Thermal	100 K	$3.02 \cdot 10^{-4}$	$7.21 \cdot 10^{-4}$	$4.37 \cdot 10^{-5}$	$1.43 \cdot 10^{-4}$
	300 K	$2.79 \cdot 10^{-1}$	$7.12 \cdot 10^{-1}$	$2.77 \cdot 10^{-2}$	$9.85 \cdot 10^{-2}$
Amorphous	100 K	$8.56 \cdot 10^{-5}$	$8.56 \cdot 10^{-5}$	$1.19 \cdot 10^{-4}$	$5.20 \cdot 10^{-4}$
	300 K	$6.76 \cdot 10^{-2}$	$5.82 \cdot 10^{-2}$	$1.08 \cdot 10^{-1}$	$3.71 \cdot 10^{-2}$

The diffusion coefficients for triplet exciton transport are summarized in Table 5.3. The simulations in crystalline rubrene follow the same trend as the electronic couplings discussed in Chapter 4, yielding triplet diffusion constants that are approximately five orders of magnitude smaller than those obtained for singlets. Furthermore, triplet transport is restricted to the b- and c-directions of the crystal, consistent with the vanishing couplings along the a-axis (a- and ac-dimers). This anisotropy is in line with experimental findings [103].

Triplet exciton dynamics with couplings obtained from thermally averaged distortions show a slight reduction of the diffusion constants in the bc-plane, reflecting the decrease of TEET couplings caused by structural distortions. Conversely, as observed for singlet transport, increasing the simulation temperature from 100 K to 300 K produces a systematic enhancement of diffusion by nearly three orders of magnitude. In contrast, triplet transport in the amorphous solid is effectively suppressed, since TEET couplings remain close to zero. For this reason, no triplet transport simulations were carried out in the amorphous phase.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

TABLE 5.3: Diffusion coefficients (D) in cm^2/s of the simulations of ten triplet excitons of the crystal and thermal average TEET couplings at 100 K and 300 K. $D_{a,b,c}$ represent the directional diffusion coefficient over a, b and c axes of the crystal.

	Temperature	D	D_a	D_b	D_c
Crystal	100 K	$5.30 \cdot 10^{-9}$	0.0	$7.36 \cdot 10^{-9}$	$8.52 \cdot 10^{-9}$
	300 K	$2.55 \cdot 10^{-6}$	0.0	$6.89 \cdot 10^{-6}$	$7.75 \cdot 10^{-7}$
Thermal	100 K	$4.34 \cdot 10^{-9}$	0.0	$5.32 \cdot 10^{-9}$	$7.71 \cdot 10^{-9}$
	300 K	$1.59 \cdot 10^{-6}$	0.0	$4.51 \cdot 10^{-6}$	$2.50 \cdot 10^{-7}$

5.4.3 Exciton dynamics beyond diffusion

All processes related to triplet conversion in the crystalline solid have been simulated with the couplings calculated in the thermally activated structures. The crystalline solid with the couplings calculated at 0 K and the amorphous solid are discarded due to the zero value of the TF/SF couplings. In Figure 5.7, the evolution of the number of excitons in the solid throughout the simulation can be observed, starting from 20 triplet excitons and 10 singlet excitons. Additional simulations can be found in Appendix C.3. Note that the x-axis is not related to the real time of the processes since each exciton contain its own temporal evolution, as explained in Chapter 3. From this visual representation, it can be observed that all the processes involved in TF/SF (Eqs. (2.15) and (2.16)) occur in the simulation since the number of singlet excitons decreases, while the number of triplet excitons increases. Similarly, during the simulation the triplet excitons interact and they are converted to singlets.

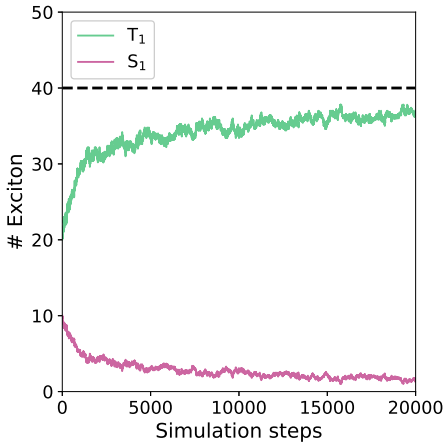


FIGURE 5.7: Simulation of 10 trajectories of TF/SF starting with 10 S_1 and 20 T_1 excitons. The exciton numbers represent the qualitative exciton density evolution.

However, a trend can be observed that reduces the number of singlets in the system by converting them to triplets; that is, in the crystalline solid, the SF process is favoured over the TF process. These results are in agreement with experimental studies that indicate that the SF process occurs in the crystalline solid and that it can be suppressed by transforming the crystalline solid into an amorphous one [33, 123]. Experimentally, it has been shown that the SF process does not occur in amorphous solids [33]. In those amorphous solids that have been observed, it has been due

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

to co-crystallinity regions that maintain shorter intermolecular distances and larger orbital overlap that enable this process [124]. Theoretically, these experimental results can be rationalised by arguing that in the crystalline solid, SF is favored because the transformation $S_1 \rightarrow {}^1TT \rightarrow T_1 + T_1$ is spin-allowed, while the interaction of two triplets statistically only results in singlet spin once in nine times. On the other hand, in an amorphous solid, I have shown that TF/SF processes are allowed despite the intermolecular distance. However, this intermolecular distance prohibits triplet transport in the solid and therefore, the TF process is not observed because the triplets do not interact, and the SF process is blocked because the ${}^1TT$ state cannot dissociate into triplets due to the zero value of the TEET coupling.

5.5 Conclusions

In summary, the analysis of the amorphous rubrene solid reveals that molecules are spherically distributed at an average distance of 10 Å, which implies that singlet transport is allowed in all directions. SEET coupling is reduced due to molecular distortions in the amorphous solid and the relative orientations of the transition dipole moments. Furthermore, the intermolecular distance of 10 Å severely hinders triplet transport, as orbital overlap rapidly decreases with distance, resulting in vanishing TEET couplings. Similarly, the triplet-pair binding energy (J) tends to zero in rubrene dimers extracted from the amorphous assembly. However, the SF/TF coupling is greater than 1 meV because the contributions of the CT states allow these processes to occur at large distances. This concludes that the SF/TF process can theoretically occur in the amorphous rubrene solid, but that the zero value of the TEET coupling prevents both triplet dissociation and singlet formation through triplet interactions.

On the other hand, I have shown that DT models are useful tools for parameterizing the couplings of processes occurring in crystalline solids. However, in the amorphous solid, since there is no ordering, predictive models must be used. I have shown that the KRR model is an effective tool for predicting SEET couplings because there is a direct structure-coupling relationship, whereas the model fails for couplings that depend on orbital overlap.

Finally, simulations of exciton dynamics in crystalline rubrene considering couplings from thermal vibrations predict the coexistence of SF and TF processes, in agreement with experimental observations. However, SF emerges as the dominant mechanism, driving the system toward triplet generation. This predominance arises because all TT states, formed by the interaction of two triplets or an S_1 state, can dissociate into two separate triplets when the TEET coupling is sufficiently strong, whereas only the singlet-coupled ${}^1TT$ states can recombine to form S_1, and not all triplet pairs possess this spin configuration.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Chapter 6

Final Remarks

"In the end, we only regret the chances we didn't take."

Taric - League of Legends

In this thesis, I have developed and applied a comprehensive computational framework to investigate the processes of singlet transport, triplet transport, and SF/TF in rubrene solids. In particular, I have analyzed the SEET and TEET couplings in the orthorhombic crystalline form of rubrene by examining the dimers that constitute the unit cell. The computed coupling values were correlated with the structural, physical, and electronic properties of the dimers, providing insight into the mechanisms governing exciton transport. Furthermore, I have studied SF/TF electronic couplings in the crystalline lattice and evaluated their temperature dependence, thereby elucidating how thermal vibrations influence exciton dynamics. To enable these analyses, I also developed a computational protocol for calculating the relevant couplings in rubrene dimers using the RASCI method.

This protocol was subsequently extended to study SF and TF processes in amorphous rubrene, where the structural disorder gives rise to a wide distribution of coupling strengths. Owing to this complexity, a statistical analysis was carried out to obtain average coupling values and to compare the computational results with available experimental data.

A significant part of this work has been devoted to the design and implementation of a software package, SEDyn, for simulating exciton dynamics in molecular solids. The development of SEDyn allowed the adaptation of KMC algorithms traditionally used for crystalline materials to more complex and structurally diverse systems, including amorphous solids. Moreover, the program

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

was equiped with ML capabilities to predict electronic couplings on the fly during the simulation, thereby improving computational efficiency and scalability.

Using SEDyn, I performed simulations of singlet and triplet exciton transport in both crystalline and amorphous rubrene solids. Diffusion constants were computed to quantify transport efficiency under different conditions, including couplings evaluated at 0 K and those incorporating temperature effects. Finally, simulations of exciton dynamics in crystalline rubrene using thermally modulated couplings revealed the coexistence of SF and TF processes, with singlet fission emerging as the dominant pathway, consistent with experimental observations.

The results of the study of the crystalline solid reveal that singlet transport is permitted in all directions, while TEET coupling is most efficient along the *b*-axis. These results are in agreement with experimental measurements. Furthermore, I have shown that SF/TF coupling is zero in the crystalline solid, while temperature-induced distortions activate these processes predominantly through the CT mechanism. These thermal distortions are key to understanding the photophysical processes and exciton transport in the rubrene crystal.

On the other hand, the study of the amorphous rubrene solid concludes that the singlet transport process is permitted in all directions, while triplet transport is prohibited because the intermolecular distance is approximately 10 Å, thereby decreasing the orbital overlap. In these distorted structures, the SF/TF coupling is sufficiently large for this process to occur. However, the prohibited triplet transport prevents the interaction between them through TF, such as the splitting of the TT state formed from a singlet. Therefore, I conclude that the limiting process for the amorphous solid to form SF/TF is triplet transport. All these results are consistent with experimental observations, which highlight that only amorphous solids with a certain pseudocrystallinity can give rise to SF and TF processes, since they are able to hold efficient triplet transport.

The simulations of exciton dynamics in crystalline rubrene reveal that both SF and TF processes occur when the thermal effects are taken into account. Furthermore, the results confirm experimental findings showing that crystalline rubrene efficiently generates and collects triplets by SF, while the TF process, although energetically allowed, place a secondary role.

SEDyn is a computational tool designed to simulate photophysical processes and exciton transport in molecular solids, leveraging ML models. This new technology opens the door to realistic simulations of amorphous materials, the rational design of new materials, and the development of a theoretical complement to experimental research. The software will continue to be developed, implementing new ML models, parallelizing trajectories using MPI, and even proposing the design of an algorithm to parallelize excitons on a GPU.

In future works, the protocol could be used to calculate SF/TF processes in other molecular solids, both crystalline and amorphous, and in systems with more than one compound. Furthermore, rubrene derivatives featuring different functional groups that yield crystalline solids with larger intermolecular distances could be explored, such as aliphatic *t*-butyl rubrene [31] and extend the results of this work.

I conclude this thesis with the satisfaction of having developed a tool that could be useful to the scientific community, and of having theoretically contrasted experimental observations on crystalline rubrene. In addition, the proposed protocol for calculating SF/TF couplings in distorted amorphous structures paves the way for exploring new materials that were previously too complex or computationally demanding for traditional approaches.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
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Appendix A

Supplementary Material for Chapter 3

A.1 Advanced Simulation

This example uses the same system as in Section 3.4. In this simulation a second state is declared and the internal conversion between both states is included. Besides, there is no initial number of the new state, which will allow the simulation to convert from the first state to the second and the user will be able to observe this transition.

```
! ******
! SYSTEM
! ******

$run
    max_step 10000
    trajectories 2
    cutoff 3.5, 3.5, 3.5
    boundaries 1, 1, 1
    temperature 298
$$

! ******
! SCHEMA
! ******

$schema
    system standard
    trajectory standard
$$

! ********
! ANALYSIS
! ********

$analysis
    diff_coeff S, T
    diff_coeff_subtraj_length 100
$$

! *********
! EXCITON S
! *********

$state
    label S
    initial_num 4
    size 1
```

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REGISTRO TELEMÁTICO
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30/10/2025 11:18
EHU2025E051403


```
    elec_func const 2.23
$$

$process
    label S
    type fermigoldenrule
    interaction_state gs
    target_states gs, S
    elec_func const 0.1
$$

$process
    label S
    type fermigoldenrule
    interaction_state gs
    target_states T, gs
    elec_func const 0.001
$$

$transition
    label S
    interaction_state gs
    elec_func const 0.1
$$

$transition
    label S
    interaction_state T
    elec_func const 0.1
$$

! *********
! EXCITON T
! *********

$state
    label T
    initial_num 0
    size 1
    elec_func const 1
$$

$process
    label T
    type fermigoldenrule
    interaction_state gs
    target_states gs, T
    elec_func const 0.5
$$

$transition
    label T
    interaction_state gs
    elec_func const 0.1
$$

$transition
    label T
    interaction_state S
    elec_func const 0.1
$$
```

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REGISTRO TELEMÁTICO
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30/10/2025 11:18
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```
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2
3                ========================================
4
5                             SEDyn (3.0.0)
6
7                ========================================
8
9                   ---Solid State Exciton Dynamics---
10
11
12 Authors : A. Diaz-Andres
13
14 SEDyn begins on Mon Aug 04, 15:35:57, 2025
15
16 Number of OpenMP threads: 7
17
18
19 Checking system consistency
20 ---------------------------
21 Number of molecules in the system: 1000
22 Number of states in the system: 2
23 Number of processes for state T: 1
24 Number of processes for state S: 2
25
26
27 Simulation general information
28 ------------------------------
29 Cell size: [27, 27, 27]
30 Cutoff radius: [3.5, 3.5, 3.5]
31 Periodic Boundary Conditions: [1, 1, 1]
32 Temperature: 298
33 Number of steps: 10000
34 Number of trajectories: 2
35
36
37 Neighbour linking
38 -----------------
39 Max number of pairs to evaluate: 499500
40 Number of neighbours to link: 5116
41 ... Neighbours writen to archive file
42
43 Linking elapsed time: 0h 0m 0s
44
45
46 Trajectory : 0
47 ------------
48 Number of states initialized: 4
49 Number of molecules occupied: 4
50
51     Step |   S |   T |
52  ------------------------
53        0 |   4 |   0 |
54     1000 |   4 |   0 |
55     2000 |   4 |   0 |
56     3000 |   4 |   0 |
57     4000 |   4 |   0 |
58     5000 |   4 |   0 |
59     6000 |   4 |   0 |
60     7000 |   4 |   0 |
61     8000 |   4 |   0 |
62     9000 |   4 |   0 |
```


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REGISTRO TELEMÁTICO
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30/10/2025 11:18
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```
 ------------------------

Trajectory elapsed time: 0h 2m 39s


Trajectory : 1
------------
Number of states initialized: 4
Number of molecules occupied: 4

    Step |   S |   T |
 ------------------------
       0 |   4 |   0 |
    1000 |   4 |   0 |
    2000 |   3 |   1 |
    3000 |   3 |   1 |
    4000 |   3 |   1 |
    5000 |   3 |   1 |
    6000 |   3 |   1 |
    7000 |   3 |   1 |
    8000 |   3 |   1 |
    9000 |   3 |   1 |
 ------------------------

Trajectory elapsed time: 0h 2m 36s


Analysis
--------

** State : S

Diffusion Coefficient ...  8.3607e+01 cm^2/s
  x:  9.5273e+01  y:  7.3088e+01  z:  8.2461e+01


** State : T

Diffusion Coefficient ...  6.5271e+03 cm^2/s
  x:  9.6915e+03  y:  4.6374e+03  z:  5.2525e+03



  SEDyn simulation time: 0h 5m 15s


               ----------------------------------------

                  Tranquility lies in human's heart

               ----------------------------------------
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Appendix B

Supplementary Material for Chapter 4

B.1 Symmetric RASCI Couplings

The couplings obtained considering S_0S_1 the target state reveals that the anion-S_1 interaction localized in the same center is repulsive (negative values) for the stacked and herringbone dimers. Moreover, the cation-S_1 interaction result in higher values which, in fact, are positive for the stacked dimer (Table B.1). On the other hand, for large intermolecular distances (*a*-dimer and *ac*-dimer) these values tend to zero. This comes in agreement with the decrease of orbital overlap at longer distances, thus, reducing the electronic couplings.

TABLE B.1: Electronic couplings (in meV) computed at the RASCI and C-DFT level/cc-pVDZ level through Boys diabatization, and CT energies, E(CT) = (E(AC) + E(CA))/2 (in eV), computed at the C-DFT level/cc-pVDZ level for the four crystalline rubrene dimers.

	stacked	herringbone	*a*-dimer	*ac*-dimer
$\langle {}^1\mathrm{TT}\vert H\vert \mathrm{S_0S_1}\rangle$	0.0	0.0	0.2	0.0
$\langle {}^1\mathrm{TT}\vert H\vert \mathrm{AC}\rangle$	0.0	0.0	2.4	0.6
$\langle {}^1\mathrm{TT}\vert H\vert \mathrm{CA}\rangle$	0.0	0.0	2.4	0.6
$\langle \mathrm{AC}\vert H\vert \mathrm{S_0S_1}\rangle$	90.7	-8.1	1.6	0.5
$\langle \mathrm{CA}\vert H\vert \mathrm{S_0S_1}\rangle$	-173.1	-25.7	2.2	0.4
E(CT)	2.65	3.03	3.87	2.71

B.2 Median Electronic Couplings

TABLE B.2: Electronic couplings (in meV) computed for the crystal structure (cryst.) and median of 200 snapshots from MD (TM: thermal median) for SEET, TEET, triplet-triplet binding (*J*), and ${}^1\mathrm{TT}/\mathrm{S_1}$ (TF/SF).

	SEET		TEET		*J*		TF/SF	
dimer	cryst.	TM	cryst.	TM	cryst.	TM	cryst.	TM
stacked	18.2	27.4	7.2	6.3	7.5	6.4	0.0	65.3
heringbone	30.6	34.1	0.3	0.2	0.0	0.0	0.0	1.1
a-dimer	17.8	19.3	0.0	0.0	0.0	0.0	0.0	0.0
ac-dimer	9.8	10.5	0.0	0.0	0.0	0.0	0.0	0.0

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B.3 Electronic Coupling Distributions

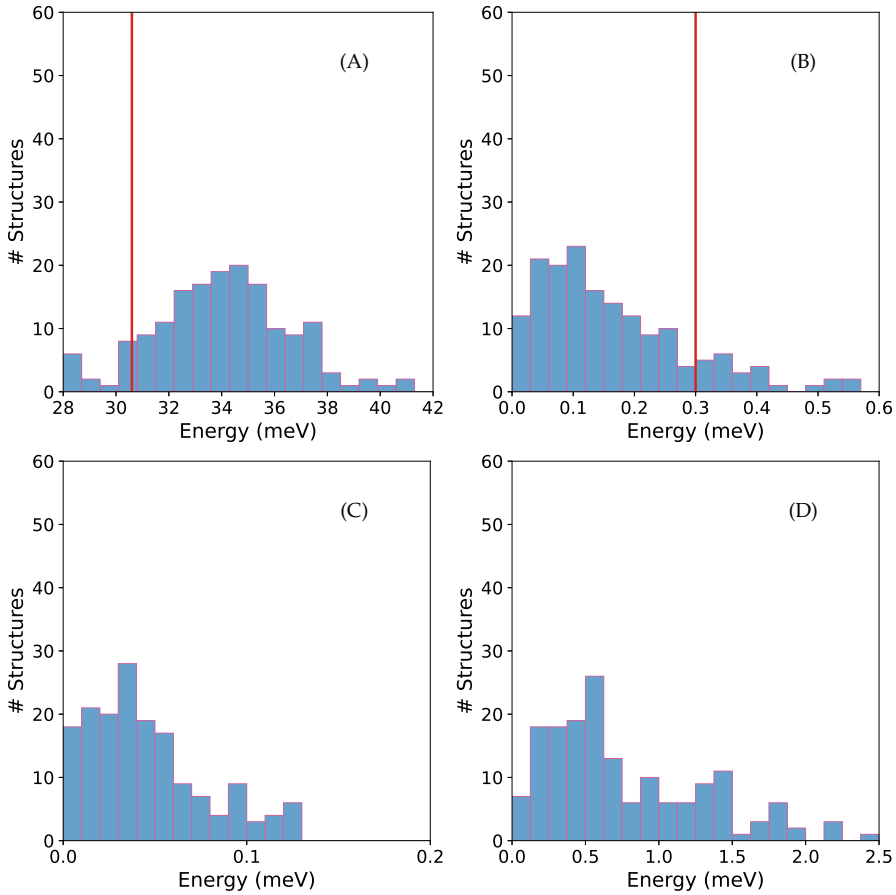


FIGURE B.1: Distribution of absolute electronic couplings (in meV) in herringbone dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) *J*, and (D) TF/SF.

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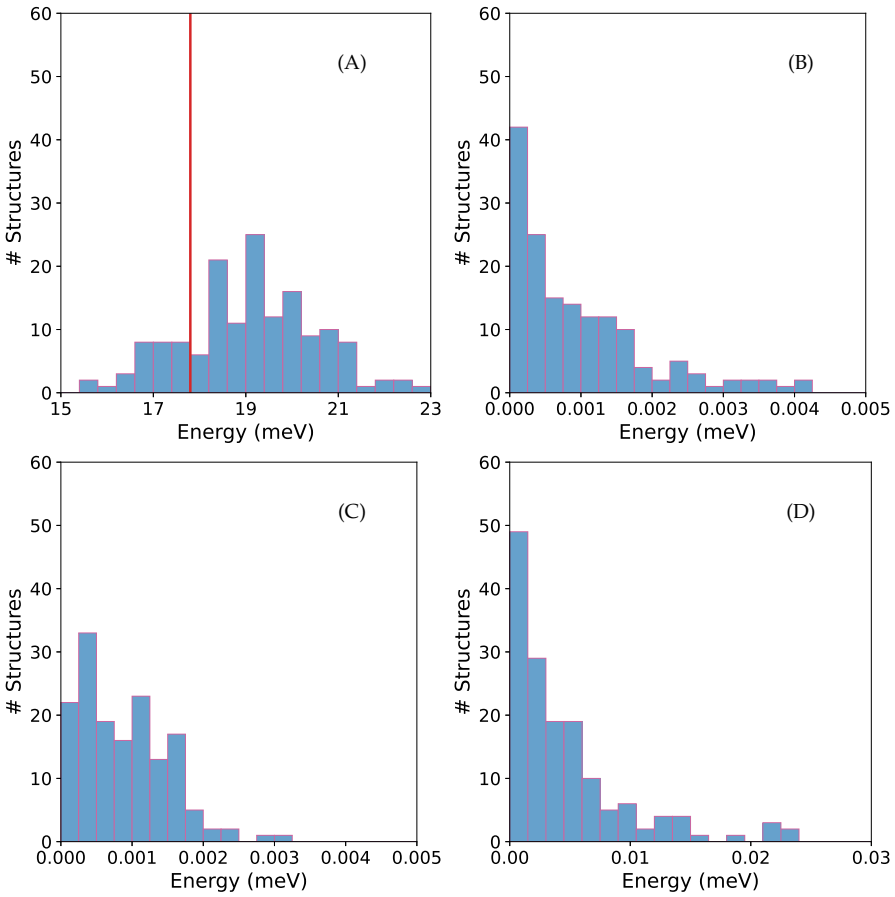


FIGURE B.2: Distribution of absolute electronic couplings (in meV) in *a*-dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) *J*, and (D) TF/SF.

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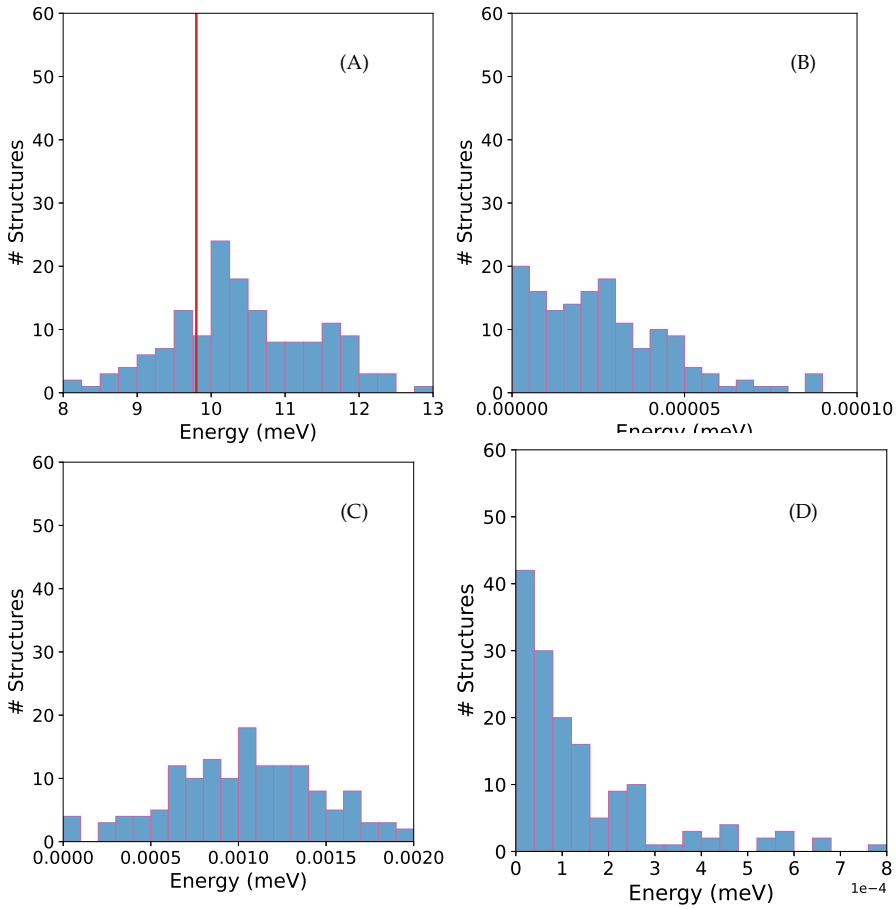


FIGURE B.3: Distribution of absolute electronic couplings (in meV) in *ac*-dimers computed from 200 MD frameworks. (A) SEET, (B) TEET, (C) *J*, and (D) TF/SF.

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B.4 Additional PCA of SEET Couplings

The PCA for herringbone (Figure B.4), *a*-dimer (Figure B.5) and *ac*-dimer (Figure B.6) shows that there is no clear molecular orbital energies to SEET coupling relationships. This shows that the molecular orbital interaction, present in stacked dimers, impacts the SEET coupling. This agrees with the study of different methods to calculate SEET. In fact, the difference between HEG and other methods for stacked dimers is a result of the orbital interaction, this is, the orbtial energies to SEET coupling relationship described by PC3.

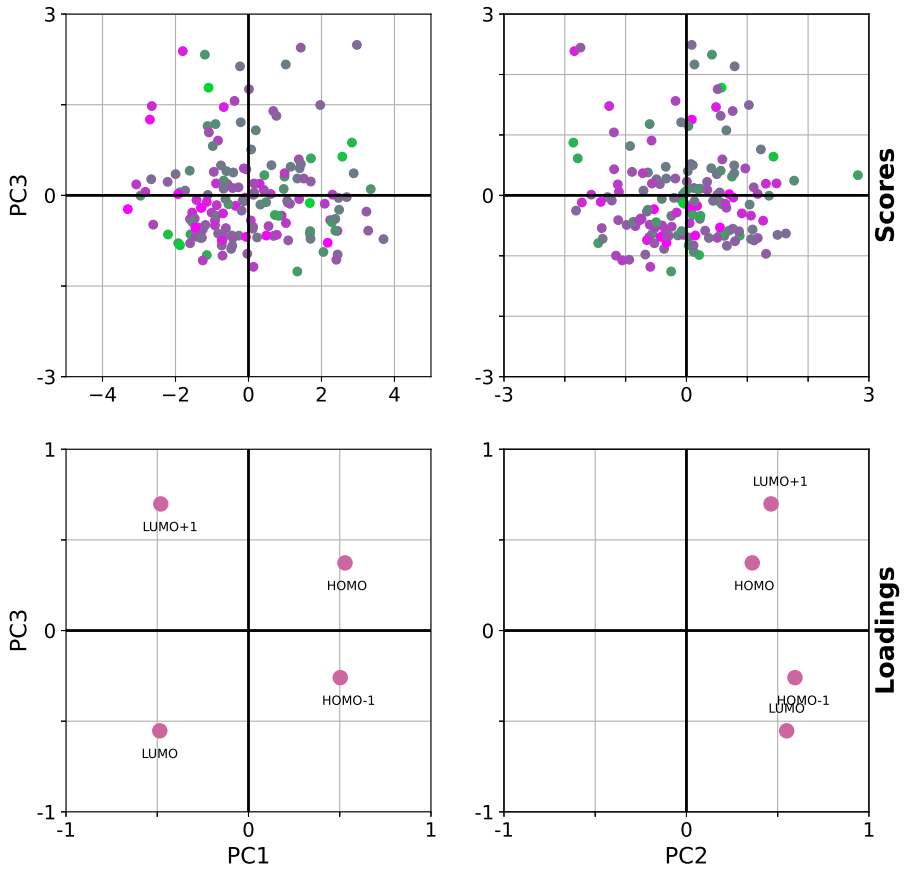


FIGURE B.4: PCA analysis of SEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks.

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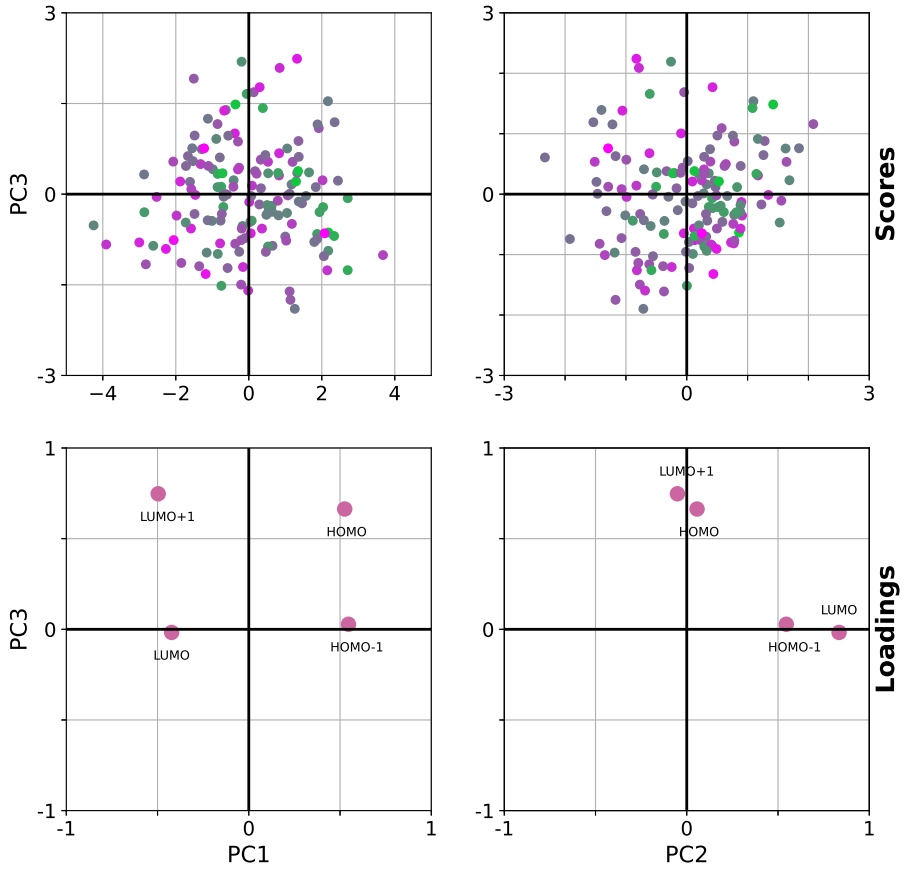


FIGURE B.5: PCA analysis of SEET coupling (in meV) in *a*-dimers computed from 200 MD frameworks.

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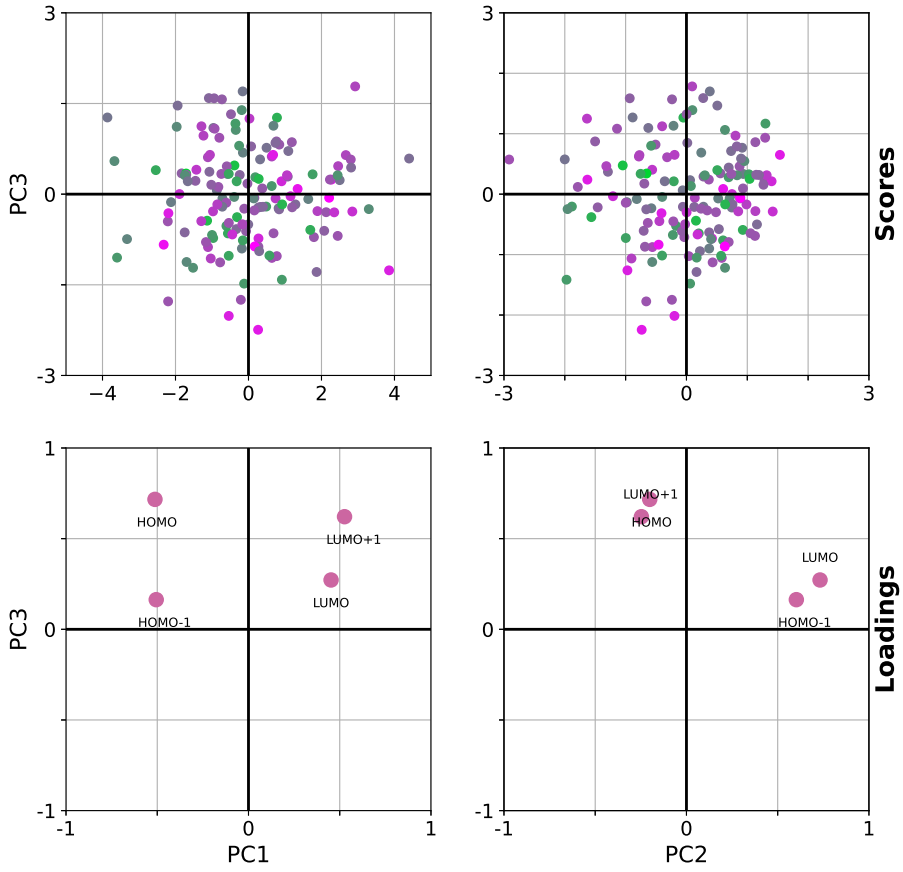


FIGURE B.6: PCA analysis of SEET coupling (in meV) in *ac*-dimers computed from 200 MD frameworks.

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B.5 PCA Analysis of TEET Couplings

The PCA analysis of the electronic couplings of TEET finds the inverse correlation for stacked dimers (Figure B.7) with respect to the SEET couplings (Figure 4.5). However, no correlation between the PCs and other structures is found. These results agree with the analysis of the SEET couplings.

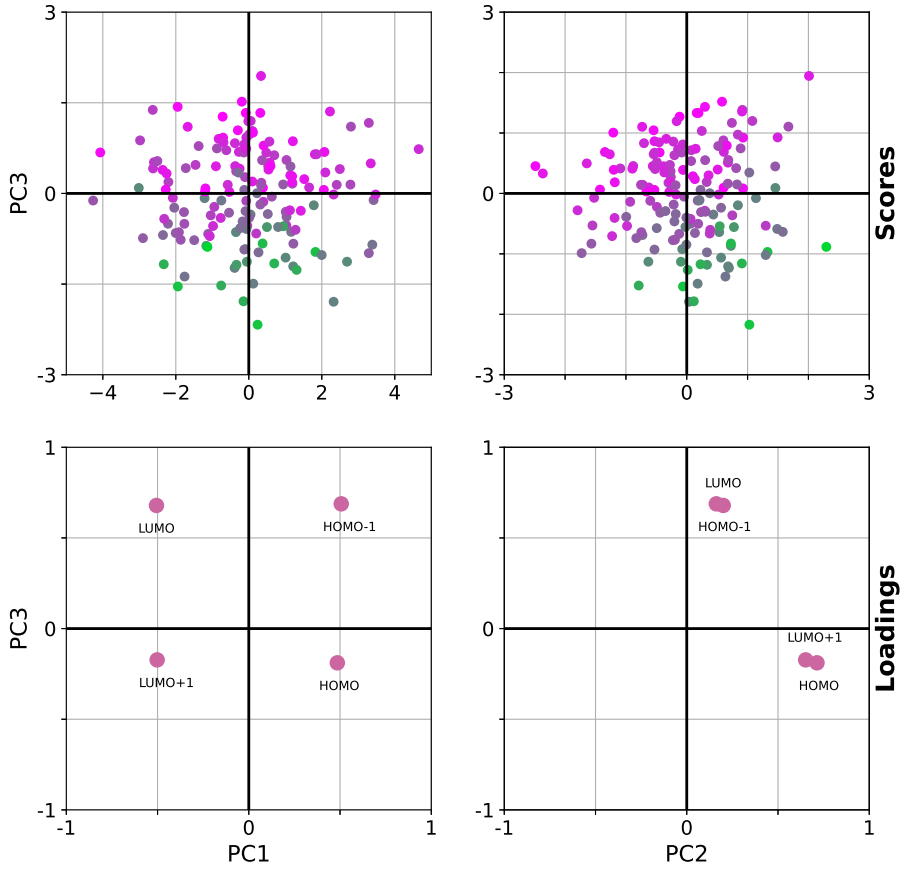


FIGURE B.7: PCA analysis of TEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks.

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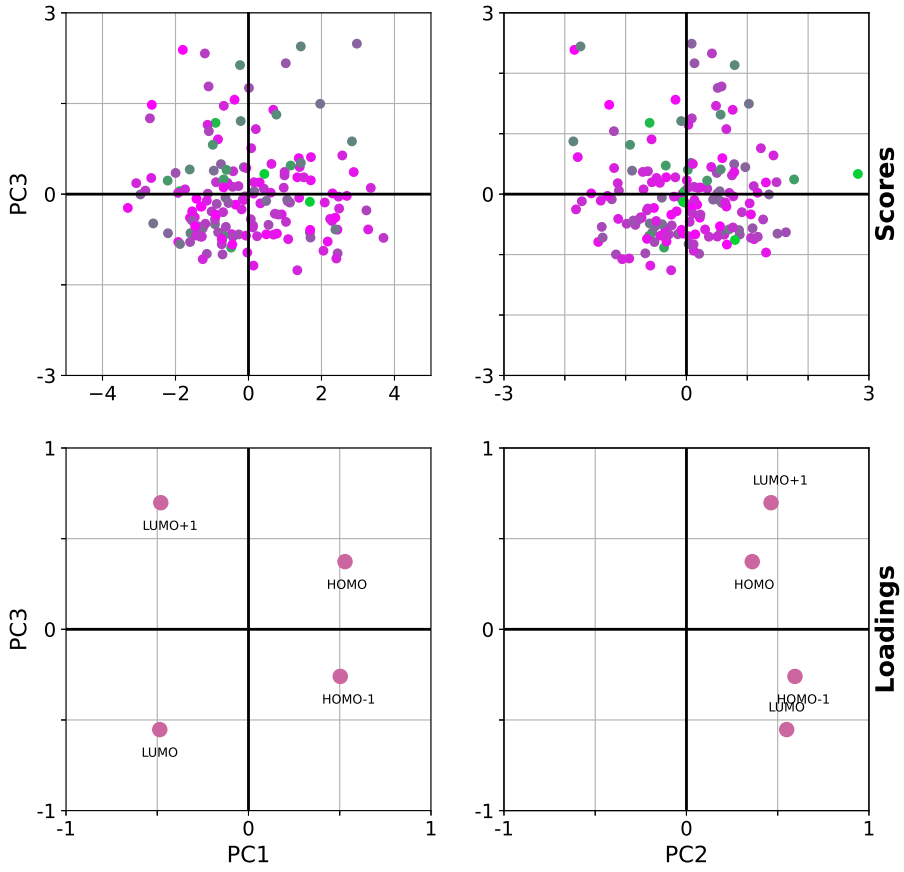


FIGURE B.8: PCA analysis of TEET coupling (in meV) in herringbone dimers computed from 200 MD frameworks.

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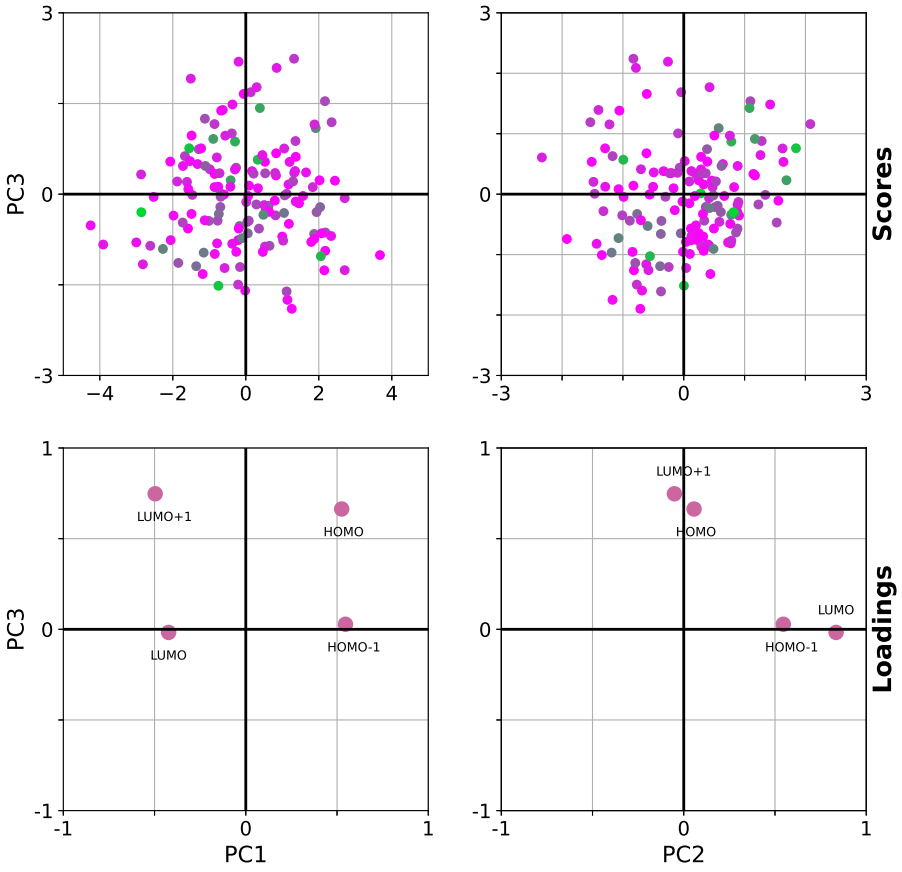


FIGURE B.9: PCA analysis of TEET coupling (in meV) in *a*-dimers computed from 200 MD frameworks.

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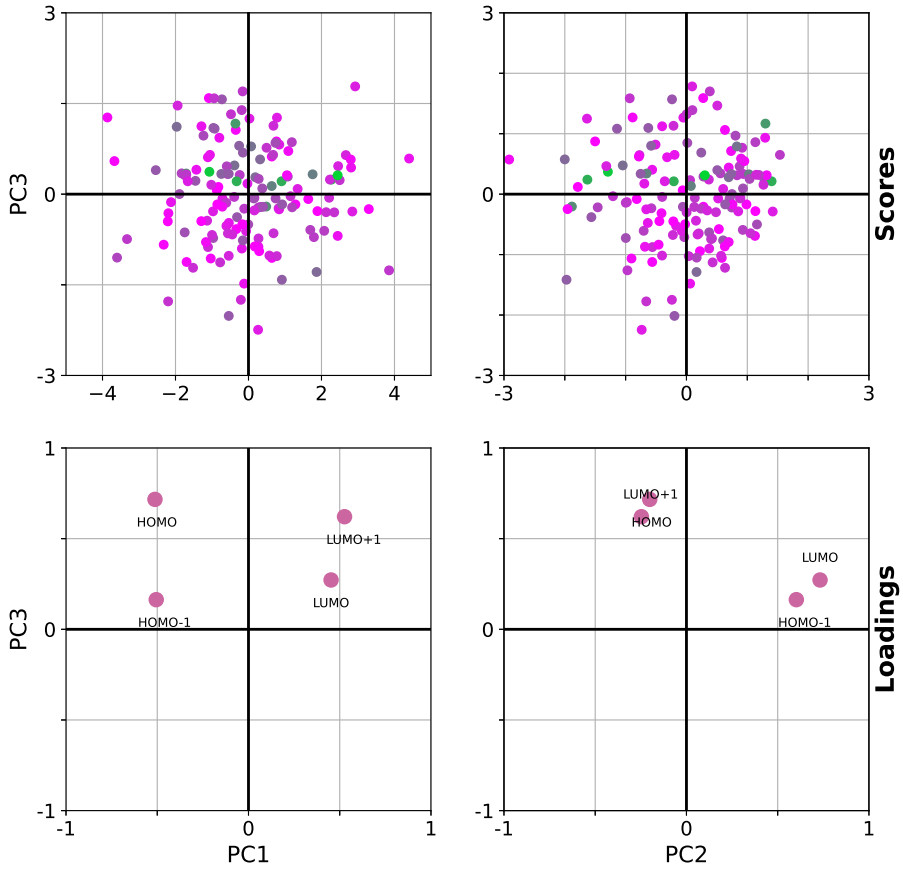


FIGURE B.10: PCA analysis of TEET coupling (in meV) in *ac*-dimers computed from 200 MD frameworks.

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Appendix C

Supplementary Material for Chapter 5

C.1 MSE and R^2 Values of the Discrete Exploration

TABLE C.1: MSE values of the discrete exploration for the optimization of the hyperparameters of the KRR model. Value in purple cell indicate the minimum with $\lambda = 10^{-3}$ and $\sigma = 100$.

λ/σ	10000	2000	1000	200	100	20	10
0	229.36	150.06	118.13	86.80	77.16	19.92	16.10
1	85.77	84.37	83.14	43.33	12.21	6.20	6.22
2	84.23	73.77	47.56	6.82	6.33	6.20	7.99
3	75.28	10.45	6.97	6.22	6.18	9.77	12.16
4	11.70	6.62	6.39	6.32	6.55	17.37	12.46
5	6.67	6.35	6.28	6.98	10.13	18.05	12.46
6	6.46	6.33	6.56	11.54	31.52	18.05	12.46
7	6.38	6.68	7.53	24.05	38.49	18.05	12.46
8	6.52	14.60	15.26	26.38	38.58	18.05	12.46

TABLE C.2: R^2 values of the discrete exploration for the optimization of the hyperparameters of the KRR model. Value in purple cell indicate the minimum with $\lambda = 10^{-3}$ and $\sigma = 100$.

λ/σ	10000	2000	1000	200	100	20	10
0	-1.7238	-0.7820	-0.4029	-0.0308	0.0837	0.7634	0.8088
1	-0.0186	-0.0019	0.0127	0.4855	0.8549	0.9264	0.9261
2	-0.0003	0.1239	0.4352	0.9190	0.9248	0.9264	0.9051
3	0.1060	0.8759	0.9172	0.9261	0.9266	0.8840	0.8557
4	0.8611	0.9214	0.9241	0.9249	0.9222	0.7937	0.8521
5	0.9208	0.9246	0.9254	0.9171	0.8797	0.7857	0.8520
6	0.9233	0.9249	0.9221	0.8630	0.6257	0.7856	0.8520
7	0.9242	0.9207	0.9106	0.7144	0.5429	0.7856	0.8520
8	0.9226	0.8266	0.8188	0.6867	0.5418	0.7856	0.8520

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C.2 Direct and Indirect TF/SF Contributions in Amorphous Solid

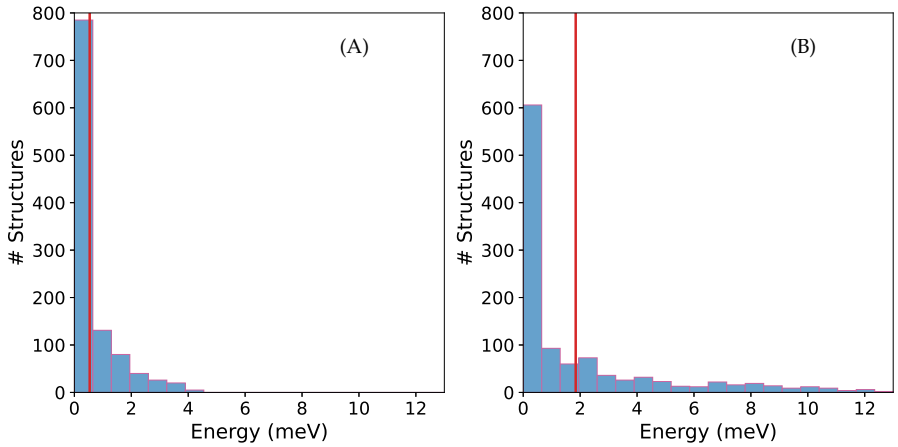


FIGURE C.1: Distribution of absolute direct (A) and indirect (B) contributions of the TF/SF electronic couplings (in meV) extracted from 9400 MD frameworks of amorphous dimers. Vertical red lines indicate mean values.

C.3 Additional Simulations

TTA process in simulations starting with only triplets is not observed because the movement of the T_1 is restricted to mainly one direction. Therefore, if the spawned triplets are not located in the same row of molecules in that direction, they are not likely to interact.

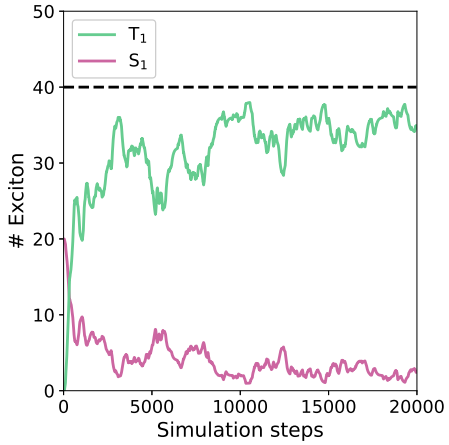


FIGURE C.2: Simulation of TF/SF starting with 20 S_1 excitons.

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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

Bibliography

(1) Lim, J. W. *Materials* **2024**, *17*, 3698.

(2) Nowsherwan, G. A.; Ali, Q.; Ali, U. F.; Ahmad, M.; Khan, M.; Hussain, S. S. *Organics* **2024**, *5*, 520–560.

(3) Kang, K.; Byeon, I.; Kim, Y. G.; Choi, J.-r.; Kim, D. *Laser Photonics Rev.* **2024**, *18*, 2400547.

(4) Deori, U.; Nanda, G. P.; Murawski, C.; Rajamalli, P. *Cell Signal.* **2024**, *15*, 17739–17759.

(5) Solak, E. K.; Irmak, E. *RSC Adv.* **2013**, *13*, 12244–12269.

(6) Guo, Z.; Zhang, J.; Liu, X.; Wang, L.; Xiong, L.; Huang, J. *Adv. Funct. Mater.* **2023**, *33*, 2305508.

(7) Xu, Y.; Liu, D.; Wang, M. *Chem. Eur. J.* **2023**, e202301294.

(8) Zhang, X.; Jiang, J.; Feng, B.; Song, H.; Shen, L. *J. Mater. Chem. C* **2023**, *11*, 12453–12465.

(9) Chen, Z.; Obaid, S. N.; Lu, L. *Opt. Mater. Express* **2019**, *9*, 3843–3856.

(10) Côté, A. P.; Benin, A. I.; Ockwig, N. W.; O'Keeffe, M.; Matzger, A. J.; Yaghi, O. M. *Science* **2005**, *310*, 1166–1170.

(11) Ding, S.-Y.; Wang, W. *Chem. Soc. Rev.* **2013**, *42*, 548–568.

(12) Ostroverkhova, O. *Chem. Rev.* **2016**, *22*, 13279–13412.

(13) Kasha, M. *Discuss. Faraday Soc.* **1950**, *9*, 14–19.

(14) Lakowicz, J. R., *Principles of Fluorescence Spectroscopy*; Springer: 2006.

(15) Casanova, D. *Chem. Rev.* **2018**, *118*, 7164–7207.

(16) Sanders, S. N.; Pun, A. B.; Parenti, K. R.; Kumarasamy, E.; Yablon, L. M.; Sfeir, M. Y.; Campos, L. M. *Chem* **2019**, *5*, 1988–2005.

(17) Maity, N.; Kim, W.; Panjwani, N. A.; Kundu, A.; Majumder, K.; Kasetty, P.; Mishra, D.; Bittl, R.; Nagesh, J.; Dasgupta, J.; Musser, A. J.; Patil, S. *Nat. Commun.* **2022**, *13*, 5244.

(18) Parker, C. A.; Hatchard, C. G. *Trans. Faraday Soc.* **1961**, *57*, 1894–1904.

(19) Phan Huu, D. K. A.; Saseendran, S.; Dhali, R.; Franca, L. G.; Stavrou, K.; Monkman, A.; Painelli, A. *J. Am. Chem. Soc.* **2022**, *144*, 15211–15222.

(20) Northrop Jack Oliver Donald B., M. I. *J. Vis. Exp* **2022**, e63433.

(21) Bai, S.; Zhang, P.; Beratan, D. N. *J. Phys. Chem. C* **2020**, *124*, 18956–18960.

(22) Saha, S.; Mackintosh, M. J.; Thompson, L. M.; Kozlowski, P. M. *J. Phys. Chem. A* **2025**, *129*, 967–977.

(23) Liu, S.; Wu, H.; Zhang, X.; Hu, W. *Front. Phys.* **2020**, *16*, 13304.

(24) Huang, L.; Liao, Q.; Shi, Q.; Fu, H.; Ma, J.; Yao, J. *J. Mater. Chem.* **2010**, *20*, 159–166.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

(25) Fielitz, T. R.; Holmes, R. J. *Crystal Grouth & Design* **2016**, *16*, 4720–4726.

(26) Ma, L.; Zhang, K.; Kloc, C.; Sun, H.; Soci, C.; Michel-Beyerle, M. E.; Gurzadyan, G. G. *Phys. Rev. B* **2013**, *87*, 201203.

(27) Sutton, C.; Tummala, N. R.; Beljonne, D.; Brédas, J.-L. *Chem. Mater.* **2017**, *29*, 2777–2787.

(28) Maslennikov, D. R.; Maimaris, M.; Ning, H.; Zheng, X.; Mondal, N.; Bruevich, V. V.; Pratik, S. M.; Dong, Y.; Tisch, J. W. G.; Musser, A. J.; Podzorov, V.; Bredas, J.-L.; Coropceanu, V.; Bakulin, A. A. *J. Am. Chem. Soc.* **2025**, *147*, 23536–23544.

(29) Ma, L.; Zhang, K.; Kloc, C.; Sun, H.; Michel-Beyerle, M. E.; Gurzadyan, G. G. *Phys. Chem. Chem. Phys.* **2012**, *14*, 8307–8312.

(30) Chakkamalayath, J.; Kamat, P. V. *J. Am. Chem. Soc.* **2024**, *146*, 18095–18103.

(31) Baronas, P.; Kreiza, G.; Naimovičius, L.; Radiunas, E.; Kazlauskas, K.; Orentas, E.; Juršėnas, S. *J. Phys. Chem. C* **2022**, *126*, 15327–15335.

(32) Li, J.; Chen, Z.; Zhang, Q.; Xiong, Z.; Zhang, Y. *Org. Electron.* **2015**, *26*, 213–217.

(33) Finton, D. M.; Wolf, E. A.; Zoutenbier, V. S.; Ward, K. A.; Biaggio, I. *AIP Advances* **2019**, *9*, 095027.

(34) Sher, P.-H.; Chen, C.-H.; Chiu, T.-L.; Lin, C.-F.; Wang, J.-K.; Lee, J.-H. *J. Phys. Chem. C* **2019**, *123*, 3279–3284.

(35) Wu, T.; Ni, W.; Gurzadyan, G. G.; Sun, L. *RSC Adv.* **2021**, 4639–4645.

(36) Refaely-Abramson, S.; da Jornada, F. H.; Louie, S. G.; Neaton, J. B. *Phys. Rev. Lett.* **2017**, *119*, DOI: 10.1103/PhysRevLett.119.267401.

(37) Patra, R.; Das, M. *J. Phys. Chem. A* **2024**, *128*, 7375–7383.

(38) Bortz, A.; Kalos, M.; Lebowitz, J. *J. Comput. Phys.* **1975**, *17*, 10–18.

(39) Voter, A. F. In *Radiation Effects in Solids*, 2007, pp 1–23.

(40) Martin, P.; Gaitero, J. J.; Dolado, J. S.; Manzano, H. *ACS Earth Space Chem.* **2021**, *5*, 516–524.

(41) De Sousa, L. E.; da Silva Filho, D. A.; de Sousa, R. T.; de Oliveira Neto, P. H. *Sci. Rep.* **2018**, *8*, 14066.

(42) Yang, W.; Pursglove De Castro, C. S.; Karuthedath, S.; Firdaus, Y.; Alshehri, N.; Chen, S.; Rosas Villalva, D.; Petoukhoff, C. E.; Dahman, A.; Baran, D.; Anthopoulos, T. D.; Laquai, F.; Gorenflot, J. *Adv. Energy Mater.* **2025**, 2405322.

(43) Dirac, P. A. M. *Proc. R. Soc. London A* **1927**, *114*, 243–265.

(44) Fermi, E., *Nuclear Physics*; University Of Chicago Press: 1974.

(45) Marcus, R. A. *J. Chem. Phys.* **1956**, *24*, 966–978.

(46) Anzola, M.; Sissa, C.; Painelli, A.; Hassanali, A. A.; Grisanti, L. *J. Chem. Theory Comput.* **2020**, *16*, 7281–7288.

(47) Cave, R. J.; Newton, M. D. *Chem. Phys. Lett.* **1996**, *249*, 15–19.

(48) Voityuk, A. A.; Rösch, N. *J. Chem. Phys.* **2002**, *12*, 5607–5616.

(49) Hsu, C.-P.; You, Z.-Q.; Chen, H.-C. *J. Phys. Chem. C* **2008**, *112*, 1204–1212.

(50) Head-Gordon, M.; Grana, A. M.; Maurice, D.; White, C. A. *J. Phys. Chem.* **1995**, *99*, 14261–14270.

(51) You, Z.-Q.; Hsu, C.-P. *J. Chem. Phys.* **2010**, *133*, 074105.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

(52) Subotnik, J. E.; Cave, R. J.; Steele, R. P.; Shenvi, N. *J. Chem. Phys.* **2009**, *130*, 234102.

(53) Subotnik, J. E.; Vura-Weis, J.; Sodt, A. J.; Ratner, M. A. *J. Phys. Chem. A* **2010**, *114*, 8665–8675.

(54) Smith, M. B.; Michl, J. *Annu. Rev. Phys. Chem.* **2013**, *64*, DOI: 10.1146/annurev-physchem-040412-110130.

(55) Gray, V.; Moth-Poulsen, K.; Albinsson, B.; Abrahamsson, M. *Coord. Chem. Rev.* **2018**, *362*, 54–71.

(56) Gray, V.; Dzebo, D.; Abrahamsson, M.; Albinsson, B.; Moth-Poulsen, K. *Phys. Chem. Chem. Phys.* **2014**, *16*, 10345–10352.

(57) Piland, G. B.; Burdett, J. J.; Kurunthu, D.; Bardeen, C. J. *J. Phys. Chem. C* **2013**, *117*, DOI: 10.1021/jp309286v.

(58) Bayliss, S. L.; Chepelianskii, A. D.; Sepe, A.; Walker, B. J.; Ehrler, B.; Bruzek, M. J.; Anthony, J. E.; Greenham, N. C. *Phys. Rev. Lett.* **2014**, *112*, 238701.

(59) Wang, R.; Zhang, C.; Zhang, B.; Liu, Y.; Wang, X.; Xiao, M. *Nat. Commun.* **2015**, 8602.

(60) Tapping, P. C.; Huang, D. M. *J. Phys. Chem. C* **2016**, *120*, 25151–25157.

(61) Bossanyi, D. G.; Sasaki, Y.; Wang, S.; Chekulaev, D.; Kimizuka, N.; Yanai, N.; Clark, J. *JACS Au* **2021**, *1*, 2188–2201.

(62) Small, D. W.; Head-Gordon, M. *Phys. Chem. Chem. Phys.* **2011**, *13*, 19285–19297.

(63) Feng, X.; Luzanov, A. V.; Krylov, A. I. *J. Phys. Chem. Lett.* **2013**, *4*, 3845–3852.

(64) Casanova, D.; Head-Gordon, M. *Phys. Chem. Chem. Phys.* **2009**, *11*, 9779–9790.

(65) Casanova, D. *WIREs: Comput. Mol. Sci.* **2022**, *12*, e1561.

(66) Kähler, S.; Cebreiro-Gallardo, A.; Pokhilko, P.; Casanova, D.; Krylov, A. I. *J. Phys. Chem. A* **2023**, *127*, 8459–8472.

(67) Jurchescu, O. D.; Meetsma, A.; Palstra, T. T. M. *Acta Crystallogr., Sect. B: Struct. Sci.* **2006**, *62*, 330–334.

(68) Groom, C. R.; Bruno, I. J.; Lightfoot, M. P.; Ward, S. C. *Acta Crystallogr., Sect. B: Struct. Sci.* **2016**, *72*, 171–179.

(69) Renaud, N.; Sherratt, P. A.; Ratner, M. A. *J. Phys. Chem. Lett.* **2013**, *4*, 1065–1069.

(70) Mirjani, F.; Renaud, N.; Gorczak, N.; Grozema, F. C. *J. Phys. Chem. C* **2014**, *118*, 14192–14199.

(71) Phillips, J. C. et al. *J. Chem. Phys.* **2020**, *153*, DOI: 10.1063/5.0014475.

(72) Case, D. A. et al. Amber, **2025**.

(73) Case, D. A. et al. *J. Chem. Inf. Model.* **2023**, *63*, 6183–6191.

(74) Case, D. A. et al. *J. Chem. Inf. Model.* **2025**, *65*, 7835–7843.

(75) Pizzirusso, A.; Savini, M.; Muccioli, L.; Zannoni, C. *J. Mater. Chem.* **2011**, *21*, DOI: 10.1039/C0JM01284J.

(76) Pizzirusso, A.; Di Pietro, M. E.; De Luca, G.; Celebre, G.; Longeri, M.; Muccioli, L.; Zannoni, C. *Comp. Phys. Comm.* **2014**, *15*, DOI: 10.1002/cphc.201400082.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

(77) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. *J. Comput. Chem.* **2009**, *30*, 2157–2164.

(78) Epifanovsky, E. et al. *J. Chem. Phys.* **2021**, *155*, 084801.

(79) Frenkel, J. *Phys. Rev.* **1931**, *37*, 17–44.

(80) Förster, T. *Ann. Phys.* **1948**, *437*, 55–75.

(81) Carreras, A. Kimonet, https://github.com/abelcarreras/kimonet, **2019**.

(82) Heiber, M. C. *Journal of Open Source Software* **2020**, *5*, 2307.

(83) Ali, S.; Qayyum, S. *ScienceOpen Preprints* **2021**, DOI: 10.14293/S2199-1006.1.SOR-.PP5RV1O.v1.

(84) Stroustrup, B., *The C++ Programming Language*; Addison-Wesley Professional: 2013.

(85) Dagum, L.; Menon, R. *IEEE: Compute. Sci. Eng.* **1998**, *5*, 46–55.

(86) Sanderson, C.; Curtin, R. *Math. Comput. Appl.* **2019**, *3*, DOI: 10.3390/mca24030070.

(87) Sanderson, C.; Curtin, R. In *2025 17th International Conference on Computer and Automation Engineering (ICCAE)*, 2025, pp 303–307.

(88) Xianyi, Z.; Qian, W.; Yunquan, Z. In *2012 IEEE 18th International Conference on Parallel and Distributed Systems*, 2012, pp 684–691.

(89) Wang, Q.; Zhang, X.; Zhang, Y.; Yi, Q. In *SC '13: Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis*, 2013, pp 1–12.

(90) Fine, T. L., *LAPACK Users' Guide*; Society for Industrial and Applied Mathematics: 1999.

(91) The HDF Group, Hierarchical Data Format, version 5, https://github.com/HDFGroup/hdf5.

(92) Rupp, M.; Tkatchenko, A.; Müller, K.-R.; von Lilienfeld, O. A. *Phys. Rev. Lett.* **2012**, *108*, 058301.

(93) Montavon, G.; Hansen, K.; Fazli, S.; Rupp, M.; Biegler, F.; Ziehe, A.; Tkatchenko, A.; von Lilienfeld, O. A.; Müller, K.-R. In *Proceedings of the 26th International Conference on Neural Information Processing Systems - Volume 1*, 2012, pp 440–448.

(94) Blockeel, H.; Devos, L.; Frénay, B.; Nanfack, G.; Nijssen, S. *Front. Artif. Intell.* **2023**, *6*, 2624–8212.

(95) Allerbo, O. *Electronic Journal of Statistics* **2025**, *19*, 1231–1285.

(96) Fine, T. L., *Feedforward Neural Network Methodology*; Springer-Verlag: 1999.

(97) Dubey, S. R.; Singh, S. K.; Chaudhuri, B. B. *Neurocomputing* **2022**, *503*, 92–108.

(98) Nair, V.; Hinton, G. E. In *Proceedings of the International Conference on Machine Learning*, 2010, pp 807–814.

(99) Diaz-Andres, A.; Tonnelé, C.; Casanova, D. *J. Chem. Theory Comput.* **2024**, *20*, 4297.

(100) Dexter, D. L. *J. Chem. Phys.* **1953**, *21*, 836–850.

(101) Da Silva Filho, D. A.; Kim, E.-G.; Brédas, J.-L. *Adv. Mater.* **2005**, *17*, 1072–1076.

(102) Reese, C.; Bao, Z. *Adv. Mater.* **2007**, *19*, 4535–4538.

Localizatzailea / Localizador	IVVH2KPLYEB6YSVKU5MS6UST6E	Data / Fecha	27/10/2025 15:46:21
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REGISTRO TELEMÁTICO
Sarreren Erregistro Orokorra / Registro General de Entradas
30/10/2025 11:18
EHU2025E051403

(103) Irkhin, P.; Biaggio, I. *Phys. Rev. Lett.* **2011**, *107*, 017402.

(104) Seki, K.; Yoshida, T.; Yago, T.; Wakasa, M.; Katoh, R. *J. Phys. Chem. C* **2021**, *125*, 3295–3304.

(105) Casanova, D. *J. Comput. Chem.* **2013**, *34*, 720–730.

(106) Tamura, H.; Huix-Rotllant, M.; Burghardt, I.; Olivier, Y.; Beljonne, D. *Phys. Rev. Lett.* **2015**, *115*, 107401.

(107) Smith, M. B.; Michl, J. *Chem. Rev.* **2010**, *110*, 6891–6936.

(108) Miyata, K.; Kurashige, Y.; Watanabe, K.; Sugimoto, T.; Takahashi, S.; Tanaka, S.; Takeya, J.; Yanai, T.; Matsumoto, Y. *Nat. Chem.* **2017**, *9*, 983–989.

(109) Ryno, S. M.; Risko, C.; Brédas, J.-L. *ACS Appl. Mater. Interfaces* **2016**, *8*, 14053–14062.

(110) Takeya, J.; Yamagishi, M.; Tominari, Y.; Hirahara, R.; Nakazawa, Y.; Nishikawa, T.; Kawase, T.; Shimoda, T.; Ogawa, S. *Appl. Phys. Lett.* **2007**, *90*, 102120.

(111) Liu, Y.; Yang, X.; Ye, L.; Ma, H.; Zhu, H. *Aggregate* **2023**, *4*, e347.

(112) Volek, T. S.; Armstrong, Z. T.; Sowa, J. K.; Wilson, K. S.; Bohlmann Kunz, M.; Bera, K.; Koble, M.; Frontiera, R. R.; Rossky, P. J.; Zanni, M. T.; Roberts, S. T. *J. Phys. Chem. Lett.* **2023**, *14*, 11497–11505.

(113) Van der Lee, A.; Polentarutti, M.; Roche, G. H.; Dautel, O. J.; Wantz, G.; Castet, F.; Muccioli, L. *J. Phys. Chem. Lett.* **2022**, *13*, 406–411.

(114) Bittner, E. R.; Silva, C. *Nat. Commun.* **2014**, *5*, 3119.

(115) Liu, T.; Troisi, A. *Adv. Funct. Mater.* **2014**, *24*, 925–933.

(116) Shuai, Z.; Geng, H.; Xu, W.; Liao, Y.; André, J.-M. *Chem. Soc. Rev.* **2014**, *43*, 2662–2679.

(117) Yavuz, I.; Martin, B. N.; Park, J.; Houk, K. N. *J. Am. Chem. Soc.* **2015**, 2856–2866.

(118) Irkhin, P.; Najafov, H.; Podzorov, V. *Sci. Rep.* **2015**, *5*, 15323.

(119) Tamura, H.; Azumaya, K.; Ishikita, H. *J. Phys. Chem. Lett.* **2019**, *10*, 7623–7628.

(120) Herkstroeter, W. G.; Merkel, P. B. *J. Photochemistry* **1981**, *16*, 331–342.

(121) Amashukeli, X.; Winkler, J. R.; Gray, H. B.; Gruhn, N. E.; Lichtenberger, D. L. *J. Phys. Chem. A* **2002**, *106*, 7593–7598.

(122) Chen, K.; Kunkel, C.; Reuter, K.; Margraf, J. T. *Digital Discovery* **2022**, *1*, 147–157.

(123) Bossanyi, D. G.; Sasaki, Y.; Wang, S.; Chekulaev, D.; Kimizuka, N.; Yanai, N.; Clark, J. *J. Mater. Chem. C* **2022**, *10*, 4684–4696.

(124) Chen, C.-H.; Sher, P.-H.; Chen, C.-P.; Choi, W.-K.; Bardeen, C. J.; Chiu, T.-L.; Wang, J.-K.; Lee, J.-H. *J. Phys. Chem. C* **2020**, *124*, 25729–25737.

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